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Chapter 1 AMINO ACIDS OF BIOLOGICAL IMPORTANCE

1-Proteins containing all the essential amino acids have:

a) High biological value
b) Moderate biological value
c) Low biological value
d) Low molecular weight

(End of module exam 2023)

2-Two sulfur containing amino acids:

a) Cysteine and serine.
b) Cysteine and threonine.
c) Methionine and serine.
d) Methionine and cysteine.

(End of module exam 2023)

3-Binding of 2 Cysteines together results in formation of:

a) Methionine
b) Cystine
c) Homocysteine
d) Arginine

(End of module exam 2023)

4-Amino acids that are formed in the body at a rate enough for adults but not enough for growing infants:

a) Essential amino acids
b) Semi-essential amino acids
c) Non-essential amino acids
d) Glucogenic amino acids

(End of module exam 2023)

5-One of the following is a pure ketogenic amino acid:

a) Tyrosine
b) Methionine
c) Leucine
d) Tryptophan

(End of module exam 2023)

6-A zwitter ion is:

a) Amino acid at acidic medium
b) Amino acid at alkaline medium
c) Amino acid at isoelectric point
d) None of the above

(End of module exam 2023)

7-Which of the following is NOT an essential amino acid?

a) Tryptophan
b) Methionine
c) Lysine
d) Serine

(End of module exam 2021)

8-The two amino acids that contain sulfur atoms are:

a) Cysteine and methionine
b) Cysteine and serine
c) Methionine and threonine
d) Methionine and serine

(End of module exam 2021, Final exam 2019 - September sitting)

9-One of the following is a hydroxyl-containing amino acid:

a) Serine
b) Aspartate
c) Valine
d) Arginine

(End of module exam 2021)

10-The semi-essential (half-essential) amino acid is:

a) Glutamate
b) Tyrosine
c) Arginine
d) Cysteine

(End of module exam 2021)

11-The two amino acids that are purely ketogenic are:

a) Lysine and leucine
b) Lysine and isoleucine
c) Leucine and isoleucine
d) Lysine and valine

(Final exam 2021)

12-Sulphur containing amino acids are

a) Cysteine and methionine
b) Cysteine and threonine
c) Methionine and threonine
d) Cysteine and serine

(End of the module 2019)

13-Amino acids with hydroxyl groups are

a) Serine and alanine
b) Alanine and valine
c) Serine and threonine
d) Valine and isoleucine

(End of the module 2019)

14-The only half-essential (semi-essential) amino acid is:

a) Aspartic acid
b) Arginine
c) Leucine
d) Methionine

(End of the module 2019)

15-Sulphur containing amino acid is:

a) Methionine
b) Leucine
c) Valine
d) Asparagine

(End of the module 2020)

16-Which of the following is NOT an aromatic amino acid?

a) Threonine
b) Phenylalanine
c) Tryptophan
d) Tyrosine

(End of the module 2020)

17-One of the following is NOT an effect of denaturation:

a) Increased viscosity
b) Loss of the primary, secondary, tertiary structure of proteins.
c) Increased digestibility
d) Loss of biologic activity

(End of the module 2020)

18-The Zwitter ion is:

a) The dipolar ion
b) Positively charged ions
c) Negatively charged ions
d) Non charged ions

(End of the module 2020)

19-One of the following is not a hydroxy-containing amino acids:

a) Serine
b) Threonine
c) Valine
d) Tyrosine

(Final exam 2019)

20-The 21st amino acid is:

a) hydroxy lysine
b) hydroxyl proline.
c) selenocysteine
d) citrulline

(End of module exam 2021)

21-Two amino acids can be covalently joined through:

a) Hydrogen bond.
b) Hydrophobic interaction.
c) Peptide bond.
d) Ionic bond.

(Old Depart. book)

22-Which of the following amino acids carries a positive charge on its side chain at pH 7.4:

a) Aspartate.
b) Lysine.
c) Asparagine.
d) Glutamate.

(Old Depart. book)

23-Proteins are:

a) Organic compounds with low molecular weight
b) Organic compounds with high molecular weight
c) Inorganic compounds with low molecular weight
d) Inorganic compounds with high molecular weight

(Depart. Questions book)

24-Synthesis of proteins commonly require:

a) 100 different amino acids
b) 300 different amino acids
c) 20 different amino acids
d) 30 different amino acids

(Depart. Questions book)

25-Which of the following is an example of α-amino acids?

a) Phenylalanine
b) Proline
c) Hydroxyproline
d) Glutathione

(Depart. Questions book)

26-Which of the following is an imino acid?

a) Alanine
b) Glutamate
c) Proline
d) Serine

(Depart. Questions book)

27-Amino acids can be classified by:

a) Physical classification
b) Pathological classification
c) Chemical classification
d) Quantitative classification

(Depart. Questions book)

28-Glycine is:

a) Neutral aliphatic amino acid
b) Basic aliphatic amino acid
c) Acidic aliphatic amino acid
d) Acidic aromatic amino acid

(Depart. Questions book)

29-Histidine is an example of:

a) Aromatic amino acid
b) Heterocyclic amino acid
c) Branched-chain amino acid
d) Non-branched-chain amino acid

(Depart. Questions book)

30-Selenocysteine contains:

a) Cysteine in place of lysine
b) Cysteine in place of methionine
c) Selenium in place of sulfur
d) Selenium in place of serine

(Depart. Questions book)

31-Valine is:

a) Aliphatic branched amino acid
b) Aliphatic non-branched amino acid
c) Aromatic amino acid
d) Heterocyclic amino acid

(Depart. Questions book)

32-Which of the following amino acids contain a hydroxyl group?

a) Arginine
b) Serine
c) Cysteine
d) Methionine

(Depart. Questions book)

33-An amino acid containing a hydroxyl group is

a) Threonine
b) Methionine
c) Glutathione
d) Hydroxyproline

(Depart. Questions book)

34-Which of the following amino acids contain a sulfur atom?

a) Serine
b) Threonine
c) Methionine
d) Aspartate

(Depart. Questions book)

35-Which of the following is formed when 2 cysteine are conjugated by a disulfide bond?

a) Homo cysteine
b) Acetyl cysteine
c) Cystine
d) Homo cystine

(Depart. Questions book)

36-An amino acid containing an acidic group:

a) Arginine
b) Glycine
c) Pyruvate
d) Aspartate

(Depart. Questions book)

37-An amino acid containing a basic group:

a) Lysine
b) Proline
c) Glycine
d) Alanine

(Depart. Questions book)

38-An amino acid containing an aromatic ring:

a) Histidine
b) Proline
c) Alanine
d) Phenylalanine

(Depart. Questions book)

39-An example of heterocyclic amino acid:

a) Histidine
b) Cysteine
c) Tyrosine
d) Methionine

(Depart. Questions book)

40-Serine is an amino acid with:

a) Non-polar group
b) Uncharged polar group
c) Charged polar group
d) Hydrophobic group

(Depart. Questions book)

41-Arginine is an:

a) Amino acid with uncharged polar group
b) Amino acid with charged polar group
c) Amino acid with non-polar group
d) Amino acid with hydrophobic group

(Depart. Questions book)

42-Amino acids are nutritionally classified into:

a) Essential, non-essential, and half-essential amino acids
b) Ketogenic, glucogenic, and mixed amino acids
c) Aliphatic, aromatic, and heterocyclic amino acids
d) Non-polar, uncharged polar, and charged polar amino acids

(Depart. Questions book)

43-Valine is:

a) Essential amino acid
b) Non-essential amino acid
c) Semi-essential amino acid
d) Mixed amino acid

(Depart. Questions book)

44-The only half-essential amino acid is:

a) Alanine
b) Arginine
c) Asparagine
d) Aspartate

(Depart. Questions book)

45-Which of the following is a non-essential amino acid?

a) Lysine
b) Leucine
c) Arginine
d) Tyrosine

(Depart. Questions book)

46-Amino acids are classified metabolically into:

a) Ketogenic, glucogenic, and mixed amino acids
b) Acidic, alkaline, and neutral amino acids
c) Aliphatic, aromatic, and heterocyclic amino acids
d) Essential, non-essential, and semi-essential amino acids

(Depart. Questions book)

47-Lysine is:

a) Pure ketogenic amino acid
b) Pure glucogenic amino acid
c) Pure galactogenic amino acid
d) Glucogenic and ketogenic amino acid

(Depart. Questions book)

48-Phenylalanine is:

a) Pure galactogenic amino acid
b) Pure glucogenic amino acid
c) Pure ketogenic amino acid
d) Glucogenic and ketogenic amino acid

(Depart. Questions book)

49-Amino acid is:

a) Positively charged in acidic medium
b) Positively charged in alkaline medium
c) Negatively charged in acidic medium
d) Uncharged in alkaline medium

(Depart. Questions book)

50-The Isoelectric Point (IEP) of all monoamino-monocarboxylic amino acids is at pH:

a) 10.02
b) 6.02
c) 2.02
d) 30.02

(Depart. Questions book)

51-Peptide bond is formed between the carboxylic group of one amino acid:

a) With the carboxylic group of the other
b) With the amino group of the other
c) With the hydroxylic group of the other
d) With the amidic group of the other

(Depart. Questions book)

52-Amino acids are linked together by:

a) Ionic non-covalent bond
b) Hydrogen non-covalent bond
c) Hydrogen covalent bond
d) Peptide covalent bond

(Depart. Questions book)

53-Which of the following is a branched amino acid?

a) Cysteine.
b) Isoleucine.
c) Proline.
d) Histidine.

(Old Depart. Questions book)

54-Which of the following amino acids is known as half-cystine residue?

a) Cysteine.
b) Isoleucine.
c) Valine.
d) Histidine.

(Old Depart. Questions book)

55-Serine and threonine are polar amino acids due to:

a) Reactive hydroxyl group in the side chain.
b) Reactive thiol group in the side chain.
c) Reactive amide group in the side chain.
d) Reactive carboxyl group in the side chain.

(Old Depart. Questions book)

56-Amino acids are ampholytes because they can function as either a(n):

a) Acid or base.
b) Neutral molecule or ion.
c) Polar or nonpolar.
d) Strand (chain) or branched molecule.

(Old Depart. Questions book)

57-Which of the following is an α-imino acid?

a) Serine.
b) Threonine.
c) Valine.
d) Proline.

(Old Depart. Questions book)

58-Which of the following amino acids contain a heterocyclic ring?

a) Alanine.
b) Valine.
c) Cysteine.
d) Histidine.

(Old Depart. Questions book)

59-All of the amino acids that are found in proteins contain an amino group, except:

a) Proline.
b) Histidine.
c) Glycine.
d) Serine.

(Old Depart. Questions book)

60-All of the following are aliphatic amino acids except:

a) Glycine
b) Alanine
c) Proline
d) Lysine

(Old Depart. Questions book)

61-Lysine is a:

a) Basic ketogenic amino acid
b) Ketogenic and glucogenic amino acid
c) Acidic glucogenic amino acid
d) Non essential amino acid

(Old Depart. Questions book)

62-Proteins of high biological value:

a) Contain all essential amino acids
b) Are poor in essential amino acids
c) Are of plant source
d) Contain amino acid glycine

(Old Depart. Questions book)

63-The following are basic amino acids:

a) Lysine, tryptophan, arginine.
b) Alanine, histidine, glycine.
c) Histidine, lysine, arginine.
d) Arginine, lysine, valine.

(Old Depart. Questions book)

64-Branched chain amino acids include:

a) Glutamine, aspargine, valine.
b) Valine, leucine, threonine.
c) Leucine, isoleucine, valine.
d) Valine, isoleucine, methionine.

(Old Depart. Questions book)

65-A nonessential amino acid containing sulfur is:

a) Methionine.
b) Threonine.
c) Valine.
d) Cysteine.

(Old Depart. Questions book)

66-During denaturation of proteins, all of the following are disrupted except:

a) Primary structure
b) Secondary structure
c) Tertiary structure
d) Quaternary structure

(Old Depart. Questions book)

67-Denaturation of proteins leads to all the following except:

a) Increased viscosity
b) decreased solubility
c) Loss of biological activity.
d) Increased digestibility.

CANCELLED

(Old Depart. Questions book)

68-All of the following are important in maintenance of the secondary, tertiary and quaternary structure of proteins EXCEPT:

a) hydrogen bonds
b) peptide bonds
c) hydrophobic interactions
d) ionic bonds

(Old Depart. Questions book)

69-All of the amino acids that are found in proteins contain an amino group, Except:

a) Proline
b) Histidine
c) Glycine
d) Serine.

(Old Depart. Questions book)

70-Which amino acid can form disulphide bonds?

a) Glycine
b) Proline.
c) Glutamate.
d) Cysteine.

(Old Depart. Questions book)

71-An amino acid which contains a disulphide bond is:

a) Lysine
b) Methionine
c) Homocysteine
d) Cystine

(Old Depart. Questions book)

72-All of the following are polar amino acids Except:

a) Alanine
b) Glutamate
c) Arginine
d) Serine

(Old Depart. Questions book)

73-Acidic amino acids include:

a) Arginine and glutamate
b) Aspartate and glutamate.
c) Aspartate and lysine
d) Aspartate and asparagines

(Old Depart. Questions book)

74-At the isoelectric point (IEP), the amino acids:

a) Have a net positive charge
b) Have a net negative charge
c) Have dipolar ions (Zwitterions)
d) Exist only in uncharged form

(Exam 2026)

75-Examples of essential amino acids include:

a) Valine and Methionine
b) Alanine and Phenylalanine
c) Lysine and Serine
d) Cysteine and Glycine

(Exam 2026)

76-The following are purely ketogenic amino acids:

a) Leucine and Lysine
b) Isoleucine and Tyrosine
c) Valine and Threonine
d) Phenylalanine and Tryptophan

(Exam 2026)

77-Hydroxyproline and hydroxylysine are formed:

a) During transcription
b) Before incorporation into the polypeptide chain
c) By hydroxylation after protein synthesis
d) Only in high biological value proteins

(Exam 2026)

78-One of the following is amide of acidic amino acid:

a) Asparagine
b) Alanine
c) Methionine
d) Valine

(Exam 2024)

79-Which of the following is formed when 2 cysteine are conjugated by disulfide bond?

a) Homo cysteine
b) Acetyl cysteine
c) Cystine
d) Homo cysteine

(Exam 2024)

80-Phenylalanine is:

a) Pure galactogenic amino acid
b) Pure glucogenic amino acid
c) Pure ketogenic amino acid
d) Glucogenic and ketogenic amino acid

(Exam 2024)

81-Which one of the following is OH- containing amino acid:

a) Lysine
b) Serine
c) Aspartic acid
d) Phenylalanine

(Exam 2024)

82-Which of the following is semi-essential amino acid?

a) Alanine.
b) Valine.
c) Arginine.
d) Histidine.

(Exam 2024)

83-An example of heterocyclic amino acid:

a) Histidine
b) Cysteine
c) Tyrosine
d) Methionine

(Exam 2024)

84-Essential sulfur containing amino acid is:

a. Serine
b. Cysteine
c. Methionine
d. Cystine

85-Essential basic amino acid is:

a. Arginine
b. Glycine
c. Methionine
d. Lysine
e. Tryptophan

86-Semi essential amino acid is:

a. Arginine
b. Asparagine
c. Glutamine
d. Histidine
e. Choline

Answers

1-A	2-D	3-B	4-B	5-C	6-C	7-D	8-A	9-A
10-C	11-A	12-A	13-C	14-B	15-A	16-A	17-B	18-A
19-C	20-C	21-C	22-B	23-B	24-C	25-A	26-C	27-C
28-A	29-B	30-C	31-A	32-B	33-A	34-C	35-C	36-D
37-A	38-D	39-A	40-B	41-B	42-A	43-A	44-B	45-D
46-A	47-A	48-D	49-A	50-B	51-B	52-D	53-B	54-A
55-A	56-A	57-D	58-D	59-A	60-C	61-A	62-A	63-C
64-C	65-D	66-A	67-D	68-B	69-A	70-D	71-D	72-A
73-B	74-C	75-A	CANCELLED	77-C	78-A	79-C	80-D	81-B
82-C	83-A	84-C	85-D	86-A				

Chapter 2 PROTEINS OF BIOLOGICAL IMPORTANCE

1-All the following are interactions that stabilize the tertiary structure of proteins except:

a) Van der Waals interaction.
b) Hydrogen bonds.
c) Peptide bonds.
d) Ionic bonds.

(End of module exam 2023)

2-Among the effects of protein denaturation:

a) Increase protein solubility.
b) Increase protein viscosity.
c) Decrease protein digestibility.
d) Increase enzymes activity.

(End of module exam 2023)

3-Denaturation of proteins leads to:

a) Loss of the primary structure of proteins
b) Decreased solubility
c) Decreased viscosity
d) Increased antigenic properties

(Final exam 2022 - September, Faculty summer revision 2024)

4-The primary structure of protein represents:

a) Linear sequence of amino acids joined by peptide bonds
b) 3-dimensional structure of protein
c) Helical structure of protein
d) β-pleated sheet structure

(End of module exam 2021)

5-Which amino acid is most likely to disrupt an alpha helix?

a) Glycine
b) Leucine
c) Proline
d) Valine

(End of module exam 2021)

6-Which bonds do not maintain the tertiary structure of proteins?

a) Hydophobic interactions
b) Van der Waals interactions
c) Disulphide bonds
d) Peptide bonds

(End of module exam 2021)

7-α-helices, β-pleated sheets and β-bends are examples of protein's:

a) Primary structure
b) Secondary structure
c) Tertiary structure
d) Quaternary structure

(End of module exam 2021,Final exam 2021)

8-All of the following bonds stabilize the tertiary structure of a protein Except:

a) Vander Waal's
b) Hydrogen
c) Peptide
d) Ionic

(End of the module 2019)

9-One of the following about protein denaturation is NOT true:

a) It is loss of primary, secondary, tertiary and quaternary structures.
b) It causes loss of biological activity of proteins.
c) It causes increased viscosity of proteins.
d) Strong acids can cause protein denaturation.

(End of the module 2019)

10-Which of the following amino acids is most likely to disrupt an alpha helix?

a) Proline
b) Leucine
c) Glycine
d) Valine

(End of the module 2020)

11-Primary structure of a protein is stabilized by:

a) Hydrogen bonds
b) Peptide bonds
c) Disulfide bonds
d) All of the above

(Depart. book)

12-The effects of denaturation on proteins include:

a) Loss of the primary, secondary, tertiary and quaternary structure of proteins
b) Increased solubility
c) Loss of biologic activity e.g. inactivation of enzymes
d) Decreased digestibility (Depart. book)

13-Which one of the following statements about protein structure is correct?

a) Proteins consisting of one polypeptide chain can have quaternary structure
b) The quaternary structure is stabilized by covalent bonds among the subunits
c) The denaturation of proteins always leads to irreversible loss of secondary and tertiary structure
d) The specific sequence of amino acids along the polypeptide chain is determined genetically

(Depart. book)

14-Secondary structure of proteins contains:

a) Two or more polypeptide chains
b) α-helical or ß-pleated sheets
c) Ionic bonds (salt bridges) between NH3+ of basic A.A & COO- of acidic amino acids
d) Van der Waals interactions

(Faculty Summer revision 2024)

15-Proteins are formed of:

a) 10 or less amino acids
b) 30 or less amino acids
c) 50 or more amino acids
d) From 30 to 40 amino acids

CANCELLED

(Depart. Questions book)

16-When 5 amino acids are linked together by peptide bonds they are called:

a) Tripeptide
b) Polypeptide
c) Oligopeptide
d) Protein

(Depart. Questions book)

17-When 30 amino acids are linked together, they form:

a) Protein
b) Polypeptide
c) Tripeptide
d) Oligopeptide

(Depart. Questions book)

18-Proteins which are formed of 1 polypeptide chain have:

a) Tertiary, quaternary, and pentagonal structures
b) Tertiary, quaternary, and hexagonal structures
c) Primary, secondary, and tertiary structures
d) Primary, secondary, tertiary, and quaternary structures

(Depart. Questions book)

19-Proteins which are formed of 2 or more polypeptide chains have:

a) Tertiary, quaternary, and pentagonal structures
b) Tertiary, quaternary, and hexagonal structures
c) Primary, secondary, and tertiary structures
d) Primary, secondary, tertiary, and quaternary structures.

(Depart. Questions book)

20-Proteins in their native state are characterized by:

a) 1-dimensional structure
b) 2-dimensional structure
c) 3-dimensional structure
d) 5-dimensional structure

(Depart. Questions book)

21-In the primary structure, the polypeptide chain starts on the left side by:

a) N-terminus amino acid
b) C-terminus amino acid
c) S-terminus amino acid
d) OH- terminus amino acid

(Depart. Questions book)

22-In the primary structure, the polypeptide chain ends on the right side by:

a) N-terminus amino acid
b) C-terminus amino acid
c) S-terminus amino acid
d) OH- terminus amino acid

(Depart. Questions book)

23-The synthesis of the polypeptide chain starts:

a) From the N-terminus to the C-terminus
b) From the C-terminus to the N-terminus
c) From the S-terminus to the N-terminus
d) From the S-terminus to the C-terminus

(Depart. Questions book)

24-The primary structure of proteins:

a) Determines the secondary and tertiary structures
b) Is the final structure of the proteins
c) Is the functional structure of the proteins
d) Includes the α-helix structure

(Depart. Questions book)

25-Secondary structure of proteins includes mainly:

a) α-helix or α-pleated sheets
b) α-helix or β-pleated sheets
c) σ-helix or β-pleated sheets
d) α-helix or β-globular sheets

(Depart. Questions book)

26-In the α-helix protein structure:

a) The polypeptide chain coils along its long axis
b) The polypeptide chain is perpendicular to its axis
c) The polypeptide chains line side by side
d) The polypeptide chains cross each other

(Depart. Questions book)

27-In the β-pleated sheet structure, the polypeptide chains:

a) Cross each other to form a sheet
b) Cross each other to form coils
c) Line side by side to form coils
d) Line side by side to form a sheet

(Depart. Questions book)

28-The α-helix structure is stabilized by:

a) Inter-chain hydrogen bonds
b) Intra-chain hydrogen bonds
c) Inter-chain ionic bonds
d) Intra-chain ionic bonds

(Depart. Questions book)

29-In the α-helix structure, the intra-chain hydrogen bonds are:

a) Formed between NH and C=O groups
b) Formed between NH and R-groups
c) Formed between C=O and R-groups
d) Formed between NH and SH groups

(Depart. Questions book)

30-In the α-helix structure, the R-groups project:

a) Inwards of the helix
b) Outwards of the helix
c) Upwards of the helix
d) Downwards of the helix

(Depart. Questions book)

31-What is the effect of the presence of the tryptophan R-groups in the helical structure?

a) It Enforces it
b) It Stabilizes it
c) It Disturbs it
d) It has no effect

(Depart. Questions book)

32-Aspartate and histidine disturb the α-helical structure by their:

a) Ionic bonds and hydrogen bonds
b) Ionic bonds and peptide bonds
c) Hydrogen bonds and ring structures
d) Ionic bonds and ring structures

(Depart. Questions book)

33-How many adjacent strands combine to form the β-pleated sheet structure?

a) From 10 to 20 adjacent strands
b) From 20 to 100 adjacent strands
c) From 20 to 50 adjacent strands
d) From 2 to 5 adjacent strands

(Depart. Questions book)

34-In the β-pleated sheet, the side chains are:

a) Above or below the sheet plane
b) Inside or outside the sheet plane
c) Forward or backward to the sheet plane
d) Away or near to the sheet plane

(Depart. Questions book)

35-Adjacent polypeptide chains running in opposite directions are called:

a) Parallel β-pleated sheet
b) Anti-parallel β-pleated sheet
c) Perpendicular β-pleated sheet
d) Separate β-pleated sheet

(Depart. Questions book)

36-Which of the following terms refer to adjacent polypeptide chains running in the same direction?

a) Parallel β-pleated sheet.
b) Anti-parallel β-pleated sheet.
c) Perpendicular β-pleated sheet.
d) Separate β-pleated sheet.

(Depart. Questions book)

37-The β-pleated sheet formed by different chains is stabilized by:

a) Inter-chain hydrogen bonds
b) Intra-chain hydrogen bonds
c) Inter-chain ionic bonds
d) Intra-chain ionic bonds

(Depart. Questions book)

38-β-pleated sheets formed between different regions of the same chain are stabilized by

a) Inter-chain hydrogen bonds
b) Intra-chain hydrogen bonds
c) Inter-chain ionic bonds
d) Intra-chain ionic bonds

(Depart. Questions book)

39-Collagen protein contains:

a) α-helix and α-pleated sheet structures
b) α-helix and β-pleated sheet structures
c) α-helix and β-pleated sheet structures
d) Its own characteristic secondary structure

(Depart. Questions book)

40-Tertiary structure is the folding of the polypeptide chain to:

a) Higher 2-dimensional structure
b) Higher 3-dimensional structure
c) Higher 4-dimensional structure
d) Higher 5-dimensional structure

(Depart. Questions book)

41-Which of the following tends to associate together in the hydrophobic interactions in the tertiary structure?

a) nonpolar side chains of neutral amino acids
b) nonpolar side chains of acidic amino acids
c) polar side chains of acidic amino acids
d) polar side chains of alkaline amino acids

(Depart. Questions book)

42-The electrostatic bonds in the tertiary structures are formed between:

a) Uncharged groups of amino acids
b) Oppositely charged groups of amino acids side chains
c) The same charged groups of amino acids
d) 2 sulfur groups of amino acids

(Depart. Questions book)

43-Which of the following is involved in hydrogen bond formation in the tertiary structure?

a) The hydroxyl group of aspartate
b) The hydroxyl group of serine
c) The amide group of threonine
d) The sulfur group of methionine

(Depart. Questions book)

44-The Van der Waal interaction is:

a) strong bond in the tertiary structure
b) weak bond in the tertiary structure
c) strong bond in the secondary structure
d) weak bond in the primary structure

(Depart. Questions book)

45-Disulfide bond can be between:

a) alanine
b) tyrosine
c) cysteine
d) serine

(Depart. Questions book)

46-Which of the following structures is essential for hemoglobin activity?

a) Primary structure
b) Secondary structure
c) Tertiary structure
d) Quaternary structure

(Depart. Questions book)

47-Denaturation is a specific property of:

a) Proteins
b) Carbohydrates
c) Lipids
d) Water

(Depart. Questions book)

48-Albumin coagulation is due to the formation of:

a) Ionic bonds
b) Carboxyl cross linkage
c) Glycoprotein cross linkage
d) Disulfide cross linkage

(Depart. Questions book)

49-Proteins are classified according to their conformation classified into:

a) Simple and compound proteins
b) Glycoproteins and lipoproteins
c) Fibrous and globular proteins
d) Essential and nonessential proteins

(Depart. Questions book)

50-Collagen and elastin are:

a) Fibrous proteins
b) Globular proteins
c) Spherical proteins
d) Cylindrical proteins

(Depart. Questions book)

51-Hemoglobin and myoglobin are:

a) Fibrous proteins
b) Globular proteins
c) Apoproteins proteins
d) Cylindrical proteins

(Depart. Questions book)

52-A-72-years-old male patient was brought to the neurology clinic by his son. The son said that 2 months ago his father started to forget recent information and cannot track the dates nor current locations. The doctor asked for CSF analysis and brain CT. The CSF analysis showed abnormal aggregation of p- amyloid misfolded proteins. The patient was diagnosed as Alzheimer's disease patient.
The misfolded proteins are due to decline in:

a) Mitochondria
b) Chaperones
c) Nucleus
d) Plasma membrane.

(Depart. Questions book)

53-When a peptide bond is formed there is removal of:

a) CO_2
b) H_2O
c) NH_3
d) H^+

(Old Depart. Questions book)

54-2ry structure of proteins is stabilized by:

a) Ionic bond.
b) Hydrophobic interaction.
c) Hydrogen bonds.
d) Peptide bond.

(Old Depart. Questions book)

55-The tertiary structure of proteins is stabilized by the following, except:

a) Hydrogen bonds.
b) Hydrophobic interactions.
c) Peptide bonds.
d) Disulfide bonds.

(Old Depart. Questions book)

56-The four-subunit structure of the Hb represents protein's:

a) Primary structure.
b) Secondary structure.
c) Tertiary structure.
d) Quaternary structure.

(Old Depart. Questions book)

57-A protein can be unfolded by a process called:

a) Renaturation.
b) Denaturation.
c) Oxidation.
d) Reduction.

(Old Depart. Questions book)

58-The protein bonds that are not lost by denaturation is:

a) Peptide bonds
b) Hydrogen bonds
c) Disulfide bonds
d) Electrostatic bond

(Old Depart. Questions book)

59-Alzheimer and Parkinson's disease are age related defects in action of:

a) Ubiquitin.
b) lysosomal proteases.
c) Proteosomes.
d) Chaperones.

(Old Depart. Questions book)

60-Tertiary structure is maintained by:

a) Hydophobic interactions
b) Hydrogen bond.
c) Disulphide bond
d) All of the above.

(Old Depart. Questions book)

61-All of the below mentioned amino acids can participate in hydrogen bonding Except:

a) Serine
b) Cysteine.
c) Threonine
d) Valine.

(Old Depart. Questions book)

62-Which of the following amino acids is mostly likely to disrupt an alpha helix?

a) Tryptophan.
b) Proline.
c) Leucine.
d) Glycine.

(Old Depart. Questions book)

63-Denaturation of protein is due to disruption of the bonds stabilizing all the following EXCEPT:

a. The primary structure
b. The secondary structure
c. The tertiary structure
d. The quaternary structure

64-One of the following amino acids interfere with formation of right handed helix:

a. Glycine
b. Alanine
c. Proline
d. Methionine

CANCELLED

65-All are true regarding α helix EXCEPT:

a. Formed of 1 folded polypeptide chain
b. Connected by interchain hydrogen bond
c. Insoluble in water
d. Each peptide can form 2 hydrogen bonds

66-All are true regarding β pleated sheets EXCEPT:

a. Formed of 2 to 5 folded polypeptide chains
b. Connected by interchain hydrogen bonds
c. May be present in the parallel or anti-parallel form
d. Leads to globular protein

67-One of the following amino acids prevents formation of right-handed helix:

a. Alanine
b. Glycine
c. Methionine
d. Tryptophan

CANCELLED

68- The level of protein structure which describes amino acid sequence of the polypeptide chain is

a) Primary structure
b) Secondary structure
c) Tertiary structure
d) Quaternary structure

(Exam 2026)

69-The ring structure of ..... can disrupt the α-helical structure of the polypeptide chain:

a) Glycine
b) Alanine
c) Proline
d) Serine

(Exam 2026)

70- Chaperones are characterized by:

a) Being enzymes that synthesize new proteins
b) Promoting abnormal protein folding and aggregation
c) Promoting proper protein folding
d) Showing increased level in Alzheimer's disease

(Exam 2026)

71-Denaturation of proteins leads to:

a) Loss of the primary structure of protein
b) Decreased digestibility by enzymes
c) Decreased viscosity
d) Loss of biological activity

72-Decline in which of the following leads to abnormal protein folding and various protein-aggregation diseases as Parkinson's disease:

a) Collagen
b) Chaperones
c) Elastin
d) Cholesterol

73-Denaturation of proteins leads to one of the following:

a) Increased viscosity.
b) Increased solubility.
c) Activation of enzymes.
d) Decrease digestibility.

Answers

1-C	2-B	3-B	4-A	5-C	6-D	7-B	8-C	9-A
10-A	11-B	12-C	13-D	14-B	CANCELLED	16-C	17-B	18-C
19-D	20-C	21-A	22-B	23-A	24-A	25-B	26-A	27-D
28-B	29-A	30-B	31-C	32-D	33-D	34-A	35-B	36-A
37-A	38-B	39-D	40-B	41-A	42-B	43-B	44-B	45-C
46-D	47-A	48-D	49-C	50-A	51-B	52-B	53-B	54- C
55-C	56-D	57-B	58-A	59-D	60-D	61-D	62-A&B	63-A
CANCELLED	65-B	66-D	CANCELLED	68-A	69-C	70-C	71-D	72-B
73-A								

Chapter 3 CARBOHYDRATES OF BIOLOGICAL IMPORTANCE

1-Sorbitol is produced by reduction of:

a) Galactose
b) Glucose
c) Mannose
d) Ribose

(End of module exam 2023)

2-Fructose and mannose are:

a) Anomers
b) Aldose-ketose isomers
c) Epimers
d) Enantiomers

(End of module exam 2023)

3-Reduction of monosaccharides produces:

a) Sugar acid
b) Deoxy-sugar
c) Amino sugar
d) Sugar alcohol

(End of module exam 2023)

4-Which of the following is sulfate free GAG?

a) Heparin
b) Hyaluronic acid
c) Insulin
d) Dextrin

(End of module exam 2023, Final exam 2019)

5-The polysaccharide which is used as dermal filler is:

a) Cellulose
b) Dermatan sulfate
c) Hyaluronic acid
d) Heparin

(End of module exam 2023)

6-_____ and _____ are used as sweetener:

a) Dulcitol and mannitol
b) Sorbitol and dulcitol
c) Mannitol and sorbitol
d) Mannitol and ribitol

(End of module exam 2023)

7-Mucopolysaccharidosis are caused by.....:

a) Genetic increase of lysosomal enzymes which break down GAGs
b) Genetic deficiency of lysosomal enzymes which break down GAGs
c) Genetic deficiency of lysosomal enzymes which synthesize GAGs
d) Genetic increase of lysosomal enzymes which synthesize GAGs

(End of module exam 2023)

8-The polysaccharide which is used to prevent constipation:

a) Heparin
b) Glycogen
c) Cellulose
d) Inulin

(End of module exam 2023)

9-All of the following are functions of cellulose except:

a) It increases the bulk of the food
b) It is a storage form of energy in muscles
c) It stimulates intestinal motility
d) It prevents constipation

(Final exam 2022 - September, Faculty summer revision 2024)

10-One of the following is not a disaccharide:

a) Lactose
b) Maltose
c) Glycogen
d) Sucrose

(End of module exam 2021)

11-Mannitol is a/an:

a) Sugar alcohol
b) Amino sugar
c) Sugar acid
d) Deoxy-sugar

(End of module exam 2021, Final exam 2019 - September sitting)

12-Keratan sulfate is:

a) A homopolysaccaride
b) An Anticoagulant
c) Important for cell-cell interaction
d) Important for transparency of the cornea

(End of module exam 2021)

13-The following are Aldo-hexoses:

a) Glucose and fructose
b) Fructose and ribose
c) Glucose and Galactose
d) Ribose and glucose

(End of module exam 2021)

14-One of the following is a non- reducing disaccharide:

a) Isomaltose
b) Sucrose
c) Lactose
d) Maltose

(End of module exam 2021)

15-Glucose and galactose are:

a) Anomers
b) Optical isomers
c) Aldose-Ketose Isomers
d) Epimers

(End of module exam 2021)

16-Choose the incorrect statement:

a) Glycogen is the storage form of carbohydrates in animals
b) Cellulose prevents constipation
c) Heparin proteoglycan is an important anticoagulant
d) Hyaluronic acid is sulphate-containing glycosaminoglycan

(End of module exam 2021)

17-Heparin is a/an:

a) Non- reducing disaccharide
b) Anticoagulant
c) Aldose
d) Homopolysaccharide

(End of module exam 2021)

18-Glucose and mannose are:

a) Anomers
b) Aldose-Ketose Isomers
c) Optical isomers
d) Epimers

(End of the module 2019)

19-The sugar alcohol derived from glucose is:

a) Sorbitol.
b) Inositol.
c) Dulcitol.
d) Lipositol.

(End of the module 2019)

20-Which of the following is NOT a disaccharide?

a) Lactose
b) Maltose
c) Hyaluronic acid
d) Sucrose

(End of the module 2019)

21-The sugar found in RNA is:

a) Ribose.
b) Deoxyribose.
c) Ribulose.
d) Erythrose.

(End of the module 2019)

22-Heparin is:

a) A non- reducing disaccharide
b) An Aldose
c) A homopolysaccharide
d) An anticoagulant

(End of the module 2019)

23-A heteropolysaccharide among the following is:

a) Inulin
b) Cellulose
c) Heparin
d) Dextrin

(End of the module 2020)

24-Dulcitol is the alcohol of:

a) Glucose
b) Galactose
c) Mannose
d) Glyceraldehyde

(End of the module 2020)

25-The following are Aldohexoses:

a) Glucose, fructose
b) Fructose, ribose
c) Glucose, galactose
d) Ribose, glucose

(End of the module 2020)

26-All the following are reducing disaccharides except:

a) Isomaltose
b) Lactose
c) Sucrose
d) Maltose

(End of the module 2020, Final exam 2019)

27-The polysaccharide that prevents constipation:

a) Heparin
b) Glycogen
c) Cellulose
d) Inulin

(End of the module 2020)

28-A sugar alcohol is

a) Mannitol.
b) Fructose.
c) Galactose
d) L- ascorbic acid.

(Final exam 2019)

29-One of the following by hydrolysis does not give glucose units only:

a) Maltose.
b) Isomaltose.
c) Starch
d) Lactose

(Final exam 2019 - September sitting)

30-All the following are homopolysaccharides except:

a) Starch.
b) Hyaluronic acid.
c) Cellulose.
d) Glycogen.

(Final exam 2019 - September sitting)

31-When two carbohydrates are epimers:

a) One is a pyranose, the other a furanose
b) One is an aldose, the other a ketose
c) They differ in length by one carbon
d) They differ only in the configuration around one carbon atom

(Old Depart. book)

32-Which of following is an anomeric pair?

a) D-glucose and L-glucose
b) D-glucose and D-fructose
c) α-D-glucose and β-D-glucose
d) α-D-glucose and β-L-glucose

(Old Depart. book)

33-The precursor of all aldoses is:

a) Dihydroxyacetone
b) Glyceraldehyde
c) Ribose
d) Erythrose

(Depart. Questions book)

34-Which of the following is an aldotriose?

a) Glyceraldehyde
b) Erythrose
c) Ribose
d) Glucose

(Depart. Questions book)

35-Which of the following is an aldose?

a) Ribulose
b) Fructose
c) Erythrulose
d) Glucose

(Depart. Questions book)

36-Which of the following is an aldose sugar?

a) Glyceraldehyde
b) Ribulose
c) Erythrulose
d) Dihydroxyacetone

(Depart. Questions book)

37-Concerning glucose:

a) It is an aldopentose
b) It is a reducing disaccharide
c) It is a polysaccharide
d) It is oxidized to form glucuronic acid

(Depart. Questions book)

38-Which of the following is the simplest ketose?

a) Erythrulose
b) Ribulose
c) Dihydroxyacetone
d) Fructose

(Depart. Questions book)

39-A ketohexose sugar is:

a) Dihydroxyacetone
b) Ribulose
c) Fructose
d) Glucose

(Depart. Questions book)

40-One of the following is a ketose sugar is:

a) Glyceraldehyde
b) Ribose
c) Erythrose
d) Dihydroxyacetone

(Depart. Questions book)

41-Concerning ketoses:

a) All ketoses have two terminal primary alcohol groups and two ketone groups
b) All ketoses have one terminal primary alcohol group and one ketone group
c) All Ketoses have two terminal primary alcohol groups and one ketone group
d) All Ketoses have one terminal primary alcohol group and two ketone groups

(Depart. Questions book)

42-Which of the following is a keto tetrose?

a) Xylose
b) Erythrulose
c) Fructose
d) Sedoheptulose

(Depart. Questions book)

43-Which of the following is a keto pentose?

a) Fructose
b) Erythrulose
c) Ribulose
d) Dihydroxyacetone

(Depart. Questions book)

44-D-Glucose and D-Fructose are:

a) Polysaccharides
b) Hexoses
c) Aldoses
d) Disaccharides

(Depart. Questions book)

45-Most of the naturally occurring monosaccharides are of which type?

a) D type
b) C type
c) L type
d) E type

(Depart. Questions book)

46-Cyclization creates a carbon with two possible orientations of the hydroxyl group around it:

a) D and L forms
b) α and β forms
c) α and γ forms
d) δ and β forms

(Depart. Questions book)

47-Compounds having the same molecular formula but differ in their functional groups are known as:

a) Functional group isomers
b) Anomers
c) Optical isomers
d) Epimers

(Depart. Questions book)

48-Which one of the following is a carbohydrate with 6 carbon atoms and a keto group as the functional group?

a) Glyceraldehyde
b) Dihydroxyacetone
c) Fructose
d) Galactose

(Depart. Questions book)

49-Two sugars which differ from one another only in the configuration around a single carbon atom are termed:

a) Epimers
b) Anomers
c) Optical isomers
d) Stereoisomers

(Depart. Questions book)

50-Which of the following is an epimer of glucose?

a) Galactose
b) Fructose
c) Arabinose
d) Xylose

(Depart. Questions book)

51-D-glucose and D-mannose are epimers at carbon:

a) 1
b) 3
c) 2
d) 5

(Depart. Questions book)

52-D-glucose and D-galactose are epimers at carbon:

a) 1
b) 2
c) 4
d) 5

(Depart. Questions book)

53-D-glucose and L-glucose are:

a) Epimers
b) Aldose-Ketose isomers
c) Anomers
d) Enantiomers

(Depart. Questions book)

54-Fructose and Glucose are:

a) Anomers
b) Aldose-ketose isomers
c) Epimers
d) Enantiomers

(Depart. Questions book)

55-Fructose is the functional group isomer of:

a) Sucrose
b) Ribose
c) Glucose
d) Erythrose

(Depart. Questions book)

56-The sugar abundantly present in honey is:

a) Maltose
b) Fructose
c) Ribulose
d) Lactose

(Depart. Questions book)

57-The sugar found in DNA is:

a) Xylose
b) Ribose
c) Deoxyribose
d) Ribulose

(Depart. Questions book)

58-Sugar alcohol of glucose is:

a) Sorbitol
b) Inositol
c) Dulcitol
d) Mannitol

(Depart. Questions book)

59-Which of the following is a sugar alcohol?

a) Fructose
b) Galactose
c) Ribose
d) Sorbitol

(Depart. Questions book)

60-Which of the following is a monosaccharide derivative?

a) Sucrose
b) Cellulose
c) Heparin
d) Glucosamine

(Depart. Questions book)

61-Glucose reduction produces:

a) Sorbitol
b) Mannitol
c) Dulcitol
d) Ribitol

(Depart. Questions book)

62-Glycerol is a:

a) Glycoside
b) Glycosaminoglycans
c) Sugar alcohol
d) Protein
e) Deoxysugar

(Depart. Questions book)

63-Dulcitol is a sugar alcohol of:

a) Glucose
b) Mannose
c) Galactose
d) Fructose

(Depart. Questions book)

64-Sugar alcohol of Mannose is:

a) Sorbitol
b) Ribitol
c) Mannitol
d) Dulcitol

(Depart. Questions book)

65-Sorbitol and dulcitol are:

a) Glycosides
b) Sugar alcohols
c) Sugar acids
d) Amino sugars

(Depart. Questions book)

66-Ribitol is a component of which vitamin?

a) Vitamin A
b) Vitamin C
c) Vitamin D
d) Vitamin B2

(Depart. Questions book)

67-In amino sugars, an amino group replaces the hydroxyl group on which carbon?

a) First carbon
b) Second carbon
c) Third carbon
d) Fourth carbon

(Depart. Questions book)

68-Which of the following is present in several antibiotics?

a) Amino sugars
b) Sugar acids
c) Sugar alcohol
d) Glycosides

(Depart. Questions book)

69-Nucleosides are examples of:

a) Amino sugars
b) Sugar acids
c) Sugar alcohol
d) Glycosides

(Depart. Questions book)

70-Sucrose hydrolysis produces:

a) Galactose and mannose
b) Glucose and fructose
c) Glucose and galactose
d) Glucose and mannose

(Depart. Questions book)

71-Monosaccharide units linked by α 1, 4-glucosidic linkage in:

a) Maltose
b) Sucrose
c) Isomaltose
d) Lactose

(Depart. Questions book)

72-The hydrolysis of which of the following yields fructose:

a) Lactose
b) Cellulose
c) Sucrose
d) Starch

(Depart. Questions book)

73-One of following is a non-reducing disaccharide:

a) Isomaltose
b) Lactose
c) Sucrose
d) Maltose

(Depart. Questions book)

74-One of the following gives glucose and fructose on hydrolysis:

a) Lactose
b) Isomaltose
c) Sucrose
d) Maltose

(Depart. Questions book)

75-Galactose units are produced by hydrolyzing one of the following:

a) Maltose
b) Isomaltose
c) Sucrose
d) Lactose

(Depart. Questions book)

76-A disaccharide on hydrolysis produces glucose units only is:

a) Sucrose
b) Maltose
c) Lactose
d) Dextrin

(Depart. Questions book)

77-Concerning Lactose:

a) It is called Malt sugar
b) It is hydrolyzed into glucose and galactose
c) It is hydrolyzed by lactase enzyme or by alkali
d) It is formed by a 1,6-glucosidic linkage

(Depart. Questions book)

78-A carbohydrate formed by β 1,4-galactosidic linkages is:

a) Maltose
b) Cellulose
c) Lactose
d) Sucrose

(Depart. Questions book)

79-Oligosaccharides contain:

a) 3 to 10 monosaccharide units per molecule
b) More than 10 monosaccharide units per molecule
c) 5 to 10 monosaccharide units per molecule
d) 2 monosaccharide units per molecule

(Depart. Questions book)

80-Which of the following is a polymer of fructose?

a) Starch
b) Dextrin
c) Inulin
d) Cellulose

(Depart. Questions book)

81-Which of the following is the chief storage form of carbohydrates in chlorophyll-containing plants?

a) Starch
b) Glycogen
c) Dextrin
d) Cellulose

(Depart. Questions book)

82-Starch is a:

a) Homopolysaccharide
b) Monosaccharide
c) Disaccharide
d) Heteropolysaccharide

(Depart. Questions book)

83-Starch and Glycogen are polymers of:

a) D-Glucose
b) Fructose
c) Mannose
d) Galactose

(Depart. Questions book)

84-Which of the following is the main product of digestion of starch by amylase?

a) Maltose
b) Sucrose
c) Fructose
d) Lactose

(Depart. Questions book)

85-Which of the following is the carbohydrate storage form in animals?

a) Starch
b) Glycogen
c) Glucose
d) Inulin

(Depart. Questions book)

86-A polysaccharide which is called animal starch is:

a) Glycogen
b) Starch
c) Inulin
d) Dextrin

(Depart. Questions book)

87-Glycogen is mainly present in:

a) Skeletal muscles and liver
b) Brain
c) Spleen
d) Kidney

(Depart. Questions book)

88-Which of the following is a heteropolysaccharide?

a) Inulin
b) Cellulose
c) Heparin
d) Dextrin

(Depart. Questions book)

89-One of the following is heteropolysaccharides:

a) Starch
b) Hyaluronic acid
c) Cellulose
d) Glycogen

(Depart. Questions book)

90-Which of the following is a heteropolysaccharide?

a) Cellulose
b) Keratan sulfate
c) Dextrin
d) Inulin

(Depart. Questions book)

91-Which of the following is a glycosaminoglycans?

a) Inulin
b) Heparin
c) Glucosamine
d) Galactosamine

(Depart. Questions book)

92-Heparan sulfate is:

a) Homopolysaccharide
b) Anticoagulant
c) Important for cell-cell interaction
d) Important for transparency of the cornea

(Depart. Questions book)

93-The polysaccharide which is used to prevent blood clotting is:

a) Heparin
b) Keratan sulfate
c) Cellulose
d) Heparan sulfate

(Depart. Questions book)

94-Which of the following is present in high concentration in embryonic tissues:

a) Keratan sulfate
b) Heparan sulfate
c) Heparin
d) Hyaluronic acid

(Depart. Questions book)

95-The conjugation of GAGs with a protein core produces:

a) Glycon
b) Aglycon
c) Proteoglycans
d) Amino sugar

(Depart. Questions book)

96-A disaccharide which contains glucose and fructose is:

a) Lactose.
b) Isomaltose.
c) Sucrose.
d) Maltose

(Old Depart. Questions book)

97-A polysacchharide which is stored in animals is:

a) Glycogen
b) Starch
c) Inulin
d) Dextrin

(Old Depart. Questions book)

98-One of the following polysaccarides does not give glucose units by hydrolysis:

a) Glycogen
b) Starch
c) Inulin
d) Cellulose

(Old Depart. Questions book)

99-Which of the following is ketose?

a) Galactose
b) Fructose
c) Mannose
d) Glucose

(Old Depart. Questions book)

100-α and β forms of D-glucose are:

a) Disaccharides.
b) Polysaccharides.
c) Anomers.
d) Mirror image isomers.

(Old Depart. Questions book)

101-A pentose sugar is:

a) Dihydroxyacetone
b) Ribulose
c) Fructose
d) Glucose

(Old Depart. Questions book)

102-The following is not a monosaccharide derivative:

a) Sorbitol.
b) Glucuronic acid.
c) Dihydroxyacetone.
d) Glycerol.

(Old Depart. Questions book)

103-As regards glucose, one of the following is not correct:

a) It is aldohexoses
b) It is a disaccharide.
c) It has asymmetric carbon atom.
d) It has mirror image isomers.

(Old Depart. Questions book)

104-As regards glucuronic acid one of the following is not correct:

a) It is a sugar acid.
b) Derived from glucose.
c) By hydrolysis gives glucose.
d) Enters in GAGs formation.

(Old Depart. Questions book)

105-D-glyceraldhyde and L-Glyceraldhyde are:

a) Epimers.
b) Hexoses.
c) Anomers
d) Enantiomers

(Old Depart. Questions book)

106-All the following contain asymmetric carbon atoms except:

a) Glucose.
b) Ribose.
c) Dihydroxyacetone.
d) Glyceraldehyde.
e) Fructose

(Old Depart. Questions book)

107-All the following are disaccharides except:

a) Isomaltose.
b) Maltose.
c) Aggrecan.
d) Sucrose.

(Old Depart. Questions book)

108-The polysaccharide which is used as anticoagulant is:

a) Heparin.
b) Glycogen.
c) Cellulose.
d) Heparan sulfate.

(Old Depart. Questions book)

109-One of the following is a polysaccharide:

a) Isomaltose.
b) Lactose.
c) Aggrecan.
d) Sucrose.

(Old Depart. Questions book)

110-As regard sucrose all of the following is right except:

a) It is reducing disaccharide.
b) Hydrolysed into glucose and fructose.
c) It has free anomeric carbon.
d) A and C.

(Old Depart. Questions book)

111-The ketose sugars are all of the following except:

a) Glyceraldehyde
b) Ribulose
c) Erythrulose
d) Dihydoxyacetone

(Old Depart. Questions book)

112-A carbohydrate formed by β 1-4 galactosidic linkages is:

a) Glycogen
b) Cellulose
c) Lactose
d) Inulin

(Old Depart. Questions book)

113-A disaccharide made up of two glucose units is:

a) Sucrose
b) Maltose
c) Lactose
d) Dextrin

(Old Depart. Questions book)

114-A carbohydrate found only in milk:

a) Glucose
b) Galactose
c) Lactose
d) Maltose

(Old Depart. Questions book)

115-Sugar alcohol derived from galactose is:

a) Sorbitol.
b) Mannitol
c) Dulcitol.
d) Ribitol

(Old Depart. Questions book)

116-One of the following polysaccharides is not easily digested:

a) Starch.
b) Glycogen.
c) Cellulose.
d) Dextrin

(Old Depart. Questions book)

117-The mother compound of all aldoses is:

a) Dihydroxyacetone
b) Glyceraldehyde
c) Ribose
d) Erythrose

(Old Depart. Questions book)

118-The simplest ketose is:

a) Erythrulose.
b) Ribulose.
c) Dihydoxyacetone
d) Fructose

(Old Depart. Questions book)

119-A sugar alcohol is:

a) Ribose
b) Ribitol
c) Ribulose
d) Deoxy ribose

(Old Depart. Questions book)

120-Reduction turns monosaccharides into:

a) Sugar acid
b) Deoxy sugar
c) Amino sugar
d) Sugar alcohol

(Old Depart. Questions book)

121-Carbon atom attached to 4 different atoms or groups is called:

a. Annomeric carbon
b. Asymmetric carbon.
c. Epimeric carbon.
d. Carbonyl group.

122-The largest Monosaccharide in the body is:

a. Glucose
b. Galactose
c. Fructose
d. sedoheptulose

123-The most important epimer of glucose is:

a. Galactose
b. Fructose
c. Arabinose
d. Xylose

124-α-D-glucose and β-D-glucose are:

a. Enantiomers
b. Epimers
c. Annomers
d. Keto-aldo pairs

125-The sugar found in milk is:

a. Galactose
b. Glucose
c. Fructose
d. Lactose

126-The sugar found in semen is:

a. Galactose
b. Glucose
c. Fructose
d. Lactose

127-One of the following is functional group isomers:

a. Glucose and Galactose
b. Glyceraldehyde and ribose
c. Dihydroxyacetone and fructose
d. Ribose and Ribulose

128-Sugar alcohols are obtained from the sugars by:

a. Reduction
b. Oxidation
c. Dehydration
d. Hydrolysis

129-Reduction of fructose gives:

a. Glycerol and Ribitol
b. Sorbitol and mannitol
c. Dulcitol and sorbitol
d. Mannitol and dulcitol

130-One of the following is deoxy sugar:

a. Glucuronic acid
b. Sorbitol
c. Deoxy ribose
d. Glucosamine

131-The mother compound of all aldoses is:

a) Dihydroxyacetone
b) Glyceraldehyde
c) Galactose
d) Glucose

(End of module exam 2026)

132- Reduction of monosaccharides produces:

a) Sugar acid
b) Deoxy sugar
c) Amino sugar
d) Sugar alcohol

(End of module exam 2026)

133- ..... and .....are aldopentoses:

a) Ribose and xylose
b) Glucose and mannose
c) Ribulose and erythrose
d) Fructose and glucose

(End of module exam 2026)

134- A natural product that might be involved in the pathogenesis of diabetic complications is:

a) Mannitol
b) Ribitol
c) Sorbitol
d) Dulcitol

(End of module exam 2026)

135- Glucose and galactose are hydrolytic products of:

a) Sucrose
b) Maltose
c) Lactose
d) Ribitol

(End of module exam 2026)

136- Glycogen is a:

a) Homopolysaccharide
b) Monosaccharide
c) Disaccharide
d) Heteropolysaccharide

(End of module exam 2026)

137- A known glycosaminoglycan is:

a) Inulin
b) Hyaluronic acid
c) Glucosamine
d) Galactosamine

(End of module exam 2026)

138- The glycosaminoglycan that binds to heparin and augments its action is:

a) Dermatan sulfate
b) Keratan sulfate
c) Chondroitin sulfate
d) Hyaluronic acid

(End of module exam 2026)

139-Fructose is the functional group isomer of:

a) Sucrose
b) Ribose
c) Glucose
d) Erythrose

(End of module exam 2024)

140-Which of the following is present in several antibiotics?

a) Amino sugars
b) Sugar acids
c) Sugar alcohol
d) Glycosides

(End of module exam 2024)

141-The polysaccharide which is used as anticoagulant is:

a) Heparin
b) Keratan sulfate
c) Hyaluronic acid
d) Heparan sulfate

(End of module exam 2024)

142-One of the following is responsible for corneal transparency:

a) Heparin.
b) Glycogen.
c) Cellulose
d) Keratan sulfate

(End of module exam 2024)

143-The only sulfate free GAG is:

a) Heparin.
b) Hyaluronic acid.
c) Inulin.
d) Dextrin.

(End of module exam 2024)

144-Cellulose:

a) Is a branched chain heteropolysaccharide.
b) Is formed by α1-4 glucosidic bonds .
c) Is digested by amylase in human intestine.
d) Increases the bulk of stool and stimulate intestinal contractions.

(End of module exam 2024)

145-A carbohydrate which is a fructan is:

a) Glycogen
b) Dextrin
c) Inulin
d) Cellulose.

(End of module exam 2024)

Answers

1-B	2-B	3-D	4-B	5-C	6-C	7-B	8-C	9-B
10-C	11-A	12-D	13-C	14-B	15-D	16-D	17-B	18-D
19-A	20-C	21-A	22-D	23-C	24-B	25-C	26-C	27-C
28-A	29-D	30-B	31-D	32-C	33-B	34-A	35-D	36-A
37-D	38-C	39-C	40-D	41-C	42-B	43-C	44-B	45-A
46-B	47-A	48-C	49-A	50-A	51-C	52-C	53-D	54-B
55-C	56-B	57-C	58-A	59-D	60-D	61-A	62-C	63-C
64-C	65-B	66-D	67-B	68-A	69-D	70-B	71-A	72-C
73-C	74-C	75-D	76-B	77-B	78-C	79-A	80-C	81-A
82-A	83-A	84-A	85-B	86-A	87-A	88-C	89-B	90-B
91-B	92-C	93-A	94-D	95-C	96-C	97-A	98-C	99-B
100-C	101-B	102-C	103-B	104-C	105-D	106-C	107-C	108-A
109-C	110-D	111-A	112-C	113-B	114-C	115-C	116-C	117-B
118-C	119-B	120-D	121-B	122-D	123-A	124-C	125-D	126-C
127-D	128-A	129-B	130-C	131-B	132-D	133-A	134-C	135-C
136-A	137-B	138-A	139-C	140-A	141-A	142-D	143-B	144-D
145-C								

Chapter 4 LIPIDS OF BIOLOGICAL IMPORTANCE

1-The decrease of which of the following phospholipid is associated with Alzheimer disease?

a) Cardiolipins
b) Lecithin
c) Plasmalogens
d) Cephalin

(End of module exam 2023)

2-Linoleic acid is an example for:

a) Omega-3 fatty acid
b) Omega-6 fatty acid
c) Monoenoic fatty acid
d) Saturated fatty acid

(End of module exam 2023)

3-An example of acyclic eicosanoids is:

a) Prostaglandins
b) Thromboxane
c) Leukotrienes
d) Prostacyclins

(End of module exam 2023)

4-Snake venom causes hemolysis of RBCs due to activation of the following enzyme:

a) Cholesteryl ester esterase
b) Hormone sensitive lipase
c) Lecithinase
d) Phosphatase

(End of module exam 2023)

5-Lecithin/sphingomyelin ratio of amniotic fluid in a pregnant female is important as it evaluates:

a) Fetal kidney maturity
b) Fetal lung maturity
c) Fetal brain maturity
d) Fetal liver maturity

(End of module exam 2023 , End of module exam 2021)

6-Which of the following are second messengers

a) Phosphatidyl choline
b) Phosphatidyl serine
c) Inositol-triphosphate and diacylglycerol
d) Plasmalogens

(End of module exam 2023)

7-The importance of phospholipids as constituent of cell membrane is because possess:

a) Hydrophobic heads and tails
b) Hydrophobic tails and hydrophilic heads
c) Large cyclic structure
d) Hydrophobic heads and hydrophilic tails

(End of module exam 2023)

8-Which of the following is considered a glycolipid?

a) sphingomyelin
b) lecithin
c) Plasmalogens
d) Cerebrosides

(End of module exam 2023)

9-Cholesterol is a precursor for the following:

a) Vitamin E
b) Vitamin C
c) Vitamin D
d) Vitamin B12

(End of module exam 2023)

10-Essential fatty acids are:

a) Linoleic and arachidonic acids
b) α-linolenic and timnodonic acids
c) Linoleic and α-linolenic acids
d) Oleic and linoleic acids

(End of module exam 2021)

11-Which of the following is not a derivative of cholesterol?

a) Steroid hormones
b) Bile salts
c) Prostaglandins
d) Vitamin D

(End of module exam 2021)

12-One of the following is not an eicosanoid:

a) Leukotrienes
b) Prothrombin
c) Prostacyclins
d) Prostaglandins

(End of module exam 2021)

13-An example of a saturated fatty acid is:

a) Palmitic acid
b) Oleic acid
c) Linoleic acid
d) Arachidonic acid

(End of module exam 2021)

14-Which of the following is a second hormone messenger?

a) Phospholipase C
b) Inositol-trisphosphate (IP3)
c) Lecithin
d) Cerebroside

(End of module exam 2021)

15-Which of the following is not a phospholipid?

a) Sphingomyelin
b) Lecithin
c) Cephalin
d) Cerebroside

(End of module exam 2021)

16-Which of the following is a male sex hormone?

a) Estrogen
b) Testosterone
c) Progesterone
d) Cortisol

(End of module exam 2021)

17-A fatty acid which is not synthesized in the body and has to be supplied in the diet is:

a) Palmitic acid
b) Linoleic acid
c) Timnodonic acid
d) Palmitoleic acid

(End of the module 2019, Final exam 2019 – September sitting)

18-An example of a saturated fatty acid is:

a) Stearic acid
b) Oleic acid
c) Linoleic acid
d) Arachidonic acid

(End of the module 2019)

19-Which of the following is not a phospholipid

a) Sphingomyelin
b) Lecithin
c) Cephalin
d) Cerebroside

(End of the module 2019)

20-The importance of estimating Lecithin/Sphingomyelin ratio (L/S ratio) in pregnant females is to evaluate:

a) Fetal heart rate
b) Fetal lung maturity
c) Fetal head size
d) Expected date of delivery

(End of the module 2019)

21-In humans, a dietary essential fatty acid is:

a) Palmitic acid
b) Stearic acid
c) Oleic acid
d) Linoleic acid

(End of the module 2020)

22-According to the chemical and nutritional classifications of fatty acids, we can classify palmitic acid as:

a) Saturated and non-essential
b) Monounsaturated and non-essential
c) Saturated and essential
d) Polyunsaturated and non-essential

(End of the module 2020, End of the module 2019)

23-As regards cholesterol one of the following is incorrect:

a) It is the precursor of steroid hormones
b) It is excreted in the form of bile salts
c) It is an important constituent of cell membranes
d) Its plasma level is 200-250 mg/dl

(End of the module 2020)

24-Which out of the following is NOT a glycerophospholipid?

a) Cardiolipin
b) Sphingomyelin
c) Lecithin
d) Plasmalogen

(End of the module 2020)

25-One of the following is NOT an eicosanoid:

a) Leukotrienes
b) Prostaglandins
c) Prostacyclins
d) Progesterone

(End of the module 2020)

26-Which of the following is a phospholipid:

a) Lecithin
b) Cholesterol
c) Sterol
d) Steroid

(Final exam 2019)

27-The activation of phospholipase-C associated with a cell membrane initiates a chemical signal most directly involving:

a) Arachidonic acid
b) cAMP
c) Inositol triphosphate
d) Prostacyclin

(Depart. book)

28-Aspirin inhibits prostaglandin production by inhibiting:

a) Lipoxygenase
b) Cyclooxygenase
c) Phospholipase A_2
d) Prostaglandin H synthase

(Depart. book)

29-Lipoxygenase is required for synthesis of:

a) Prostaglandins
b) Leukotrienes
c) Thromboxanes
d) Prostacyclins

(Depart. book)

30-Which one of the following statements concerning choline is true?

a) It is an essential nutrient
b) It is a constituent of cardiolipin
c) It is a constituent of ceramide
d) It is a constituent of lung surfactant

CANCELLED

(Old Depart. book)

31-Which of the following is an example for saturated fatty acid?

a) Oleic acid
b) Stearic acid
c) Linoleic acid
d) Arachidonic acid

(Depart. Questions book)

32-Which of the following is the most common saturated fatty acid in human tissues?

a) Palmitic acid
b) Timnodonic acid
c) Arachidonic acid
d) Linoleic acid

(Depart. Questions book)

33-Which of the following is an example for monounsaturated fatty acid?

a) Palmitic acid
b) Arachidonic acid
c) Oleic acid
d) Stearic acid

(Depart. Questions book)

34-The main dietary source for trans fatty acids is:

a) Vegetables
b) Margarine
c) Ground nuts
d) Fruits

(Depart. Questions book)

35-Which of the following is an omega-3 fatty acid?

a) Linoleic acid
b) α-Linolenic acid
c) Nervonic acid
d) Palmitic acid

(Depart. Questions book)

36-Which of the following is an essential fatty acid?

a) α-Linolenic acid
b) Arachidonic acid
c) Oleic acid
d) Palmitic acid

(Depart. Questions book)

37-Stearic acid is considered as:

a) Saturated and non-essential fatty acid
b) Monounsaturated and non-essential fatty acid
c) Saturated and essential fatty acid
d) Polyunsaturated and non-essential fatty acid

(Depart. Questions book)

38-Deficiency of essential fatty acids leads to:

a) Opacity of the cornea
b) Fatty liver
c) Spleen enlargement
d) Renal failure

(Depart. Questions book)

39-Which of the following fatty acids is a precursor for eicosanoids?

a) Stearic acid
b) Palmitic acid
c) Arachidonic acid
d) Lignoceric acid

(Depart. Questions book)

40-The number of double bonds in arachidonic acid is:

a) 1
b) 2
c) 3
d) 4

(Depart. Questions book)

41-How many carbon atoms do eicosanoids have?

a) 18
b) 20
c) 22
d) 24

(Depart. Questions book)

42-Which of the following is a simple lipid?

a) Lecithin
b) Fatty acid
c) Triacylglycerol
d) Steroids

(Depart. Questions book)

43-What is the other name of triglycerides?

a) Sphingolipid
b) Lecithin
c) Neutral fats
d) Waxes

(Depart. Questions book)

44-If a fatty acid is esterified with long-chain monohydroxy alcohol higher than glycerol, the result is:

a) Waxes
b) Sphingolipids
c) Phospholipids
d) Neutral fats

(Depart. Questions book)

45-Which of the following is considered as wax?

a) Triacylglycerol
b) Cholesteryl esters
c) Vitamin A esters
d) Lecithin

(Depart. Questions book)

46-Cholesteryl esters are examples of:

a) Waxes
b) Neutral fats
c) Compound lipids
d) Derived lipids

(Depart. Questions book)

47-Which of the following is called phosphatidic acid?

a) Diacylglycerol phosphate
b) Diphosphatidyl glycerol
c) Lipositol
d) Cephalin

(Depart. Questions book)

48-Hydrolysis of phosphatidic acid yields:

a) 1,2-diacylglycerol and phosphoric acid
b) 1,3-diacylglycerol and phosphoric acid
c) 1,4-diacylglycerol and phosphoric acid
d) 1,5-diacylglycerol and phosphoric acid

(Depart. Questions book)

49-Phosphatidic acid is present in:

a) Lecithin and cephalin
b) Lecithin and cerebrosides
c) Gangliosides and cephalin
d) Lecithin and gangliosides

(Depart. Questions book)

50-What is the other name of lecithin?

a) Phosphatidylcholine
b) Phosphatidylserine
c) Cardiolipin
d) Sphingomyelins

(Depart. Questions book)

51-Lung surfactant is formed mainly of:

a) Dipalmitoyl-lecithin
b) Phosphatidylethanolamine
c) Ceramide
d) Phosphatidylinositol

(Depart. Questions book)

52-Which of the following lipids is an important component of the inner mitochondrial membrane?

a) Lecithin
b) Cephalin
c) Cardiolipins
d) Ceramide

(Depart. Questions book)

53-Which of the following is a type of ether phospholipids?

a) Gangliosides
b) Sulfolipids
c) Plasmalogens
d) Cerebrosides

(Depart. Questions book)

54-Which compounds form ceramide?

a) Glycerol and glucose
b) Fatty acid and choline
c) Fatty acid and serine
d) Fatty acids and sphingosine

(Depart. Questions book)

55-Which of the following is connected to phosphocholine to form sphingomyelin?

a) Pyruvate
b) Serine
c) Ceramide
d) Glucose

(Depart. Questions book)

56-Sphingomyelin is considered as:

a) Simple lipid
b) Phospholipid
c) Glycolipid
d) Derived lipid

(Depart. Questions book)

57-The importance of phospholipids as a constituent of the cell membrane is because they possess:

a) Hydrophobic heads and tails
b) Hydrophobic tails and hydrophilic heads
c) Large cyclic structure
d) Hydrophobic heads and hydrophilic tails

(Depart. Questions book)

58-Which of the following is considered glycolipid?

a) Sphingomyelin
b) Lecithin
c) Plasmalogens
d) Cerebrosides

(Depart. Questions book)

59-Which of the following is a derived lipid?

a) TAG
b) Fatty acid
c) Lecithin
d) Cardiolipin

(Depart. Questions book)

60-Steroids are examples of which type of lipid?

a) Neutral fats
b) Waxes
c) Derived lipids
d) Compound lipids

(Depart. Questions book)

61-A steroid that moderates membrane fluidity:

a) Estradiol
b) Cholesterol
c) Testosterone
d) Progesterone

(Depart. Questions book)

62-Free cholesterol contains:

a) 24 carbon atoms
b) 25 carbon atoms
c) 26 carbon atoms
d) 27 carbon atoms

(Depart. Questions book)

63-Bile acids are derived from:

a) Cholesterol
b) Amino acids
c) Fatty acids
d) Bilirubin

(Depart. Questions book)

64-One of the following is not a phospholipid:

a) Sphingomyelin
b) Lecithin
c) Cephalin
d) Cerebroside

(Old Depart. Questions book)

65-An example of monounsaturated fatty acids is:

a) Oleic acid
b) Arachidonic acid
c) Palmitic acid
d) Linolenic acid

(Old Depart. Questions book)

66-Due to their amphipathic characteristics, phospholipids form:

a) lipid layer
b) lipid bilayer
c) lipid trilayer
d) lipid tetralayer

(Old Depart. Questions book)

67-Which of the following is omega-3 polyunsaturated fatty acid?

a) Linoleic acid
b) α-Linolenic acid
c) γ-Linolenic acid
d) Arachidonic acid
e) Oleic acid

(Old Depart. Questions book)

68-A steroid which moderates membrane fluidity is:

a) cholesterol
b) testosterone
c) estradiol
d) progesterone

(Old Depart. Questions book)

69-The enzyme that catalyzes the splitting of PIP2 into two molecules of inositol triphosphate (IP3) and diacylglycerol in cell signaling, is:

a) phosphokinase C
b) phospholipase C
c) phosphodiesterase C
d) lipokinase

CANCELLED

(Old Depart. Questions book)

70-Which of the following is a second messenger?

a) Inositol triphosphate
b) Diacyl glycerol
c) Phospholipase C
d) Both (a) and (b)

CANCELLED

(Old Depart. Questions book)

71-Which of the following fatty acids is NOT a glycerophospholipid?

a) Phosphatidic acid
b) Plasmalogen
c) Cephalin
d) Sphingol

CANCELLED

(Old Depart. Questions book)

72-One of the following is an unsaturated fatty acid:

a) Arachidonic acid
b) Oleic acid
c) Palmitic acid
d) Palmitoleic acid

(Old Depart. Questions book)

73-Ceramide consists of:

a) Glycerol and FA.
b) Sphingosine and galactose.
c) Sphingosine and FA.
d) Glycerol and galactose

(Old Depart. Questions book)

74-One of the following is not a glycolipid:

a) Cerebrosides
b) Sulpholipids
c) Lecithin.
d) Gangliosides

(Old Depart. Questions book)

75-Saturated FA which contains 18C is:

a) Palmitoleic acid.
b) Oleic acid.
c) Lignoceric acid.
d) Stearic acid.

(Old Depart. Questions book)

76-FA which contains one double bond and ω7 is:

a) Nervonic acid.
b) Palmitoleic acid.
c) Linoleic acid.
d) Oleic acid.

(Old Depart. Questions book)

77-Linoleic acid is characterized by:

a) It contains 20 C.
b) It contains 3 double bonds.
c) Not formed inside the body.
d) It is an ω3 FA.

(Old Depart. Questions book)

78-α-linolenic acid is characterized by:

a) It can be formed in our bodies.
b) It is diethenoid.
c) It is ω3 FA.
d) Can be converted to arachidonic acid.

(Old Depart. Questions book)

79-Deficiency of essential fatty acids leads to:

a) Fatty liver.
b) Alfa 1 anti-trypsin deficiency.
c) Marfan syndrome.
d) Microcytic anemia.

(Old Depart. Questions book)

80-Which of the following is not steroid?

a) Vitamin D3
b) Testosterone.
c) Prostacyclin.
d) Progesterone.

(Old Depart. Questions book)

81-Membrane fluidity is affected by all the following EXCEPT:

a) Temperature
b) percentage of unsaturated fatty acids
c) cholesterol
d) distribution of proteins

(Old Depart. Questions book)

82-Timnodonic will be essential in absence of:

a. Oleic
b. α Linolenic
c. Linoleic
d. Arachidonic

83-Arachidonic will be essential in absence of:

a. Oleic
b. α Linolenic
c. Linoleic
d. Timnodonic

84-All the following are cyclic eicosanoids EXCEPT:

a. Leukotriens
b. Prostaglandins
c. Thromboxanes
d. Prostacyclins

85-One of the following is a non-cyclic eicosanoid:

a. Leukotriens
b. Prostaglandins
c. Lipoxins
d. a and c

86-Most of the functions of the phospholipids are due to their:

a. Hyrophobic properties
b. Hydrophilic properties
c. Amphipathic properties
d. None of the above

87-Building unit of cholesterol is:

a. Acetyl CoA
b. Succinyl CoA
c. Malonyl CoA
d. Propionyl CoA

88-The most potent glucocorticoid is:

a. Estradiol
b. Testosterone
c. Progesterone
d. Cortisol

89-Number of carbon atoms in cholesterol is:

a. 17
b. 19
c. 27
d. 30

90- From the options, select a saturated fatty acid

a) Oleic acid
b) Palmitic acid
c) Linoleic acid
d) Arachidonic acid

(Exam 2026)

91- Among the choices, an ω6 polyunsaturated fatty acid (PUFA) is:

a) Linoleic acid
b) Oleic acid
c) Palmitic acid
d) Stearic acid

(Exam 2026)

92-This lipid is a key component of the inner mitochondrial membrane

a) Lecithin
b) Cephalin
c) Cardiolipins
d) Ceramide

(Exam 2026)

93-A well-known derivative of cholesterol is:

a) Vitamin C
b) Vitamin B
c) Vitamin E
d) Vitamin D3

(Exam 2026)

94-A lipid classified as a phospholipid is:

a) Cholesterol
b) Sphingomyelin
c) Triacylglycerol
d) Stearic acid

(Exam 2026)

95-Acyclic eicosanoids include:

a) Prostaglandins
b) Thromboxanes
c) Lipoxins
d) Prostacyclins

(Exam 2026)

96- Lung surfactant is formed mainly of:

a) Dipalmitoyl-lecithin
b) Phosphatidylethanolamine
c) Ceramide
d) Phosphatidylglycerol

(Exam 2026)

97- A phospholipid that acts as second messenger is

a) Phosphatidylcholine
b) Phosphatidylserine
c) Phosphatidylinositol
d) Phosphatidylglycerol

(Exam 2026)

98-Which of the following is a simple lipid?:

a) Lecithin
b) Proteolipids
c) Triacylglycerol
d) Cephalin

(Exam 2024)

99-Linoleic acid is an example for:

a) Omega-3 fatty acid
b) Omega-6 fatty acid
c) Monoenoic fatty acid
d) Saturated fatty acid

(Exam 2024)

100-An example of acyclic eicosanoids is::

a) Prostaglandins
b) Thromboxane
c) Leukotrienes
d) Prostacyclins

(Exam 2024)

101-Phospholipid that acting as lung surfactant is:

a) Dipalmitoyl Cephalin
b) Dipalmitoyl lecithin
c) Phosphatidyl inositol
d) Phosphatidyl serine

(Exam 2024)

102-Decrease level of which of the following phospholipid is associated with Alzheimer disease?

a) Cardiolipins
b) Lecithin
c) Plasmalogens
d) Cephalin

(Exam 2024)

103-One of the following is a hydrolytic product of Sphingomyelin:

a) Sphingosine
b) Ethanolamine
c) Serine
d) Glycerol

(Exam 2024)

104-If the fatty acid is esterified with long chain monohydroxy alcohol higher than glycerol, the resulting compound is:

a) Inositol
b) Phosphatidic acid
c) Wax
d) Cephalin

(Exam 2024)

105-One of the following is correct about Cholesterol:

a) Its increased plasma level associated with development of atherosclerosis.
b) Is present as esterified cholesterol only
c) Is the precursor of phospholipids
d) Is a 29-carbon sterol

(Exam 2024)

Answers

1-C	2-B	3-C	4-C	5-B	6-C	7-B	8-D	9-C
10-C	11-C	12-B	13-A	14-B	15-D	16-B	17-B	18-A
19-D	20-B	21-D	22-A	23-D	24-B	25-D	26-A	27-C
28-B	CANCELLED	30-D	31-B	32-A	33-C	34-B	35-B	36-A
37-A	38-B	39-C	40-D	41-B	42-C	43-C	44-A	45-B
46-D	47-A	48-A	49-A	50-A	51-A	52-C	53-C	54-D
55-C	56-B	57-B	58-D	59-B	60-C	61-B	62-D	63-A
64-D	65-A	66-B	67-B	68-A	CANCELLED	CANCELLED	CANCELLED	72-A
73-C	74-C	75-D	76-B	77-C	78-C	79-A	80-C	81-D
82-B	83-C	84-A	85-D	86-C	87-A	88-D	89-C	90-B
91-A	92-C	93-D	94-B	95-C	96-A	97-C	98-C	99-B
100-C	101-B	102-C	103-A	104-C	105-A			

Chapter 5 CHEMISTRY OF HEMEPROTEINS

1-Concerning glycated hemoglobin:

a) It is formed by non-enzymatic glycosylation of hemoglobin
b) It is formed by non-enzymatic reaction with white blood cells
c) It is decreased when control of diabetes is poor.
d) It represents about 15% of HbA in normal individuals.

(End of module exam 2023)

2-Myoglobin is formed of:

a) Four heme groups attached to four polypeptide chains
b) Two heme groups attached to one polypeptide chain
c) Three heme groups attached to three polypeptide chains
d) One heme attached to one polypeptide chain

(End of module exam 2023)

3-Fetal Hb is characterized by:

a) Faster than HbA in electrophoresis
b) Has higher affinity for O2 than HbA
c) Has higher affinity for 2,3 BPG
d) Appear after birth

(End of module exam 2023)

4-Which of the following is a hemoprotein?

a) Collagen
b) Transferrin
c) Cytochromes
d) Elastin

(End of module exam 2023)

5-In sickle cell disease (HbS), the 6th amino acid (glutamate) of β chain is replaced by:

a) Tyrosine
b) Lysine
c) Aspartic acid
d) Valine

(End of module exam 2023, End of module exam 2021)

6-In hemoglobin C disease:

a) There are defects in the synthesis of the β chain.
b) There are defects in the synthesis of the α chain.
c) Glutamic acid at position 6 of the β chain is replaced by lysine.
d) Glutamic acid at position 6 of the β chain is replaced by valine.

(End of module exam 2023)

7-In β-thalassemia major:

a) Patients carry one mutated gene
b) The expression of the gamma and delta chains increases
c) Patients suffer from mild anemia
d) Clinical manifestations appear during intrauterine life

(End of module exam 2023)

8-Regarding HbA, one of the following statements is incorrect:

a) It is the major hemoglobin in adults
b) It is formed of two polypeptide chains
c) Its α-chain is formed of 141 amino acids
d) Its β chain is formed of 146 amino acids

(End of module exam 2023)

9-Glycated hemoglobin (HbA1c):

a) Is not present in normal persons
b) Is formed by the enzymatic glycosylation of hemoglobin
c) Has glucose residue covalently attached to it
d) It reflects the mean blood glucose concentration over the preceding 2-3 weeks

(End of module exam 2021)

10-Myoglobin is characterized by the following:

a) Formed of a four polypeptide chains
b) Contains four heme per molecule
c) Has a lesser affinity for oxygen than Hb
d) It acts as a reservoir for oxygen in muscles

(End of module exam 2021)

11-Fetal hemoglobin (HbF) is characterized by:
a) It contains two alpha and two gamma chains (α2 , γ2)

b) Has lower affinity for O2 than HbA
c) Faster than HbA in electrophoresis
d) Appears after birth

(End of module exam 2021)

12-Which statement is not correct regarding globin part of hemoglobin?

a) It makes heme soluble
b) It prevents the diffusion of heme to the plasma
c) It helps oxidation of iron
d) It is responsible for the characteristic sigmoid oxygen dissociation curve

(End of module exam 2021)

13-β-thalassemia major is associated with increased:

a) HbF
b) HbA1c
c) HbF and HbA2
d) HbC

(End of module exam 2021)

14-2, 3-Bisphosphoglycerate (2, 3-BPG):

Binds to a pocket (made by the twoβ-chains) in the deoxyhemoglobin
b) Carries several positive charges
c) Bindswiththeα- chain of hemoglobin
d) Increases the hemoglobin affinity for oxygen and prevents its release to tissues

CANCELLED

(End of module exam 2021)

15-The porphyrin ring in hemoglobin molecules have in its center an atom of:

a) Magnesium
b) Iron
c) Hydrogen
d) Nickel

(End of module exam 2021)

16-Which one of the following statements concerning hemoglobin is correct?
a) HbA_2 is the most abundant hemoglobin in normal adults
b) Fetal blood has a higher affinity for oxygen than adult blood
c) The globin chain composition of HbF is $\alpha_2\delta_2$
d) HbA appears early in fetal life

(Depart. book)

17-Which one of the following statements concerning the binding of oxygen by hemoglobin is correct?

a) The Bohr effect results in a lower affinity for oxygen at higher pH values
b) Carbon dioxide increases the oxygen affinity of hemoglobin by binding to the C-terminal groups of the polypeptide chains
c) The oxygen affinity of hemoglobin increases as the percentage saturation increases
d) The hemoglobin tetramer binds four molecules of 2,3-bisphosphoglycerate

CANCELLED

(Depart. book)

18-Which one of the following is the amino acid in hemoglobin that accepts H^+ and allows hemoglobin to act as a buffer?

a) Alanine
b) Histidine
c) Serine
d) Aspartate

CANCELLED

(Depart. book)

19-Iron of Heme is characterized by:

a) Heme is present in a hydrophobic pocket between F and G helices
b) The 6th coordination of iron forms with the proximal Histidine (His E8)
c) The 5th coordination of iron forms with the distal Histidine (His E7)
d)Fourcoordinationareformedbetweenironand4 nitrogen of the porphyrin ring

(Faculty revision 2022)

20-Binding of carbon monoxide to heme at less favored angle is caused by:

a) Presence of proximal histidine
b) Presence of distal histidine
c) Solubility of heme
d) Formation of heme-O_2-heme complex

(Faculty revision 2022)

21-The efficiency of hemoglobin as an oxygen transporter is caused by:
a) The low affinity of heme to oxygen
b) The low affinity of heme to carbon monoxide
c) The cooperative binding of oxygen to heme
d) All of the above

(Faculty revision 2022)

22-As regards hemoglobinopathies, one statement is CORRECT:

a) In sickle cell anemia Valine is replaced with Glutamic acid.
b) In HbM proximal or distal Histidine of α- or β-chains are replaced with Lysine
c) In Hb C proximal or distal Histidine of α- or β-chains are replaced with tyrosine
d) β-Thalassemia is associated with increased HbF as well as HbA_2.

(Faculty revision 2022)

23-One of the following DOES NOT characterizes functions of Globin and Apomyoglobin:

a) Globin has surface non-polar amino acids
b) Reduces the affinity of Heme to bind Carbon Monoxide
c) They are responsible for the characteristic Sigmoid Oxygen Dissociation curve
d) Prevents formation of Heme-O2-Heme complex

(Faculty revision 2022)

24-As regards minor hemoglobins, one statement is incorrect:

a) HbA_2 is increased in β-thalassemia
b) HbA_2 is increased in α-thalassemia
c) HbF is composed of α_2B_2
d) Hemoglobin Gower appears during the 1st few weeks after conception

(Faculty revision 2022)

25-As regards hemoglobinopathies, one statement is CORRECT:

a) In sickle cell anemia Glutamic acid in β-chain is replaced with Valine.
b) In homozygous α-thalassemia, the fetus suffers from hydrops fetalis
c) In β-Thalassemia, manifestations of hemolytic anemia DO NOT appear immediately after birth
d) All of the above

(Faculty revision 2022)

26-As regards hemoglobinopathies, one statement is INCORRECT:

a) In HbM disease tyrosine replaces proximal or distal histidine in either α or β chain.
b) In α-thalassemia trait, patients suffer from severe anemia
c) In β-Thalassemia, manifestations of hemolytic anemia DO NOT appear immediately after birth
d) Polymerization of HbS to form insoluble fibers occurs in the deoxygenated state

(Faculty revision 2022)

27-Hemoproteins are:

a) Simple proteins
b) Conjugated proteins
c) Glycoprotein
d) Lipoproteins

(Depart. Questions book)

28-The main function of hemoglobin is:

a) Oxygen transport in blood
b) Oxygen storage in muscles
c) Hydroxylation reactions
d) Oxidation of tryptophan

(Depart. Questions book)

29-The heme ring is present in the hydrophobic pocket between:

a) E and F helices
b) A and B helices
c) C and D helices
d) G and H helices

(Depart. Questions book)

30-Importance of apomyoglobin and the globin of hemoglobin:

a) Facilitates the formation of heme-O2-heme complex
b) Prevents diffusion of heme to the plasma
c) Makes heme less soluble
d) Increase the affinity of heme to bind carbon monoxide (CO)

(Depart. Questions book)

31-The main function of myoglobin is:

a) Acting as a reservoir for oxygen in cardiac and skeletal muscles
b) Acting as a buffer in RBCs
c) Oxygen transport in blood
d) Degradation of H202

(Depart. Questions book)

32-When the partial pressure of oxygen (pO2) is 20 mmHg (low partial pressure of oxygen):

a) Myoglobin is 50% saturated
b) Myoglobin is 20% saturated
c) Myoglobin is 30% saturated
d) Myoglobin is 90% saturated

(Depart. Questions book)

33-In sickle cell anemia, the basis of the malfunction of the hemoglobin molecules is:

a) Incorrect secondary structure
b) Substitution of a single amino acid
c) Insufficient iron in the diet
d) Faulty binding of the heme groups

(Depart. Questions book)

34-Which of the following is false?

a) Heme consists of a complex organic ring structure, protoporphyrin
b) Protoporphyrin is bound to a single iron atom in its Fe+++ state
c) Iron atom has 6 coordination bonds
d) Heme is found in a number of oxygen-transporting proteins

(Depart. Questions book)

35-Myoglobin is present in:

a) Nerves
b) Blood cells
c) Muscles
d) Skin

(Depart. Questions book)

36-Myoglobin and the subunits of hemoglobin have:

a) Different primary structure
b) Different tertiary structure
c) Similar primary structure
d) Similar quaternary structure

(Depart. Questions book)

37-Hemoglobin S:

a) Results from insertion of a nucleotide
b) Results from point mutation
c) Is incompatible with life
d) Runs faster than the normal HbA in electrophoresis

(Depart. Questions book)

38-Fetal hemoglobin (HbF):

a) Consists of 2 alpha and 2delta chains (α_2 and δ_2)
b) Runs faster than adult hemoglobin in electrophoresis
c) Has a higher affinity to oxygen than adult hemoglobin
d) Is the major hemoglobin found in adults

(Depart. Questions book)

39-2,3-bisphosphoglycerate:

a) Is positively charged
b) Binds to the R form of hemoglobin
c) Binds to a pocket formed by 2 α-chains
d) Is present in the center of the deoxyhemoglobin

(Depart. Questions book)

40-Hemoglobin A_2 contains:

a) α_2, β_2
b) α_2, δ_2
c) α_2, γ_2
d) δ_2, δ_2

(Depart. Questions book)

41-In α-thalassemia:

a) The synthesis of α-globin chains is increased
b) The homozygous α-thalassemia patient shows no manifestations of the disease
c) The α-thalassemia trait suffers from mild anemia
d) The α-thalassemia major patient has 2 α-globin defective genes

(Depart. Questions book)

42-One of the following is not a hemoproteins:

a) Cytochromes.
b) Hemoglobin.
c) Catalase.
d) Kinases

(Old Depart. Questions book)

43-One of the following statements about glycosylated hemoglobin is incorrect:

a) It is formed by non-enzymatic glycosylation of hemoglobin
b) It has glucose residues attached to serine, threonine or tyrosine residues of hemoglobin.
c) It is normally about 5% of the total hemoglobin.
d) It is elevated when control of diabetes is poor.

(Old Depart. Questions book)

44-Heme is present in a pocket between:

a) F and G helices.
b) E and F helices.

d) A and B helices.

(Old Depart. Questions book)

45-Fe++ of heme can form:

a) 5 coordinations.
b) 6 coordinations.
c) 8 coordinations.
d) 5 or 6 coordinations.

(Old Depart. Questions book)

46-Proximal Histidine is:

a) HisE 7.
b) His F 6.
c) HisE 8.
d) His F 8.

(Old Depart. Questions book)

47-Apomyoglobin contains:

a) 141 aa.
b) 146 aa.
c) 151 aa.
d) 153 aa.

(Old Depart. Questions book)

48-HbA contains:

a) α2β2 .
b) α2δ2 .
c) α2γ2 .
d) α2ε2 .

(Old Depart. Questions book)

49-β-chain of Hb consists of:

a) 141 aa.
b) 146 aa.
c) 153 aa.
d) 148 aa.

(Old Depart. Questions book)

50-Hemoglobin A1c:

a) Is not present in normal persons.
b) Is formed by the enzymatic glycosylation of hemoglobin.
c) Has glucose residues attached to it.
d) Its level decreases in uncontrolled diabetes mellitus.

(Old Depart. Questions book)

51-2,3-Bisphosphoglycerate is released from hemoglobin:

a) Oxygen tension decreases.
b) Oxygen tension increases
c) Glycolysis increases
d) Glycolysis decreases

(Old Depart. Questions book)

52-Fetal Hb is characterized by:

a) Contains α2 ,γ2
b) Has lower affinity for O than HbA.
c) Faster than HbA in electrophoresis.
d) Appear after birth.

(Old Depart. Questions book)

53-In HbC , glutamic acid in 6th position of β-chain is replaced by:

a) Leucine.
b) Valine.
c) Lysine.
d) Tyrosine.

(Old Depart. Questions book)

54-Hemoglobin is formed of:

a. 1 Heme and 1 Polypeptide chain
b. 2 Heme and 2 polypeptide chains
c. 3 Heme and 3 polypeptide chains
d. 4 Heme and 4 polypeptide chains

55-Importance of distal histidine in hemoglobin is:

a. Prevents oxidation of heme to hematin
b. Form the boundary of oxygen pocket stabilizing binding of O2
c. Decreases binding of CO
d. b and c

56-2,3 BPG causes:

a. Release of O2 at tissues
b. Shift of oxygen dissociation curve to the right
c. Stabilize T form of hemoglobin
d. All the above

57-Binding of of CO2 to hemoglobin causes:

a. Carbamino hemoglobin
b. Carboxy hemoglobin
c. Cyano hemoglobin
d. All the above

58-Binding of CO to hemoglobin causes:

a. Carbamino hemoglobin
b. Carboxy hemoglobin
c. Cyano hemoglobin
d. All the above

59-Carbon monoxide is very toxic as:

a. Binds to hemoglobin at oxygen binding site
b. Affinity of heme to CO 200 times as oxygen
c. Stabilize R form and prevent release of oxygen
d. All the above

60-HbA2 is formed of:

a. α2β2
b. α2γ2
c. α2δ2
d. α2ϵ2

61-HbF is formed of:

a. α2β2
b. α2γ2
c. α2δ2
d. α2ϵ2

62-HbA is formed of:

a. α2β2
b. α2γ2
c. α2δ2
d. α2ϵ2

63-Hb Gower is formed of:

a. α2β2
b. α2γ2
c. α2δ2
d. α2ϵ2

64-Adult hemoglobin is a tetramer consisting of:

a. α2β2
b. α2δ2
c. α2γ2
d. β4

65-In Sickle cell anemia the glutamate is replaced by:

a. Valine
b. Lysine
c. Tyrosine
d. Histidine

66-One of the following decreases affinity of heme to CO from 25000 to 200 times:

a. Proximal histidine
b. Distal histidine
c. 2 Propionates
d. None of the above

67-Most of CO2 produced in tissues is transferred to lungs:

a. Bound to hemoglobin
b. In the form of bicarbonate
c. Diffuse in blood as CO
d. Bound to myoglobin

68-In HbM disease there is replacement of proximal or distal histidine by:

a. Valine
b. Serine
c. Tyrosine
d. Lysine

69-α Thalassemia may result in a silent carrier if there is a defect in:

a. 1 gene
b. 2 genes
c. 3 genes
d. 4 genes

70-Hemoglobin A1 in adults is composed of:

a) Two alpha and two gamma chains
b) Two alpha and two delta chains
c) Two alpha and two beta chains
d) Two zeta and two epsilon chains

(Exam 2026)

71-The primary function of myoglobin is to:

a) Transport oxygen in the blood
b) Act as a reservoir for oxygen in muscles
c) Transport electrons
d) Degrade hydrogen peroxide

(Exam 2026)

72-Concerning affinity of hemoglobin to oxygen:

a) The R form is the deoxy form of hemoglobin
b) The T form is the form of hemoglobin when it binds oxygen
c) The T form is the low oxygen-affinity form of hemoglobin.
d) The R form is the low oxygen-affinity form of hemoglobin.

(Exam 2026)

73-HbA1c is formed by a non-enzymatic reaction between glucose and .....:

a) The heme iron
b) The C-terminal valine of the alpha chain
c) The N-terminal valine of the beta chain
d) The distal histidine of globin

(Exam 2026)

74-The Protein Part (Globin) of Hemoglobin:

a) Facilitates the formation of heme-O2-heme complex
b) Prevents diffusion of heme to the plasma
c) Makes heme less soluble
d) Increases the affinity of heme to bind carbon monoxide (CO)

(Exam 2026)

75-In sickle cell anemia, the 6th amino acid glutamate of the β chain is replaced by:

a) Tyrosine
b) Valine
c) Glycine
d) Serine

(Exam 2026)

76-The "sticky receptors" that lead to the polymerization of HbS are exposed in:

a) Only oxy-HbS
b) Only deoxy-HbS
c) Both oxy-HbS and deoxy-HbS
d) Neither form; it is always hidden (Exam 2026)

77-The embryonic hemoglobin is:

a) HbA ($\alpha_2\beta_2$)
b) HbF ($\alpha_2\gamma_2$)
c) HbA2 ($\alpha_2\delta_2$)
d) Hb Gower-1 ($\zeta_2\varepsilon_2$)

(Exam 2026)

78-Hemoglobin is formed of:

a) One heme group attached to one polypeptide chain
b) Two heme groups attached to one polypeptide chain
c) Three heme groups attached to three polypeptide chains
d) Four heme groups attached to four polypeptide chains

(Exam 2024)

79-Myoglobin is regarded as:

a) Simple protein
b) Conjugated protein ✓ [Marked Answer]
c) Derived protein
d) Lipoprotein

(Exam 2024)

80-Transport oxygen in blood is the main function of:

a) Cytochromes
b) Hemoglobin
c) Myoglobin
d) Transferrin

(Exam 2024)

81-In β-thalassemia major:

a) Patients carry two mutated β- globin genes b) The expression of the gamma and delta chains decreases
c) Patients suffer from mild anemia
d) HbS will be formed instead of α2β2

(Exam 2024)

82-The 6th amino acid (glutamate) of β chain is replaced by valine in:

a) Sickle cell disease
b) β-Thalassemia major
c) β-Thalassemia minor
d) Hb C disease

(Exam 2024)

83-Globin part of hemoglobin is important for:

a) Allowing oxidation of iron
b) Allowing diffusion of heme to plasma
c) Making heme soluble
d) Increasing affinity of heme to bind carbon monoxide

(Exam 2024)

84-Regarding Fetal Hb, which of the following is correct?

a) It is slower than adult hemoglobin in electrophoresis.
b) It has lower affinity for O2 than adult hemoglobin.
c) It is formed of two alpha and two gamma chains
d) It appears after birth

(Exam 2024)

Answers

1-A	2-D	3-B	4-C	5-D	6-D	7-B	8-B	9-D
10-D	11-A	12-C	13-C	CANCELLED	15-B	16-B	CANCELLED	CANCELLED
19-D	20-B	21-C	22-D	23-A	24-B	25-D	26-B	27-B
28-A	29-A	30-B	31-A	32-D	33-B	34-B	35-C	36-B
37-B	38-C	CANCELLED	40-B	41-C	42-D	43-B	44-B	45-D
46-B	47-D	48-A	49-B	50-C	51-B	52-A	53-C	54-D
55-D	56-D	57-A	58-B	59-D	60-C	61-B	62-A	63-D
64-A	65-A	66-B	67-B	68-C	69-A	70-C	71-B	72-C
73-C	74-B	75-B	76-B	77-D	78-D	79-B	80-B	81-A
82-A	83-C	84-D						

Chapter 6 PROTEINS OF EXTRACELLULAR MATRIX

1-Which of the following is a fibrous protein?
a) Hemoglobin
b) Immunoglobulin
c) Collagen
d) Myoglobin

(End of module exam 2023)

2-Concerning collagen molecule, which of the following is not correct?

a) Collagen has a strong structure
b) It is formed of three polypeptides (α- chains)
c) Glycine accounts for one-third of amino acids found within collagens
d) Vitamin A is essential for hydroxylation of proline

(End of module exam 2021)

3-Marfan's syndrome is due to mutations in the gene encoding:
a) Collagen
b) Elastin
c) Fibrillin
d) Fibronectin

CANCELLED

(End of module exam 2021)

4-Collagen is rich in:

a) Glycine, proline, and hydroxyproline
b) Glutamate, proline and hydroxyproline
c) Glycine, cysteine and cystine
d) Valine, leucine and isoleucine

(End of module exam 2021)

5-One of the following is used as a marker of bone formation:

a) Chondronectin
b) Osteonectin
c) Osteocalcin
d) Proteoglycans

(End of module exam 2021)

6-Osteogenesis imperfecta is caused by:

a) Deficiency of alpha-1 antitrypsin
b) Replacement of glycine residues by amino acids with bulky side chains
c) Mutation of the gene encoding fibronectin
d) Deficiency of lysyl hydroxylase

(End of module exam 2021)

7-A thirty-year-old woman presented with progressive shortness of breath. She denied the use of cigarettes. A family history revealed that her sister had suffered from unexplained lung disease. Which one of the following etiologies most likely explains this patient's pulmonary symptoms?

CANCELLED

a) Deficiency of proline hydroxylase
b) Defici3ency of α-1-antitrypsin
c) Deficiency in dietary vitamin C
d) Decreased elastase activity
(Depart. book)

8-A teenager boy came to the clinic for a routine physical exam. The boy had rarely seen a doctor before. Upon examination, the doctor noticed a high arched palate, disproportionately long arms and fingers, and a sunken chest. He has been complaining of lack of breath while doing routine efforts, and upon listening to his heart, an aortic regurgitation murmur was detected. Careful examination of the eyes showed dislocation of the eye lens. Based on the physical exam and history, which of the following proteins is suspicious to have an inborn error of its metabolism?

CANCELLED

a) Collagen
b) Fibrillin
c) Elastin
d) Fibronectin

(Depart. book)

9-A protein rich in glycine is:
a) Collagen
b) Hemoglobin
c) Keratin
d) Fibronectin

(Depart. Questions book)

10-The number of helices in collagen molecule are:
a) 2
b) 3
c) 4
d) 1

(Depart. Questions book)

11-Each turn in the helix of tropocollagen contains:

a) 3 amino acids
b) 4 amino acids
c) 3.6 amino acids
d) 4.6 amino acids

(Depart. Questions book)

12-Each third position in the amino acid composition of collagen contains:

CANCELLED

a) Serine
b) Proline
c) Lysine
d) Glycine

(Depart. Questions book)

13-In collagen, the vitamin required for the formation of hydroxyproline is:

a) Vitamin C
b) Vitamin A
c) Vitamin D
d) Vitamin E

(Depart. Questions book)

14-In collagen, glucose and galactose residues are attached to:

a) Lysine
b) Proline
c) Hydroxylysine
d) Hydroxyproline

(Depart. Questions book)

15-A deficiency of copper effects the formation of normal collagen by reducing the activity of which of the following enzymes?

CANCELLED

a) Prolyl hydroxylase
b) Lysyl hydroxylase
c) Lysyl oxidase
d) Glucosyl transferase

(Depart. Questions book)

16-Osteogenesis imperfecta syndrome is due to mutation in:

CANCELLED

a) Collagen
b) Elastin
c) Fibrillin
d) Fibronectin

(Depart. Questions book)

17-Collagen and elastin have the following similarity:

a) Both are glycoproteins
b) Both, tropocollagen and tropoelastin, are triple helices
c) Both have hydroxyproline residues
d) Both have hydroxylysine residues

(Depart. Questions book)

18-Fibronectin has binding sites for:

a) Collagen
b) Elastin
c) Osteocalcin
d) Chondronectin

(Depart. Questions book)

19-The 2 identical polypeptide chains of fibronectin are joined by:

a) Ionic bond
b) Hydrogen bond
c) Disulfide bond
d) Hydrophobic interactions

(Depart. Questions book)

20-Marfan's syndrome results from a mutation in the gene coding:

CANCELLED

a) Collagen
b) Elastin
c) Fibrillin
d) Keratin

(Depart. Questions book)

21-Carboxylation of glutamate residues in bone tissue requires:

a) Vitamin A
b) Vitamin C
c) Vitamin D
d) Vitamin K

(Depart. Questions book)

22-Bone proteins contain mainly collagen:

a) Type I
b) Type II
c) Type III
d) Type IV

(Depart. Questions book)

23-The most abundant collagen in cartilage is:

a) Type I
b) Type II
c) Type III
d) Type IV

(Depart. Questions book)

24-The major macromolecules of the ECM are polysaccharides and ..... :

a) Lipids.
b) Fat.
c) Proteins.
d) Amino acids

(Old Depart. Questions book)

25-The most abundant structural protein is:

a) Collagen.
b) Keratin.
c) Fibronectin.
d) Elastin.

(Old Depart. Questions book)

26-Collagen:

a) Is a globular protein.
b) Is rich in leucine.
c) Is a very strong structure.
d) Consists of 2 alpha chains.

(Old Depart. Questions book)

27-With regards to collagen molecule, Which of the following is FALSE:

a) It is formed from trimers
b) It is formed from polypeptides containing α-chains
c) Glycine accounts for one-third of amino acids found within collagens
d) Vitamin A is essential for hydroxylation of proline.

(Old Depart. Questions book)

28-Conversion of preprocollagen α-chain to procollagen is associated with:

a) Removal of signal peptide.
b) Hydroxylation of prolyl and lysyl residues.
c) Glycosylation of hydroxylysyl residues.
d) All of the above.

(Old Depart. Questions book)

29-Hydroxylation of proline and lysine in a protein is effected by:

a) Vitamin B1 .
b) Vitamin B2 .
c) Vitamin B6 .
d) Vitamin C.

(Old Depart. Questions book)

30-The only protein which contains hydroxylysine is:

a) Elastin.
b) Fibrillin.
c) Collagen.
d) Fibronectin

(Old Depart. Questions book)

31-A child with tall stature, hyperextensible joints, and dislocation of retinas is found to have a mutation in the gene encoding:

a) Collagen
b) Elastin
c) Trypsin
d) Fibrillin

CANCELLED

(Old Depart. Questions book)

32-In vitamin C deficiency, which amino acid that is normally part of collagen is not synthesized?

a) Hydroxytryptophan.
b) Hydroxytyrosine.
c) Hydroxyalanine.
d) Hydroxyproline

(Old Depart. Questions book)

33-Which ECM molecule provides structure while allowing certain tissues to expand and contract (e.g., lungs during breathing, intestine during digestion)?

a) Collagen
b) Elastin
c) Fibronectin
d) Glycosaminoglycans.

CANCELLED

(Old Depart. Questions book)

34-Which of the following statements about fibronectin is TRUE?

a) Fibronectin is present in extracellular matrix and lung.
b) Fibronectin is formed from a triple helix of alpha-chains.
c) Fibronectin contains binding sites for heparin and collagen.
d) Defects in fibronectin gene lead to Marfan's syndrome.

(Old Depart. Questions book)

35-Which collagen type is most rigid and is very abundant in bones?

a) Type I
b) Type II
c) Type IV
d) Type V

(Old Depart. Questions book)

36-All of the following are true as regards osteocalcin except:

a) Bone specific.
b) Contains 3 to 5 γ-carboxyl glutamate.
c) Synthesized by osteoclasts.
d) Induced by calcitriol.

(Old Depart. Questions book)

37-Structural unit of collagen is:

a. α Chain
b. Amino acid
c. Procollagen
d. Tropocollagen

38-One of the following amino acids represents 1/3 of the amino acid sequence in collagen:

a. Glycine
b. Proline
c. Hydroxy proline
d. Alanine

39-Prolyl hydroxylase and lysyl hydroxylase require:

a. Vitamin A
b. Vitamin C
c. Vitamin K
d. Vitamin D

40-Lysyl oxidase requires:

a. Cupper
b. Fe
c. Mg
d. Mn

CANCELLED

41-In scurvy, which amino acid that is normally part of collagen is not synthesized?

a. Hydroxy Tryptophan
b. glycine
c. Hydroxy Alanine
d. Hydroxy Proline

CANCELLED

42-The most abundant protein in bones is:

a. Collagen type I
b. Collagen type II
c. Collagen type III
d. Non-collagen proteins

43-The most abundant collagen in cartilages is:

a. Type I
b. Type II
c. Type III
d. Type IV

44-Major non collagenous protein of bone is:

a. Osteocalcin
b. Osteonectin
c. Osteoblast
d. Proteoglycan

45-Major non collagenous protein of cartilage is

a. Aggrecan
b. Chondronectin
c. Osteoblast
d. Proteoglycan

46-One of the following is responsible for elastic properties of elastin:

a. Allysine
b. Desmosine
c. Fibrillin
d. Tropoelastin

47-One of the characteristic features of elastin is:

a) It is a soluble protein.
b) Its monomers are stabilized by desmosine cross-links.
c) It has high hydroxylysine content.
d) It is formed of three polypeptide chains.

(Exam 2026)

48-The predominant collagen type in cartilage is:

a) Type I
b) Type II
c) Type III
d) Type V

(Exam 2026)

49-The triple helix structure of tropocollagen is formed by:

a) Three right-handed α-helices.
b) Two right-handed and one left-handed helix.
c) Three left-handed α-helices twisted into a right-handed superhelix.
d) A single polypeptide chain coiled upon itself.

(Exam 2026)

50-The bottle-brush shape is characteristic of:

a) Tropocollagen
b) Tropoelastin
c) Aggrecan
d) Osteocalcin

(Exam 2026)

51-One of the changes that occurs in the structure of aggrecan with age is:

a) Increase chondroitin sulfate content.
b) Decrease keratan sulfate content.
c) The hyaluronic acid chain becomes longer.
d) Decrease the average size of the molecule.

(Exam 2026)

52-Collagen and elastin have the following similarity:

a) Both are glycoproteins
b) Both, tropocollagen and tropoelastin, are triple helices
c) Both have hydroxyproline residues
d) Both have hydroxylysine residues

(Exam 2026)

53-Which of the following is an extracellular matrix protein?:

a) Hemoglobin
b) Immunoglobulin
c) Elastin
d) Myoglobin

(Exam 2024)

54-Hydroxylation reactions in collagen synthesis need which of the following vitamins?

a) Vitamin A
b) Vitamin C
c) Vitamin E
d) Vitamin K

(Exam 2024)

55-Which of the following is regarded as non-collagenous cartilage protein?

a) Osteocalcin
b) Aggrecan
c) Osteonectin
d) Elastin

(Exam 2024)

56- ..... is used as a marker for bone formation:

a) Chondronectin
b) Osteocalcin (Marked in image)
c) Hemoglobin
d) Proteoglycans

(Exam 2024)

Answers

1-C	2-D	3-C	4-A	5-C	6-B	7-B	8-B	9-A
10-B	11-A	12-C	13-A	14-C	15-C	16-A	17-C	18-A
19-C	20-C	21-D	22-A	23-B	24-C	25-A	26-C	27-D
28-D	29-D	30-C	31-D	32-D	33-B	34-C	35-A	36-C
37-D	38-A	39-B	40-A	41-D	42-A	43-B	44-A	45-A
46-B	47-b	48-B	49-C	50-C	51-D	52-C	53-C	54-B
55-B	56-B							

Chapter 7 ENZYMES

1- The general mechanism involved in enzyme action is:

a) Lowering activation energy
b) Increasing activation energy
c) Decreasing pH value
d) Increasing pH value

(End of module exam 2023)

2- The effect of a competitive inhibitor is reversed by increasing the concentration of the:

a) Enzyme
b) Substrate
c) Inhibitor
d) Catalyst

(End of module exam 2023)

3- In enzyme kinetics, Km is:

a) The substrate concentration that gives half Vmax
b) The dissociation constant for the enzyme substrate complex
c) Concentration of enzyme
d) Half of the substrate concentration required to achieve Vmax

(End of module exam 2023)

4- Which of the following represents irreversible enzyme inhibition?

a) Statins
b) Allopurinol
c) Salts of Heavy metals
d) Sulfonamides

(End of module exam 2023)

5- In competitive inhibition:

a) Inhibitor has structural similarity to substrate.
b) Km is decreased
c) Vmax is decreased
d) Reaction rate is independent of substrate concentration.

(End of module exam 2023)

6- Concerning the effect of pH on the catalytic activity of the enzymes:

a) Activity increases as we go away from the optimum pH.
b) Changes of pH alter the charges on the substrate and the active site.
c) Extreme changes of pH don't affect the catalytic activity.
d) Pepsin is maximally active at alkaline pH (above 7)

(End of module exam 2023)

7- Allosteric effector influences the enzyme activity by:

a) Covalently modifying the enzyme
b) Binding to the substrate and altering its conformation
c) Competing for catalytic site with substrate
d) Binding to a specific site on the enzyme away from catalytic site

(End of module exam 2023)

8- Zymogen is:

a) a hormone
b) a vitamin
c) a proenzyme
d) a modulator

(End of module exam 2023)

9- Activation or inactivation of certain enzymes is accomplished by covalent modification of the amino acid:

a) Hydroxy lysine
b) Phenylalanine
c) Lysine
d) Serine

(End of module exam 2023)

10- Covalent modifications of enzymes involve addition of one of the following:

a) Methyl group
b) Methylene group
c) Phosphate group
d) Thiol group

(Final exam 2022 - September, Faculty summer revision 2024)

11- Covalent modification of some enzymes involves addition or removal of:

a) Nitrate
b) Sulfate
c) Phosphate
d) Coenzyme

(End of module exam 2021, End of the module 2019)

12- Using strong acids or alkalies with enzymes is a kind of:

a) Competitive inhibition
b) Non-competitive inhibition
c) Allosteric inhibition
d) Irreversible inhibition

(End of module exam 2021)

13- The effect of a competitive inhibitor is reversed by increasing the concentration of the:

a) Enzyme
b) Substrate
c) Inhibitor
d) Catalyst

(End of module exam 2021)

14- The optimum pH of most of the enzymes is:

a) Between 2 and 4
b) Between 5 and 9
c) Between 8 and 12
d) Above 12

(End of module exam 2021, End of the module 2019)

15. As regards allosteric inhibitors, one of the following is incorrect:

a) They produce conformational changes in the enzyme structure leading to decrease its activity
b) They are usually small organic molecules.
c) They increase the Km
d) They bind to the catalytic site.

(End of module exam 2021)

16- Using strong acids or alkalis with enzymes is a kind of:

a) Competitive inhibition
b) Non-competitive inhibition
c) Allosteric inhibition
d) Irreversible inhibition

(Final exam 2021, End of the module 2019)

17- In competitive inhibition, all of the following is correct Except:

a) The inhibitor has structural similarity to the substrate.
b) The inhibitor binds irreversibly at the active site.
c) Km is increased.
d) Vmax is not changed.

(End of the module 2019)

18- Which of the following statements is NOT true?

a) Enzymes can denature when the temperature gets too high
b) Enzymes are changed chemically by the end of the reaction
c) Extreme pH can denature enzymes
d) Enzymes speed up chemical reactions

(End of the module 2020)

19. Sulfonamide drugs inhibit folic acid synthesis by:

a) Irreversible enzyme inhibition
b) Enzyme denaturation
c) Competitive inhibition
d) Binding to allosteric site of enzyme

(End of the module 2020)

20. Allosteric inhibition of an enzyme involves which of the following?

a) Binding of an inhibitor to a site other than the substrate binding site
b) Binding of an inhibitor competitively to the substrate binding site
c) Binding of an inhibitor noncompetitively to the substrate binding site
d) Allosteric inhibitors increase the affinity of the enzyme to its substrate

(End of the module 2020)

21- In competitive inhibition, the inhibitor:

a) Competes with the enzyme
b) Irreversibly binds with the enzyme
c) Binds with the substrate
d) Competes with the substrate

(End of the module 2020)

22. Covalent modification of an enzyme usually involves phosphorylation / dephosphorylation of the amino acid:

a) Serine
b) Proline
c) Hydroxylysine
d) Phenylalanine

(End of the module 2020)

23- In enzyme kinetics Km is:

a) The substrate concentration that gives one half Vmax
b) The dissociation constant for the enzyme substrate comples
c) Concentration of enzyme
d) Half of the substrate concentration required to achieve Vmax

(Final exam 2019 - September sitting)

24. Case: A 56-year-old male came to the hospital with acute pain in his left big toe, he gave a history of eating about 250 gm red meat every day, uric acid in the plasma was measured and found elevated 12 mg/dl (normal 4-7 mg/dl) and he was diagnosed as acute gout.
1) Which drug is best prescribed for this patient?

a) Dicumarol
b) Sulfonamides
c) Allopurinol
d) Warfarin

(Faculty Summer revision 2024)

2) What is the mechanism of action of the drug selected above?

a) Competitive inhibition
b) Allosteric inhibition
c) Antienzyme
d) Denaturizing protein

(Depart. book)

3) Which statement is correct about drug mechanism?

a) Increased Vmax and Km.
b) No effect on Vmax and increase Km.
c) Antienzyme.
d) Denaturating proteins

(Faculty Summer revision 2024)

25- Enzymes, which are produced in inactive form in the living cells, are called:

a) Apoenzymes
b) Coenzymes
c) Antienzymes
d) Proenzymes

(Depart. book)

26. All the following statements are true regarding enzymes Except:

a) Enzymes lower activation energy
b) They alter equilibrium of the reaction
c) They accelerate the chemical reaction
d) Most of the enzymes are proteins in nature

(Depart. book)

27- Which of the diagrams illustrated below represents the effect of ATP on phosphofructoki... 1(PFK-1)

CANCELLED

Old Depart. book

28. Tamiflu and Relanza are drugs used in the treatment of avian influenza virus. Both act on an enzyme called viral neuraminidase. The graph below shows the kinetic data obtained for viral neuraminidase activity in absence and presence of Tam...

CANCELLED

Based on the kinetic data which of the following statements is correct?

a) Both drugs are competitive inhibitors of viral neuraminidase
b) Both drugs are non competitive inhibitors of viral neuraminidase
c) Tamiflu increases the Km value of the substrate compared to Relanza
d) Relanza increases the Vmax value of the substrate compared to Tamiflu

(Old Depart. book)

29- A patient presents in your office with very high levels of serum cholesterol. He states that he has tried to follow the diet and exercise regimen you gave him last year. You decide that this patient would benefit from statins. This class of drugs is effective in treating hypercholesterolemia because it has what effect?

a) Stimulates phosphorylation of the HMG-CoA reductase enzyme
b) Binds cholesterol preventing it from being absorbed by the intestine
c) Directly prevents the deposition of cholesterol on artery walls
d) Inhibits the enzyme HMG-CoA reductase

(Faculty Summer revision 2024)

30- One of the following characterizes competitive enzyme inhibitors:

a) The inhibitor competes with the enzyme
b) The inhibitors decreases Km
c) The inhibitor has NO effect on Vmax
d) Cyanide is a competitive inhibitors on Cytochrome oxidase

(Faculty Summer revision 2024)

31- Enzymes can recognize and react with a special chemical substance called:

a) Cofactor
b) Activator
c) Substrate
d) Product

(Depart. Questions book)

32- Enzymes belong to which group of biomolecules?

a) Lipids
b) Carbohydrates
c) Phospholipids
d) Proteins

(Depart. Questions book)

33. Common features of the enzymes include:

a) Produced by living cells
b) Affection of the reaction equilibrium
c) Chemically changed at the end of the reaction
d) Needed in large amount

(Depart. Questions book)

34- The enzymes are polymers of:

a) Fatty acids
b) Amino acids
c) Hexose sugars
d) Inorganic phosphate

(Depart. Questions book)

35- The active site of an enzyme:

a) Is remote from the site of substrate attachment
b) Is converted to product
c) Catalyses the reaction
d) Increases the energy of reaction

(Depart. Questions book)

36- In any reaction catalyzed by an enzyme, the reacting molecule is called the:

a) Substrate
b) Cofactor
c) Coenzyme
d) Isozyme

(Depart. Questions book)

37- In any reaction catalyzed by an enzyme, the reacting molecule is called the:

a) Substrate
b) Cofactor
c) Coenzyme
d) Isozyme

(Depart. Questions book)

38. The general mechanism in enzyme action is by:

a) Reducing activation energy
b) Increasing activation energy
c) Decreasing pH value
d) Increasing pH value

(Depart. Questions book)

39- The 'induced fit' model theory of the enzyme action implies that:

a) The active site is complementary in shape to that of substance only after interaction
b) The active site is complementary in shape to that of substance
c) Substrates induce conformational change in the active site
d) The active site is flexible and adjusts to substrate

(Depart. Questions book)

40. ..... is produced by combination of apoenzyme and cofactor.

a) Enzyme product complex
b) Enzyme substrate complex
c) Holoenzyme
d) Prosthetic group

(Depart. Questions book)

41- Factors affecting enzyme activity:

a) Temperature
b) Concentration of substrate
c) pH
d) Temperature, concentration & pH

(Depart. Questions book)

42- The pH level where the enzyme is most active is known as ..... pH.

a) Standard
b) Equilibrium
c) Optimum
d) Neutral

(Depart. Questions book)

43. Concerning effect of pH on the catalytic activity of the enzymes:

a) Activity increases as we go away from the optimum pH
b) Changes of pH alter the charges on the substrate and the active site
c) Extreme changes of pH don't affect the catalytic activity
d)Pepsin is maximally active at alkaline pH (above 7)

(Depart. Questions book)

44- The catalytic efficiency of two different enzymes can be compared by the:

a) Formation of the product
b) Km value
c) Molecular size of the enzymes
d) pH of optimum value

(Depart. Questions book)

45. If the amount of substrate is not limiting, the velocity of reaction is directly proportional to:

a) Concentration of enzymes
b) Temperature
c) Concentration of products
d) Ph

(Depart. Questions book)

46. Blocking the enzyme action by blocking its active site is called:

a) Non-competitive inhibition
b) Allosteric inhibition
c) Competitive inhibition
d) Feedback inhibition

(Depart. Questions book)

47- Enzyme inhibition caused by a substance resembling substrate molecule is called:

a) Allosteric inhibition
b) Competitive inhibition
c) Feedback inhibition
d) Non-competitive inhibition

(Depart. Questions book)

48. A competitive inhibitor:

a) Increases the Km of the enzyme
b) Decreases the Vmax of the enzyme
c) Increases the Vmax of the enzyme
d) Decreases the Km of the enzyme

(Depart. Questions book)

49- Which of the following represents irreversible enzyme inhibition?

a) Statins
b) Allopurinol
c) Mercury
d) Sulfonamides

(Depart. Questions book)

50- Iodoacetate inhibitory effect on the enzymes is considered:

CANCELLED

a) Competitive inhi
b) Allosteric inhibition
c) Feedback inhibition
d) Irreversible inhibition

(Depart. Questions book)

51. Zymogen or proenzyme is:

a) Hormone
b) Vitamin
c) Enzyme precursor
d) Modulator

(Depart. Questions book)

52- Example of pro-enzyme is:

a) Pepsinogen
b) Trypsin
c) Chymotrypsin
d) Lysine

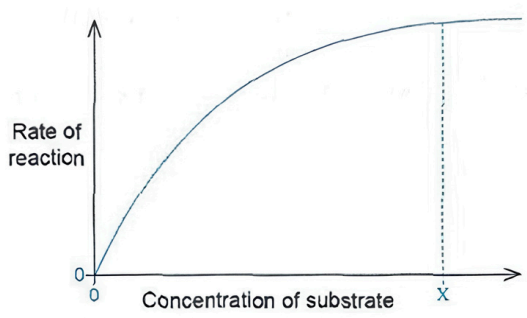
(Depart. Questions book)

53- Phosphorylation/ dephosphorylation of enzymes:

a) Occurs on specific serine and threonine residues
b) It is a non-covalent type of modification
c) It is an irreversible type of modification
d) It doesn't affect the enzyme catalytic activity

(Depart. Questions book)

54- The graph below shows an enzyme-controlled reaction. The enzyme concentration is kept constant. At concentrations of substrate greater than X:



a) The rate of the reaction is limited by enzyme concentration
b) The rate of the reaction tends toward zero
c) The substrate has an inhibitory effect
d) The products have an inhibitory effect

(Depart. Questions book)

55- All the following statements are true with regard to enzymes except:

a) Enzymes lower activation energy.
b) They alter equilibrium of the reaction.
c) They accelerate the chemical reaction.
d) Most of the enzymes are proteins in nature

(Old Depart. Questions book)

56- In competitive enzyme activity inhibition:

a) The structure of inhibitor is similar to that of the substrate
b) Inhibitor decreases apparent Km,
c) Km remains unaffected
d) Inhibitor decreases Vmax without affecting Km

(old Depart. Questions book)

57- Activation or inactivation of certain enzymes is accomplished by covalent modification of the amino acid:

a) Hydroxy lysine
b) Phenylalanine
c) Lysine
d) Serine

(old Depart. Questions book)

58- These enzymes have different structure but the same catalytic function. Frequently they are oligomers made from different polypeptide chains. These en

CANCELLED

a) Allosteric enzymes
b) Isozymes
c) Proenzymes
d) Zymogens

(Old Depart. Questions book)

59- In competitive enzyme activity inhibition, the inhibitor:

a) Increases Vmax
b) Decreases Vmax
c) Does not affect Vmax
d) Decreases Vmax and Km

(Old Depart. Questions book)

60- What type of enzyme inhibitor would have the following effects: Vmax unchanged Km increased Higher [S] needed to reach Vmax/2:

a) Competitive reversible inhibition
b) Allosteric inhibition
c) Irreversible inhibition
d) All of the above

(Old Depart. Questions book)

61- The allosteric inhibitor of an enzyme:

a) Causes the enzyme to work faster
b) Binds to the active site
c) Participates in feedback regulation
d) Denatures the enzyme

(Old Depart. Questions book)

62- When the velocity of enzyme activity is plotted against substrate concentration, which of the following is obtained?

CANCELLED

a) Hyperbolic
b) Parabola
c) Straight line with positive slope
d) Straight line with negative slope

(Old Depart. Questions book)

63- A patient presents with extreme swelling and tenderness in the joints of the extremities. Examination of synovial fluid extracted from the big toe reveals the presence of urate crystals and confirms the diagnosis of gouty arthritis. The drug allopurinol is prescribed to inhibit which of the following enzymes?

a) Adenosine deaminase
b) AMP deaminase
c) Nucleoside phosphorylase
d) Xanthine oxidase

(Old Depart. Questions book)

64- Non steroidal anti inflammatory drugs, such as aspirin act by inhibiting the activity of the enzyme:

a) Lipoxygenase
b) Cyclooxygenase
c) Phospholipase A2
d) Lipoprotein lipase

(Old Depart. Questions book)

65- An inorganic ion required for the activity of an enzyme is known as:

a) Activator
b) Cofactor
c) Coenzyme
d) None of the above

(Old Depart. Questions book)

66- An allosteric enzyme influences the enzyme activity by:

a) Competing for the catalytic site with the substrate
b) Changing the specificity of the enzyme for the substrate
c) Changing the conformation of the enzyme by binding to a site other than catalytic site
d) Changing the nature of the products formed

(Old Depart. Questions book)

67- An enzyme is a:

a) Carbohydrate
b) Lipid
c) Protein
d) Nucleic acid

(Old Depart. Questions book)

68- The best description of enzymes that they are:

a. Ribozyme
b. Zymogen
c. Biocatalysts
d. Isozymes

69- Enzymes, which are produced in inactive form in the living cells, are called:

a. Ribozyme
b. Lysozymes
c. Zymogens
d. Apoenzymes

70- Enzymes, which are RNA in nature are called:

a. Lysozymes
b. Ribozyme
c. Zymogens
d. Apoenzymes

71- Statins decrease plasma cholesterol as it is:

a. Allosteric inhibitor to HMG CoA reductase
b. Competitive inhibitor to HMG CoA reductase
c. Allosteric inhibitor to xanthine oxidase
d. Competitive inhibitor to xanthine oxidase

72- Feedback inhibition is a type of:

a. Competitive inhibition
b. Allosteric inhibition
c. Non-competitive inhibition
d. None of the above

73- In competitive enzyme activity inhibition:

a. Km is decreased
b. Km is increased
c. Vmax is increased
d. Vmax is decreased

74- Which of the following results in increase Km and lower Vmax?

a. Competitive inhibitor
b. Allosteric inhibitor
c. Allosteric activators
d. SH group inhibitors

75- Allopurinol is an inhibitor of:

a. Xanthine oxidase
b. Vitamin K reductase
c. HMG CoA reductase
d. Retinal reductase

76- Warfarin and dicumarol is inhibitor of:

a. Xanthine oxidase
b. Vitamin K reductase
c. HMG CoA reductase
d. Retinal reductase

77- Statins are inhibitors of:

a. Xanthine oxidase
b. Vitamin K reductase
c. HMG CoA reductase
d. Cyclo oxygenase

78- Sulfonamide is bacteriostatic as it prevents formation of:

a. Uric acid
b. Folic acid
c. Lactic acid
d. PABA

79- Aspirin is anti-inflammatory as it inhibits:

a. Xanthine oxidase
b. Vitamin K reductase
c. HMG CoA reductase
d. Cyclo-oxygenase

80- The catalytic activity of an enzyme is restricted to its small portion called:

a. Active site
b. passive site
c. Allosteric site
d. all of the above

81-Common features of enzymes include:

a) They accelerate the reaction and affect its equilibrium point
b) They are mainly carbohydrates in nature
c) They are not changed chemically by the end of the reaction.
d) They are not specific

(Exam 2026)

82- The inhibition of the activity of an enzyme in a pathway by the end products of this pathway is called:

a) Irreversible enzyme inhibition
b) Feedback inhibition
c) Competitive inhibition
d) Covalent modification

(Exam 2026)

83- According to the induced fit model:

a) The active site of the enzyme is not changed
b) The allosteric site of the enzyme is conformationally changed
c) The substrate induces conformational changes in the active site of enzyme
d) The substrate binds to the allosteric site of the enzyme

(Exam 2026)

84- The substrate concentration that produces half the maximal velocity is termed:

a) Zymogen
b) Cofactor
c) Km
d) Co-substrate

(Exam 2026)

85-Long-term regulation of enzyme activity can be achieved through:

a) Presence of enzyme activators
b) Presence of enzyme inhibitors
c) Covalent modification
d) Increased synthesis of the enzyme

(Exam 2026)

86- In enzyme regulation by competitive inhibition:

a) Km increases and Vmax remains unchanged
b) Km decreases and Vmax remains unchanged
c) Km increases and Vmax increases
d) Km decreases and Vmax decreases

(Exam 2026)

87- An example of irreversible enzyme inhibitors is:

a) Dicumarol
b) Allopurinol
c) Carbon monoxide
d) Sulfonamides

(Exam 2026)

88- Allosteric inhibitors affect the enzyme activity by:

a) Irreversibly modifying the enzyme
b) Binding to the substrate and altering its conformation[3]
c) Competing for catalytic site with substrate
d) Binding to a site on the enzyme away from catalytic site

(Exam 2026)

89- Covalent modification of an enzyme usually involves phosphorylation / dephosphorylation of which of the following amino acids?

a) Phenylalanine
b) Proline
c) Hydroxylysine
d) Serine

(Exam 2024)

90- Which of the following is an example of proenzyme?

a) Trypsin
b) Pepsinogen
c) Lysine
d) Glutamate

(Exam 2024)

91- Which one of the following drugs is best prescribed for treatment of gout?

a) Allopurinol
b) Warfarin
c) Dicumarol
d) Sulfonamides

(Exam 2024)

92- Regarding competitive inhibition, which of the following is correct?

a) Km is decreased.
b) Vmax is decreased.
c) The inhibitor has structural similarity to the substrate.
d) Reaction rate is independent of substrate concentration

(Exam 2024)

93- Common features of enzymes include::

a) They accelerate the reaction without affecting its equilibrium
b) They are mainly carbohydrate in nature
c) They are changed chemically by the end of the reaction.
d) They are not specific

(Exam 2024)

94-Regarding the effect of temperature on catalytic activity of enzymes, which of the following is correct?

a) Activity of the enzyme decreases by increasing the temperature
b) Activity of the enzyme virtually stops at 70^0 C due to denaturation
c) The optimum temperature for most animal enzymes is 60^0 C
d) Temperature has no effect on enzyme activity

(Exam 2024)

95- Which of the following is used as anticoagulant?:

a) Allopurinol
b) Statins
c) Dicumarol
d) Sulfonamides

(Exam 2024)

Answers

1-A	2-B	3-A	4-C	5-A	6-B	7-D	8-C	9-D
10-C	11-C	12-D	13-B	14-B	15-D	16-D	17-B	18-B
19-C	20-A	21-D	22-A	23-A	24-(1)C	24-(2)A	24-(3)B	25-D
26-B	CANCELLED	CANCELLED	29-D	30-C	31-A	32-D	33-A	34-B	35-C
36-A	37-A	38-A	39-C	40-C	41-D	42-C	43-B	44-B
45-A	46-C	47-B	48-A	49-C	CANCELLED	51-C	52-A	53-A
54-A	55-B	56-A	57-D	CANCELLED	59-C	60-A	61-C	CANCELLED
63-D	64-B	65-B	66-C	67-C	68-C	69-C	70-B	71-B
72-B	73-B	74-B	75-A	76-B	77-C	78-B	79-D	80-A
81-C	82-B	83-C	84-C	85-D	86-A	87-C	88-D	89-D
90-B	91-A	92-C	93-A	94-B	95-C			

Chapter 8

CHEMISTRY OF FREE NUCLEOTIDES

1- A nucleotide is converted into a nucleoside by the removal of a:

a) Purine base
b) Pyrimidine base
c) Pentose
d) Phosphate

(End of module exam 2021, Final exam 2021)

2- As regards methylated purines, one of the following is incorrect:

a) They include caffeine, theophylline and theobromine
b) They are di[CANCELLED]ervous system stimulators
c) They are methylated xanthine
d) It is contraindicated to take them in diet in case of gout (hyperuricemia)

(End of module exam 2021)

3- 3' phosphoadenosine 5' phosphosulphate (PAPS), acts as:

a) Sulfate donor
b) Methyl donor
c) Hydrogen donor
d) CO_2 donor

(End of module exam 2021, End of the module 2019, Final exam 2019)

4- Concerning methylated purines, which is the false statement?

a) They include caffeine, theophylline, and theobromine
b) They are m[CANCELLED]
c) It is contra[CANCELLED]em in diet in case of gout
d) They are diuretics and CNS stimulants

(Depart. book)

5- As regards cAMP, which is the false statement?

a) Formed from ATP by adenylylcyclase
b) It is converted to 5′ AMP by phosphomonoesterase
c) It activates protein kinase
d) Its level increases by epinephrine

(Depart. book)

6- Adenine is:

a) A purine base
b) A pyrimidine base
c) A nucleoside
d) A nucleotide

(Depart. Questions book)

7- Nucleoside contains:

a) Base-sugar
b) Base-phosphate
c) Base-sugar-phosphate
d) Sugar-phosphate

(Depart. Questions book)

8- The carbon of the pentose in ester linkage with the phosphate in a nucleotide structure is:

a) C1
b) C3
c) C4
d) C5

(Depart. Questions book)

9- The sugar molecule present in nucleotide is:

a) Triose
b) Tetrose
c) Pentose
d) Hexose

(Depart. Questions book)

10- Identify the purine base of nucleic acids in the following:

a) Cytosine
b) Thymine
c) Uracil
d) Adenine

(Depart. Questions book)

11- Which of the following nucleotides contains ribose sugar and not deoxyribose?

a) Thymine – pentose sugar-phosphate
b) Uracil – pentose sugar-phosphate
c) Adenine – pentose sugar-phosphate
d) Cytosine - pentose sugar-phosphate

(Depart. Questions book)

12- Group of adjacent nucleotides are joined by:

a) Phosphodiester bond
b) Peptide bond
c) Ionic bond
d) Covalent bond

(Depart. Questions book)

13- What is the pyrimidine base of nucleic acids in the following?

a) Cytosine
b) Hypoxanthine
c) Guanine
d) Adenine

(Depart. Questions book)

14- Uridine present in RNA is:

a) Nucleotides
b) Pyrimidine
c) Purine
d) Nucleoside

(Depart. Questions book)

15-Which of the following is a purine nucleoside?

a) Cytosine
b) Adenine
c) Adenosine
d) Uracil

(Depart. Questions book)

16- The nitrogenous base is linked to the ribose or deoxyribose sugar via:

a) Glycosidic bond
b) Amide bond
c) Hydrogen bond
d) Phosphoanhydride bond

(Depart. Questions book)

17- Uridylic acid is composed of:

a) Uracil only
b) Uracil and rib
c) Ribose and p
d) Uracil, ribose, and Phosphate

CANCELLED

(Depart. Questions book)

18- ATP is a:

a) Nucleoside
b) Nucleotide
c) Vitamin
d) Nucleic acid

(Depart. Questions book)

19- The chemical combination of ribose and one of the five nitrogen bases results in formation of a:

a) DNA molecule.
b) nucleotide.
c) gene.
d) nucleoside.

(Old Depart. Questions book)

20- One of the following is not a hydrogen carrier:

a) NAD+
b) FAD
c) Coenzyme A
d) NADP+

(Old Depart. Questions book)

21- Coenzyme A acts as:

a) Methyl donor
b) Sulfate donor
c) Acyl group carrier
d) Hydrogen carrier

(Old Depart. Questions book)

22- A nucleoside consists of:

a) Nitrogenous base
b) Purine or pyrimidine base + sugar
c) Purine or pyrimidine base + phosphorous
d) Purine + pyrimidine base + sugar + phosphorous

(Old Depart. Questions book)

23- A purine nucleotide is:

a) AMP
b) UMP
c) CMP
d) TMP

(Old Depart. Questions book)

24- The pyrimidine nucleotide acting as the high energy intermediate is:

a) ATP
b) UTP
c) UDP-Glc
d) CMP

(Old Depart. Questions book)

25- Gout is characterized by increased plasma levels of:

a) Urea
b) Uric acid
c) Creatine
d) Creatinine

(Old Depart. Questions book)

26- Deamination of cytosine gives:

a. Xanthine
b. Uracil
c. Hypoxanthine
d. Thymine

CANCELLED

27- Deamination of Adenine gives:

a. Xanthine
b. Uracil
c. Hypoxanthine
d. Thymine

CANCELLED

28- Deamination of Guanine gives:

a. Xanthine
b. Uracil
c. Hypoxanthine
d. Thymine

CANCELLED

29- The nucleoside of hypoxanthine is:

a. Adenosine
b. Cytosine
c. Guanosine
d. Inosine

30- Free nucleotide acts as a hormone 2nd messenger is:

a. cAMP
b. UDP
c. SAM
d. PAPS

31- Free nucleoside is:

a. cAMP
b. CDP
c. SAM
d. PAPS

32- Free nucleotide acts as a mediator for vasodilators:

a. cAMP
b. cGMP
c. SAM
d. PAPS

33- A nucleotide consists of:

a) Nitrogenous base
b) Nitrogenous base + sugar
c) Nitrogenous base + phosphate
d) Nitrogenous base + sugar + phosphate

(Exam 2026)

34- The molecule that acts as a major methyl donor in biochemical reactions is _____:

a) Adenosine monophosphate (AMP)
b) S-Adenosyl Methionine (SAM)
c) Flavin Adenine Dinucleotide (FAD)
d) Nicotinamide Adenine Dinucleotide (NAD)

(Exam 2026)

35- Uridine present in RNA is a:

a) Nucleotide
b) Pyrimidine nitrogenous base
c) Purine nitrogenous base
d) Nucleoside

(Exam 2026)

36- A nucleoside is converted to a nucleotide by the addition of:

a) Purine base.
b) Pyrimidine base.
c) Pentose.
d) Phosphate

(Exam 2024)

37- Guanosine monophosphate (GMP) is:

a) A purine base
b) A pyrimidine base
c) A nucleoside
d) A nucleotide

(Exam 2024)

Answers

1-D	2-D	3-A	4-C	5-B	6-A	7-A	8-D	9-C
10-D	11-B	12-A	13-A	14-D	15-C	16-A	17-D	18-B
19-D	20-C	21-C	22-B	23-A	24-B	25-B	26-B	27-C
28-A	29-D	30-A	31-C	32-B	33-B	34-C	35-A	36-C
37-B								

Chapter 9 CHEMISTRY OF NUCLEIC ACID

1- Chromatin is composed of:

a) Nucleic acids and proteins
b) Nitrogenous base and sugar
c) Nitrogenous base, sugar, and phosphate
d) Nucleic acids and sugars

End of module exam 2023

2- The two strands in DNA double helix are joined by:

a) Covalent bond
b) Ionic bond
c) Hydrogen bond
d) Phosphodiester bond

End of module exam 2023

3- The order of the nucleotides in any DNA strand is always written in the .....direction (direction of synthesis).

a) 3' to 1'
b) 3' to 5'
c) 1' to 3'
d) 5' to 3'

End of module exam 2023

4- Which of the following types of RNA contains unusual bases e.g., dihydrouracil:

a) mRNA
b) rRNA
c) tRNA
d) sRNA

End of module exam 2023

5- If a DNA molecule content of cytosine is 40%, the thymine content is:

a) 20%
b) 40%
c) 30%
d) 10%

(Final exam 2022 - September)

6- If a DNA molecule contains 20% adenine, what is its cytosine content?

a) 40%
b) 30%
c) 60%
d) 20%

(End of module exam 2021)

7- In the chromatid, histone proteins:

a) Are acidic proteins
b) They are negatively charged
c) They have 4 types
d) They are rich in lysine and arginine

(End of module exam 2021)

8- Regards the secondary structure of DNA, which statement is incorrect?

a) The 2 strands of DNA run antiparallel
b) The 2 strands of DNA are coiled to form a right-handed helix
c) Adenine pairs with thymine through three hydrogen bonds
d) Guanine pairs with cytosine through three hydrogen bonds

(End of module exam 2021)

9- Which of the following types of RNA carries the genetic information from DNA in the nucleus to the ribosomes?

a) mRNA
b) sRNA
c) rRNA
d) tRNA

(End of module exam 2021)

10- As regards transfer RNA (tRNA), one of the following is incorrect:

a) It contains unusual bases
b) It contains the codons
c) It contains 5 arms
d) It carries its specific amino acid to the site of protein synthesis

(End of module exam 2021)

11- Choose the correct statement out of the following:

a) The common B-form of DNA is a left-handed double helix
b) The 2 strands of DNA run parallel to each other
c) The G-C bonds are much more resistant to denaturation than A-T bonds
d) Bonds linking the base pairs in the DNA double helix are ionic bonds

(Final exam 2021)

12- Histones are rich in which of the following amino acids?

a) Lysine and arginine
b) Alanine and glutamine
c) Glycine and Glutamine
d) Arginine and Glutamine

(Final exam 2021)

13-In a DNA molecule, if the cytosine content is 40%, the adenine content will be:

a) 10%
b) 20%
c) 30%
d) 40%

(End of the module 2019)

14-Which of the following statements about tRNA molecules is FALSE?

a) There is at least one tRNA for each of the 20 amino acids.
b) Any given tRNA will normally only accept one specific amino acid.
c) A, C, G, and U are the only bases present in the tRNA molecule.
d) Although composed of a single strand of RNA, each tRNA molecule contains several short, double-helical regions.

(End of the module 2019)

15-Which of the following statements concerning the double helix structure present in DNA molecules is correct?

a) The 2 strands of DNA run parallel to each other.
b) The number of adenine bases equals guanine bases.
c) Base pairing between strands always involves one pyrimidine base and one purine base.
d) Base pairing is always A-G and C-T.

(End of the module 2019)

16- In tRNA molecule D arm is named for the presence of the base:

a) Dimethylxanthine
b) Pseudouracil
c) Dihydrouracil
d) Thymidine

End of the module 2020

17- If a DNA molecule contains 30% thymine, then what is its guanine content?

a) 10%
b) 20%
c) 30%
d) 40%

End of the module 2020

18- In DNA molecule:

a) The number of guanine bases does not equal cytosine bases
b) The number of adenine bases does not equal thymine bases
c) The number of adenine bases equals uracil bases
d) The number of guanine bases equals cytosine bases

Final exam 2019

19- The bonds that link the base pairs in the DNA double helix are:

a) Ionic bonds
b) Peptide bonds
c) Hydrogen bonds
d) Hydrophobic bonds

(Final exam 2019)

20- As regard histones of chromatid:

a) They are in contact with the major groove of DNA
b) They are negatively charged
c) They are basic proteins rich in lysine and arginine
d) There are 4 types of histones

(Depart. book)

21- Concerning DNA, which is the most correct answer:

a) It contains 3-deoxyribose
b) It is either circular or linear
c) It is present in the nucleus only
d) It contains uracil

(Depart. book)

22- In a DNA molecule, If cytosine content is 40%, the thymine content will be:

a) 20%
b) 40%
c) 30%
d) 10%

(Faculty Summer revision 2024)

23- What is the nitrogenous base present in the RNA molecule?

a) Thymine
b) Uracil
c) Xanthine
d) Hypoxanthine

(Depart. Questions book)

24- The sugar moiety present in RNA is:

a) Ribulose
b) Arabinose
c) Ribose
d) Deoxyribose

(Depart. Questions book)

25- DNA rich in G-C pairs have:

a) 1 Hydrogen bond
b) 2 Hydrogen bonds
c) 3 Hydrogen bonds
d) 4 Hydrogen bonds

(Depart. Questions book)

26- The most stabilizing force for the nucleic acids is:

a) Hydrogen bonds
b) Electrostatic bond
c) Van der Waal forces
d) Phosphodiester bonds

CANCELLED

(Depart. Questions book)

27- Purine base found in RNA is:

a) Cytosine
b) Thymine
c) Guanine
d) Uracil

(Depart. Questions book)

28- A DNA segment contains 100 Adenine and 100 cytosines. How many nucleotides are present in the segment?

a) 100
b) 200
c) 400
d) 50

(Depart. Questions book)

29- What are the building blocks of nucleic acids?

a) Nucleotides
b) Nucleosides
c) Amino acids
d) Histones

(Depart. Questions book)

30- On which of the following molecules would you find a codon?

a) Messenger RNA
b) Ribosomal RNA
c) Transfer RNA
d) DNA

(Depart. Questions book)

31- A short length of DNA has 80 thymine and 80 guanine residues. The total no. of nucleotides in DNA fragment is:

a) 40
b) 320
c) 640
d) 160

(Depart. Questions book)

32- DNA strands run ..... in relation to each other.

a) Antiparallel
b) Parallel
c) Perpendicular
d) Horizontal

(Depart. Questions book)

33- A Piece of double stranded DNA has 30% A, what will be the % of G?

a) 30%
b) 40%
c) 20%
d) 70%

(Depart. Questions book)

34- Which ratio is constant for DNA?

a) (A+G)/(T+C)
b) (A+T)/(G+C)
c) (A+C)/(U+G)
d) (A+U)/(G+C)

(Depart. Questions book)

35- During DNA supercoiling in eukaryotic cells, non-histone proteins:

a) Are not require.
b) Required from the beginning to the end of the packing
c) Required only for earlier stage of packing
d) Required for higher level of packing

(Depart. Questions book)

36- Number of hydrogen bonds between adenine and thymine is:

a) 1
b) 2
c) 3
d) 4

(Depart. Questions book)

37- Which of the following types of RNA carries the genetic information from DNA in the nucleus to the ribosomes for protein synthesis?

a) mRNA
b) rRNA
c) tRNA
d) sSRNA

(Old Depart. Questions book)

38- The melting temperature of DNA is increased by its:

a) A and T content
b) G and C content
c) Sugar content
d) Phosphate content

CANCELLED

(Old Depart. Questions book)

39- One of the following proteins is not present within the core nucleosome:

a) H1
b) H2A
c) H2B
d) H3
e) H4

(Old Depart. Questions book)

40- The bonds that link the base pairs in the DNA double helix are:

a) Ionic bonds.
b) Peptide bonds.
c) Hydrogen bonds.
d) Hydrophobic bonds.

(Old Depart. RQuestions book)

41- One of the following nucleotides is not found in DNA:

a) dUMP
b) dGMP
c) dAMP
d) dTMP

(Old Depart. Questions book)

42- As regards tRNA, all of the following statements are correct EXCEPT:

a) Forms 15% of cellular RNA.
b) It contains unusual bases.
c) Has 5 loops and an acceptor arm.
d) It has a cloverleaf appearance.

(Old Depart. Questions book)

43- The "backbone" of a nucleic acid molecule consists of an alternating sequence of:

a) Sugar and phosphate groups.
b) Sugar and base groups.
c) Phosphate and base groups.
d) Sugar, phosphate, and base groups

(Old Depart. Questions book)

44- The codon is found on ..... and the anticodon is found on ..... :

a) mRNA, tRNA
b) tRNA, mRNA
c) rRNA, tRNA
d) mRNA, rRNA

(Old Depart. Questions book)

45- The arm of tRNA which contains the unusual base dihydrouracil is the:

a) Anticodon arm.
b) D-arm
c) Extra arm.
d) T-C arm

(Old Depart. Questions book)

46- The nucleotides in the backbone of DNA are held together by bonds.

a) hydrogen
b) peptide
c) phosphodiester
d) glycosidic

(Old Depart. Questions book)

47- Which of the following is NOT a characteristic of transfer RNA?

a) It has a cloverleaf appearance.
b) It contains many unusual bases.
c) A given transfer RNA can carry any of the 20 amino acids.
d) It contains an anticodon.

(Old Depart. Questions book)

48- In RNA molecule guanine bases does not necessarily equal its cytosine bases also, its adenine bases does not necessarily equal its uracil bases since it is a:

a) Single stranded molecule.
b) Double stranded molecule.
c) Double stranded helical molecule
d) Polymer of purine ribonucleotides.

(Old Depart. Questions book)

49- One of the following concerning mitochondrial DNA is incorrect:

a) Mitochondrial DNA is present in the form of a small single stranded circular supercoil.
b) Mitochondrial DNA is maternally inherited.
c) Mutations of mitochondrial DNA are associated with certain types of myopathies.
d) Mitochondrial DNA forms 0.3 to 1% of the total cellular DNA.

(Old Depart. Questions book)

50- The melting temperature of DNA is the temperature at which:

a) Solid DNA becomes liquid
b) DNA is hydrolysed into nucleotides
c) DNA changes from relaxed into supercoiled DNA
d) Half of the helical structure of DNA is lost.

CANCELLED

(Old Depart. Questions book)

51- The nucleic acid base found in mRNA but not in DNA is:

a) Adenine
b) Cytosine
c) Guanine
d) Uracil

(Old Depart. Questions book)

52- Although each specific tRNA differs from the others in its sequence of nucleotides, all tRNA molecules contain a base paired stem that terminates in the sequence CCA at:

a) 3′ Termini
b) 5′ Termini
c) Anticodon arm
d) 5′-3′ -Termini

(Old Depart. Questions book)

53- In a DNA molecule, if the guanosine content is 40%, the adenine content will be:

a) 10%
b) 20%
c) 30%
d) 40%

(Old Depart. Questions book)

54- Left handed double helix is present in:

a) Z-DNA
b) A-DNA
c) B-DNA
d) None of these

CANCELLED

(Old Depart. Questions book)

55- The structural stability of the double helix of DNA is mainly due to:

a) Hydrogen bonding between adjacent purine bases.
b) Hydrophobic bonding between staked purine and pyrimidine nuclei.
c) Hydrogen bonding between adjacent pyrimidine bases.
d) Hydrogen bonding between purine and pyrimidine bases.

(Old Depart. Questions book)

56- The mitochondrial DNA is:

a) Like the nuclear DNA in structure.
b) Single stranded, linear.
c) Double stranded, circular.
d) Single stranded, circular.

(Old Depart. Questions book)

57- In a certain DNA molecule Adenine is 23%, percent of Guanine is:

a. 27%
b. 33%
c. 23%
d. 37%

58- In a certain DNA molecule Thymine is 13%, percent of Cytosine is:

a. 27%
b. 33%
c. 23%
d. 37%

59- Simple unit of chromatin is:

a. Nucleotide
b. Nucleosome
c. Amino acids
d. Monosaccharide

60- Thymine is present in:

a. mRNA
b. tRNA
c. rRNA
d. snRNA

61- Contains 3 nucleotides and binds to codon on mRNA:

a. Anti-codon
b. D arm
c. Thymine pseudouridine arm
d. Acceptor arm

62- Important for mRNA processing:

a. mRNA
b. tRNA
c. rRNA
d. snRNA

63- Histone proteins are rich in:

a) Glycine and Serine
b) Lysine and Arginine
c) Glutamic acid and tyrosine
d) Aspartic acid and glutamic acid

(Exam 2026)

64- In DNA double helix, the two strands are held together by:

a) Phosphodiester bonds
b) Hydrogen bonds
c) Peptide bonds
d) Ionic bonds

(Exam 2026)

65- The nucleic acid that contains codons is _____ .:

a) Messenger RNA
b) Ribosomal RNA
c) Transfer RNA
d) DNA

(Exam 2026)

66- Number of hydrogen bonds between guanine and cytosine is:

a) One
b) Two
c) Three
d) Four

(Exam 2024)

67- The anticodon is present in:

a) tRNA
b) mRNA
c) rRNA
d) DNA

(Exam 2024)

68- Histone proteins are rich in:

a) Glycine and Serine
b) Lysine and Arginine
c) Glutamic acid and tyrosine
d) Aspartic acid and glutamic acid

(Exam 2024)

Answers

1-A	2-C	3-D	4-C	5-D	6-B	7-D	8-C	9-A
10-B	11-C	12-A	13-A	14-C	15-C	16-C	17-B	18-D
19-C	20-C	21-B	22-D	23-B	24-C	25-C	CANCELLED	27-C
28-C	29-A	30-A	31-B	32-A	33-C	34-A	35-D	36-B
37-A	CANCELLED	39-A	40-C	41-A	42-C	43-A	44-A	45-B
46-C	47-C	48-A	49-A	CANCELLED	51-D	52-A	53-A	CANCELLED
55-D	56-C	57-A	58-D	59-B	60-B	61-A	62-D	63-B
64-B	65-A	66-C	67-A	68-B				

Chapter 10

DNA SYNTHESIS (REPLICATION)

1- DNA replication is:

a) Semi-conservative
b) Discontinuous
c) Unidirectional
d) Conservative

End of module exam 2023

2- After formation of replication fork:

a) Both new strands are synthesized discontinuously
b) One strand is synthesized continuously and the other discontinuously
c) Both new strands are synthesized continuously
d) RNA primer is required only for the synthesis of one new strand

(End of module exam 2023, End of the module 2020)

3- Since DNA polymerases cannot start from scratch by adding nucleotides to a free single stranded DNA template during replication, ... is required.

a) A DNA primer
b) A helicase
c) A ligase
d) An RNA primer

End of module exam 2023

4- Which of the following enzymes separates the two strands of DNA during replication?

a) Gyrase
b) Topoisomerase
c) Helicase
d) DNA polymerase

End of module exam 2023

5- Which of the following describes enzyme telomerase?

a) It is a reverse transcriptase enzyme.
b) It consists of an RNA sequence that serves as a template
c) After completion of each cycle, telomerase moves to 3' end of DNA to synthesize repetitive
d) All the above

End of module exam 2023

6- During eukaryotic DNA replication, Topoisomerase enzymes:

a) Catalyze the synthesis of an RNA primer on the lagging strand.
b) Remove the incorrectly base-paired nucleotides through a 3'-5' exonuclease activity.
c) Stabilize single-stranded DNA in the region.
d) Cut and reseal DNA in advance of the replication fork to remove supercoiling.

End of module exam 2023

7- The fragments of DNA are joined together by which of the following enzymes?

a) Endonuclease
b) DNA polymerase
c) Primase
d) DNA ligase

End of module exam 2023

8- Stability of DNA double helix is not maintained by one of the following:

a) Hydrogen bonding between complementary base pairs
b) Spiral staircase
c) Hydrophobic interactions
d) Van der Waals forces

(Final exam 2022 - September, Faculty summer revision 2024)

9- One of the following is not involved in prokaryotic replication:

a) DNA polymerase III
b) DNA polymerase I
c) Primase
d) Telomerase

CANCELLED

(End of module exam 2021, Final exam 2019)

10- As regards eukaryotic DNA polymerases, which of the following is correct?

a) Polymerase γ synthesizes mitochondrial DNA
b) Polymerase δ synthesizes the leading strand
c) Polymerase ϵ excises RNA primer and fills the gaps
d) Polymerase β synthesizes the lagging strand

(End of module exam 2021)

11- Which enzymes remove supercoiling ahead of the replication fork?

a) Helicases
b) DNA polymerases
c) Primases
d) Topoisomerases

(End of module exam 2021)

12- Okazaki fragments are formed during the synthesis of:

a) mRNA
b) tRNA
c) RNA and DNA
d) DNA

(End of module exam 2021)

13- Formation of the replication fork does not require:

a) Single strand binding (SSB) proteins
b) dnaB and dnaC
c) Primase
d) Helicase

(End of module exam 2021)

14- Which of the following statements about DNA replication is not correct?

a) Unwinding of the DNA molecule occurs as hydrogen bond break.
b) The process is known as semiconservative DNA replication because one original strand is conserved in the new DNA molecule.
c) Complementary base pairs are held together with hydrogen bonds.
d) Replication occurs as each base is paired with another base exactly like it.

(End of module exam 2021)

15- Which statement about the elongation of the leading strand during DNA replication is correct?

a) It is synthesized in 3' to 5' direction
b) It is read in 5' to 3' direction
c) It requires a short RNA primer
d) It is synthesized discontinuously

(End of module exam 2021)

16- One of the following is not correct about telomerase:

a) Is present only in eukaryotic cells
b) Contains a single stranded RNA
c) Acts as reverse transcriptase
d) Elongates the leading strand

CANCELLED

(End of module exam 2021)

17- Telomerase is NOT characterized by:

a) Telomerase is a special kind of reverse transcriptase
b) Is responsible for elongation of 3' end of telomeres of DNA strands
c) It has an internal RNA strand about 150 nucleotide long
d) Telomerase expression is inactivated in tumor cells

CANCELLED

(Final exam 2021)

18- Regarding DNA polymerases in eukaryotes, one of the following is NOT correct:

a) Polymerase α forms a complex with primase that catalyzes synthesis of RNA primers.
b) Polymerase β is concerned with DNA repair.
c) Polymerase γ is concerned with replication of mitochondrial DNA synthesis.
d) Polymerase ϵ catalyzes removal of RNA primer

(End of the module 2019)

19- Which enzymes remove positive supercoiling in replicating DNA ahead of the replication fork?

a) Helicases
b) Topoisomerases.
c) Primases
d) DNA polymerases

(End of the module 2019)

20- The following enzyme is responsible for removal of RNA primers during prokaryotic replication:

a) DNA polymerase I
b) RNA primase
c) DNA polymerase III
d) RNAse H

(End of the module 2019)

21- One of the following is NOT a characteristic of telomerase:

a) Is present in prokaryotic and eukaryotic cells.
b) Contains a single-stranded RNA.
c) Acts as reverse transcriptase.
d) Elongates the lagging strand.

(End of the module 2019)

22- Regarding DNA polymerases in eukaryotes, all of the following are true except:

a) Polymerase α forms a complex with primase that catalyzes synthesis of RNA primers.
b) Polymerase β is concerned with DNA repair.
c) Polymerase γ is concerned with replication of mitochondrial DNA synthesis.
d) Polymerase ϵ catalyzes removal of RNA primers.

(End of the module 2020)

23- Which enzymes remove supercoiling in replicating DNA ahead of the replication fork?

a) Helicases
b) DNA polymerases
c) Topoisomerases
d) Primases

(End of the module 2020, Final exam 2019 – September sitting)

24- Okazaki fragments are used to elongate:

a) the leading strand toward the replication fork
b) the lagging strand toward the replication fork
c) the leading strand in the opposite direction of the advancing replication fork
d) the lagging strand in the opposite direction of the advancing replication fork

(End of the module 2020)

25- Okasaki fragments are formed during the synthesis of:

a) mRNA
b) tRNA
c) RNA and DNA
d) DNA

(Final exam 2019)

26- Prokaryotic RNA primer is formed by:

a) DNA polymerase α.
b) DNA polymerase III.
c) Primase.
d) Telomerase.

(Final exam 2019 – September sitting)

27- After formation of the replication fork:

a) Both strands are synthesized discontinuously
b) One strand is synthesized continuously and the other discontinuously
c) Both strands are synthesized continuously
d) RNA primer is required only for the synthesis of one new strand.

(Final exam 2019 – September sitting)

28- Sigma and Rho factors are required for prokaryotic:

a) Replication
b) Transcription
c) Translation
d) Polymerisation

(Final exam 2019 – September sitting)

29- During eukaryotic DNA replication, topoisomerases:

a) Catalyze the synthesis of an RNA primer on the lagging strand
b) Remove the incorrectly base-paired nucleotides through a 3′-5′ exonuclease activity
c) Cut and reseal DNA in advance of the replication fork to eliminate supercoiling
d) Add nucleotides to the growing strand in the 5′-3′direction

(Depart. book)

30- One of the following enzymes is not required in DNA repair:

a) An endonuclease
b) An exonuclease
c) Topoisomerase
d) DNA polymerase β

(Depart. book)

31- Deoxyribonucleotides are added to the RNA primer by:

a) DNA polymerase I
b) DNA polymerase II
c) DNA polymerase III
d) DNA polymerase β

(Old Depart. book)

CANCELLED

32- Negative supercoils are introduced in DNA by:

a) Helicase
b) DNA ligase
c) DNA gyrase
d) DNA polymerase III

CANCELLED

(Old Depart. book)

33- Telomerase is characterized by all of the following except:

a) It elongates the lagging strand
b) It is a special type of RNA-dependent DNA polymerase
c) It has internal RNA strand 150 nucleotides long
d) Germ-line cells and cancer cells do NOT contain telomerase

CANCELLED

(Faculty Summer revision 2024)

34- One of the following statements is correct:

a) Telomerase expression is inactivated in tumor cells
b) High telomerase activity leads to cellular aging
c) Germ-line cells do not express telomerase
d) Differentiated cells do not express telomerase

CANCELLED

(Faculty Summer revision 2024)

35- In DNA replication, one new strand is synthesized complementary to an old (parent) strand and is referred as:

a) Semiconservative
b) Discontinuous
c) Conservative
d) DNA ligases build the new DNA strand

(Depart. Questions book)

36- DNA replication takes place in which direction?

a) 3' to 5'
b) 5' to 3'
c) Randomly
d) Vary from organism to organism

(Depart. Questions book)

37- Which of the following enzymes unwinds the DNA double helix during replication?

a) Helicase
b) Replicase
c) Polymerase
d) Ligase

(Depart. Questions book)

38- Which enzyme catalyzes the elongation of DNA molecules?

a) DNA polymerase
b) Helicase
c) DNA ligase
d) Primase

(Depart. Questions book)

39- Formation of RNA primer:

a) Precedes replication
b) Follows replication
c) Precedes transcription
d) Follows transcription

(Depart. Questions book)

40- Given that DNA polymerases are unable to synthesize DNA without a primer, what serves as the primer for these enzymes during DNA replication?

a) Five carbon sugars
b) Deoxyribose alone
c) A short RNA molecule
d) Proteins with free hydroxyl groups

(Depart. Questions book)

41- A single-stranded DNA molecule contains 20%A, 25%T, 30%G, and 25%C. When the complement of this strand is synthesized, the T content of the resulting strand will be:

a) 20%
b) 22.5%
c) 25%
d) 30%

(Depart. Questions book)

42- The process of DNA replication involves:

a) Multiple origins of replication per chromosome in eukaryotes
b) Binding of ribosomes to origins of replication
c) Continuous synthesis on both strands of the double helix
d) Conservative replication

(Depart. Questions book)

43- During eukaryotic DNA replication, single stranded binding proteins (SSBP):

a) Catalyze the synthesis of an RNA primer on the lagging strand
b) Remove the incorrectly base-paired nucleotides through a 3'-5'exonuclease activity
c) Stabilize single-stranded DNA in the region
d) Cut and reseal DNA in advance of the replication fork to eliminate supercoiling

(Depart. Questions book)

44- The short strand of ..... primer is required for the replication of DNA:

a) RNA
b) Histone
c) Protein
d) Lipid

(Depart. Questions book)

45- In eukaryotes, which of the following DNA polymerases is required for mitochondrial DNA replication?

a) DNA polymerase α
b) DNA polymerase β
c) DNA polymerase δ
d) DNA polymerase γ

(Depart. Questions book)

46- The DNA polymerase involved in synthesis of the leading strand in eukaryotes is:

a) DNA polymerase α
b) DNA polymerase δ
c) DNA polymerase ϵ
d) DNA polymerase σ

(Depart. Questions book)

47- Which statement about the elongation of the lagging strand during DNA replication is correct?

a) It is synthesized in a 3' — 5' direction
b) It progresses (grows) toward the replication fork
c) It requires a short RNA primer to proceed
d) It is synthesized continuously

(Depart. Questions book)

48- The following enzyme is responsible for removal of RNA primers during eukaryotic replication:

a) DNA polymerase α
b) RNA primase
c) DNA polymerase σ
d) RNAse H

(Depart. Questions book)

49- The gaps between Okazaki fragments are eventually sealed by:

a) Gyrase
b) DNA ligase
c) Polymerase
d) Helicase

(Depart. Questions book)

50- What is the role of topoisomerases in eukaryotic DNA replication?

a) Cut, uncoil, and reseal the double stranded DNA
b) Bind to the origin of replication sites within double stranded DNA
c) Open up the double stranded DNA at the replication fork
d) Join the Okazaki fragments together with phosphodiester bonds

(Depart. Questions book)

51- What is the function of ligase?

a) Works on the leading strand
b) Undoes the work of DNA polymerase
c) Joins the Okazaki fragments
d) Winds the leading and lagging strands together

(Depart. Questions book)

52- What is the name of the unusual repeated stretch of DNA localized at the tips of all eukaryotic chromosomes?

a) Kinetochore
b) Telomere
c) Centriole
d) Chromomere

(Depart. Questions book)

53- Reverse transcriptase activity is present in the eukaryotic:

a) DNA polymerase α
b) DNA polymerase γ
c) Telomerase
d) DNA polymerase ϵ

(Depart. Questions book)

54- DNA polymerase activity does not require one of the following:

a) A template DNA.
b) A primer with a free 5'-hydroxyl group.
c) dATP, dCTP, dGTP, and dTTP.
d) A primer with a free 3'-hydroxyl group.

(Old Depart. Questions book)

55- The direction of DNA synthesis is:

a) 5' --> 3''
b) 3' --> 5''
c) Randomly
d) Vary from organism to organism

(Old Depart. Questions book)

56- As regard eukaryotic DNA polymerases:

a) Polymerase β synthesize mitochondrial DNA.
b) Polymerase δ synthesize leading strand.
c) Polymerase ε excise RNA primer and fill the gaps.
d) Polymerase α synthesize RNA primer.

(Old Depart. Questions book)

57- In prokaryotes, DNA polymerase III is responsible for:

a) Leading DNA strand synthesis.
b) Lagging DNA strand synthesis.
c) DNA repair.
d) a and b.

(Old Depart. Questions book)

58- One of the following is not required in DNA repair:

a) An endonuclease.
b) An exonuclease.
c) Topoisomerase.
d) Repair DNA polymerase.

(Old Depart. Questions book)

59- Eukaryotes differ from prokaryote in mechanism of DNA replication in:

a) Use of DNA primer rather than RNA primer
b) Different enzyme for synthesis of lagging and leading strand
c) Discontinuous rather than semi-discontinuous replication
d) Unidirectional rather than semi-discontinuous replication

(Old Depart. Questions book)

60- The enzyme used to join nicks of DNA is:

a) DNA polymerase
b) DNA ligase
c) Endonuclease
d) Primase

(Old Depart. Questions book)

61- Okazaki fragments occur on the and are bonded together by

a) Leading strand, polymerase
b) mRNA, anticodons
c) Lagging strand, ligase
d) tRNA, polymerase

(Old Depart. Questions book)

62- Given that DNA polymerases are unable to synthesize DNA without a primer, what molecule serves as the primer for these enzymes during DNA replication?

a) Five carbon sugars
b) Deoxyribose alone
c) A short RNA molecule
d) Proteins with free hydroxyl groups

(Old Depart. Questions book)

63- Which of the following enzymes responsible for removal of RNA primers and filling of gaps by DNA in prokaryotes?

a. DNAP I
b. DNAP II
c. DNAP III
d. Primase

64-The lagging strand differs from the leading strand in:

a. Requires many RNA primers
b. Synthesized continuously
c. Synthesized towards the replication fork
d. None of the above

65- Which of the following enzymes can be described as DNA-dependent RNA polymerase?

a. DNA ligase
b. Primase
c. DNA polymerse III
d. Reverse transcriptase

CANCELLED

66- The enzyme responsible for unwinding the DNA double helix during replication is:

a) Helicase
b) Ligase
c) Primase
d) Topoisomerase

(Exam 2026)

67- During DNA replication, the supercoiling formed ahead of the replication fork is removed by:

a) Telomerase
b) Ligase
c) Topoisomerase
d) Helicase

(Exam 2026)

68- The direction of DNA synthesis is:

a) 5' → 3'
b) 3' → 5'
c) Randomly
d) Vary from organism to organism

(Exam 2026)

69- The enzyme that fills the gap left after the removal of the RNA primer during DNA replication is:

a) DNA polymerase ε
b) DNA polymerase δ
c) DNA polymerase γ
d) DNA polymerase α-primase

(Exam 2026)

70- Inhibition of DNA polymerase β in eukaryotic cells directly affects:

a) Primer formation
b) Lagging strand synthesis
c) DNA repair
d) RNA removal

(Exam 2026)

71- The enzyme that joins the ends of Okazaki fragments is:

a) DNA Helicase
b) DNA Ligase
c) Topoisomerase
d) RNase H

(Exam 2026)

72- Which of the following is a feature of DNA replication?:

a) Semi-conservative
b) discontinuous
c) Mutational
d) Conservative

(Exam 2024)

73- Which of the following enzymes unwinds the DNA double helix during replication?

a) Helicase
b) Replicase
c) Polymerase
d) Ligase

(Exam 2024)

74- After formation of replication fork

a) Both of the new strands are synthesized discontinuously
b) One strand is synthesized continuously and the other discontinuously
c) Both the new strands are synthesized continuously
d) RNA primer is required only for the synthesis of one new strand

(Exam 2024)

75- Okazaki fragments are formed during the synthesis of:

a) mRNA
b) tRNA
c) rRNA
d) DNA

(Exam 2024)

76-In eukaryotes, which of the following DNA polymerases is responsible for DNA repair?

a) DNA polymerase α
b) DNA polymerase β
c) DNA polymerase σ
d) DNA polymerase γ

(Exam 2024)

77- Which of the following is correct regarding telomerase?

a) Telomerase is a reverse transcriptase enzyme
b) Telomerase consists of a DNA sequence that serves as a template for synthesis of RNA
c) Telomerase is a DNA polymerase enzyme
d) Telomerase causes chromosomal shortening

(Exam 2024)

Answers

1-A	2-B	3-D	4-C	5-B	6-D	7-D	8-B	9-D
10-A	11--D	12-D	13-C	14-D	15-C	16-D	17-D	18-D
19-B	20-A	21-A	22-D	23-C	24-D	25-D	26-C	27-B
28-B	29-C	30-C	31-D	32-C	33-D	34-D	35-A	36-B
37-A	38-A	39-A	40-C	41-A	42-A	43-C	44-A	45-D	46-B
47-C	48-D	49-B	50-A	51-C	52-B	53-C	54-B	55-A
56-D	57-D	58-C	59-B	60-B	61-C	62-C	63-A	64-A	65-B
66-A	67-C	68-A	69-B	70-C	71-B	72-A	73-A	74-B	75-D
76-B	77-D							

Chapter 11

RNA SYNTHESIS (TRANSCRIPTION)

1- Apo B 100 gene is transcribed into mRNA that directs the synthesis of apoB-48 protein in the intestines, this results from.....

a) Addition of poly (A) tail to mRNA
b) Capping at the 5' end of mRNA
c) mRNA editing
d) mRNA splicing

(End of module exam 2023)

2- Which of the following best describes transcription?

a) RNA synthesis is always in the 5' – 3' direction.
b) RNA polymerase needs a primer to initiate transcription.
c) The sequence of the mRNA is complementary to the sequence of coding strand.
d) The RNA polymerase utilizes ribonucleoside triphosphates as ATP, GTP, CTP and TTP.

(End of module exam 2023)

3- The process of .....removes introns and joins exons from the transcript primary to form functional mRNA.

a) RNA translocation
b) RNA elongation
c) RNA splicing
d) RNA release

(End of module exam 2023)

4- Which one of the following is not required in transcription?

a) RNA polymerase
b) tRNA
c) DNA
d) A promoter

(End of module exam 2023)

5- What is a promoter?

a) A specific sequence of DNA to which DNA polymerase binds.
b) A specific sequence of DNA to which RNA polymerase binds.
c) A specific DNA sequence to which a restriction endonuclease binds.
d) A specific RNA sequence to which RNA polymerase binds.

(End of module exam 2023)

6- In eukaryotes, synthesis of mRNA is catalyzed by:

a) RNA polymerase I
b) RNA polymerase II
c) RNA polymerase III
d) DNA polymerase I

CANCELLED

(End of module exam 2023)

7- Coding information of mRNA can be changed by:

a) Capping
b) Polyadenylation
c) General transcription factors
d) mRNA editing

(Final exam 2022 – September, Faculty summer revision 2024)

8- The half-life of mRNA is determined by:

a) Basal expression elements
b) Methyl-guanosine cap
c) The length of poly(A) tail
d) AAUAAA sequence

(Final exam 2022 – September, Faculty summer revision 2024)

9- What enables RNA polymerase to recognize the promoter regions on the DNA?

a) DNA binding proteins
b) β subunit of RNA polymerase
c) Sigma (σ) Factor of RNA polymerase
d) Rho (ρ) factor

CANCELLED

(End of module exam 2021)

10- Processing of mRNA does not include:

a) Capping
b) Polyadenylation
c) Splicing
d) Trimming

(End of module exam 2021, Final exam 2019)

11- The termination signal for prokaryotic transcription is recognized by:

a) α- subunit of RNA polymerase
b) β- subunit of RNA polymerase
c) Sigma factor
d) Rho (ρ) factor

CANCELLED

(End of module exam 2021)

12- Mature mammalian mRNA does not contain:

a) A 5′ cap structure
b) A 3′ poly A tail structure
c) An intron
d) An exon

(End of module exam 2021)

13- Regarding RNA synthesis (transcription) in prokaryotes:

a) The mRNA produced is identical to the parent DNA
b) RNA polymerase needs a primer to initiate transcription
c) RNA polymerase catalyzes the synthesis of all types of RNAs
d) The RNA polymerase utilizes ATP, GTP, CTP and TTP

CANCELLED

(End of module exam 2021)

14- The half life time of mRNA is determined by:

a) The length of the poly (A) tail
b) Capping at the 5′ end
c) The number of introns
d) The number of codons

(End of module exam 2021)

15- Prokaryotic promoter is characterized by presence of:

a) TATA box
b) CAAT box
c) -35 sequence (5-TTGACA-3")
d) a and c

CANCELLED

(End of module exam 2021)

16- Which of the following statements is incorrect?

a) A spliceosome removes the introns from heterogeneous nuclear RNA (hnRNA)
b) DNA polymerase synthesizes a molecule of DNA from a DNA template
c) RNA polymerase synthesizes a molecule of RNA from an RNA template
d) DNA ligase joins segments of DNA

(End of module exam 2021)

17- Which of the following is true about transcription?

a) RNA polymerase needs a primer to initiate transcription
b) RNA polymerase binds to the promoter region
c) The messenger RNA produced is identical to the parent strand.
d) Both strands of the DNA are copied

(Final exam 2021)

18- The part of the prokaryotic RNA polymerase responsible for recognizing the promoter is the:

a) Alpha (α) subunit
b) Rho (ρ) protein
c) DNA Pol III
d) Sigma (σ) subunit

CANCELLED

(End of the module 2019)

19- The part of the prokaryotic RNA polymerase responsible for recognizing the promoter on DNA is:

a) Alpha subunit
b) Rho protein
c) Beta subunit
d) Sigma subunit

CANCELLED

(End of the module 2020)

20- In transcription:

a) The mRNA produced is identical to the parent DNA
b) A double helix containing one parent strand and one daughter strand is produced.
c) Both strands of the DNA are copied.
d) The mRNA produced is complementary to one strand of the DNA

(End of the module 2020, End of the module 2019)

21- The role of the sigma factor in bacterial RNA polymerase is:

a) to catalyze RNA synthesis.
b) to position RNA polymerase correctly on the specific promoter regions on the DNA.
c) to terminate RNA synthesis.
d) to unwind the DNA template

CANCELLED

(Final exam 2019)

22- RNA polymerases used for the transcription of genes require a _____ template.

a) rRNA
b) DNA
c) RNA
d) mRNA

(Depart. book)

23- RNA polymerase II does NOT transcribe:

a) rRNAs
b) mRNA
c) snRNAs
d) miRNAs

CANCELLED

(Depart. book)

24- The base sequence of the strand of DNA used as the template for transcription has the base sequence GATCTAC. What is the base sequence of the RNA product?

a) CTAGATG
b) GTAGATC
c) GAUCUAC
d) CUAGAUG

(Depart. book)

25- One of the following is not a function of mRNA untranslated regions:

a) mRNA stability
b) Translational efficiency
c) Activation of amino acids
d) Regulation of gene expression

(End of module exam 2023)

26- mRNA is complementary to the nucleotide sequence of:

a) Coding strand
b) Ribosomal RNA
c) tRNA
d) Template strand

(Depart. Questions book)

27- The base sequence of the strand of DNA used as the template for transcription is CTAGATG. What is the base sequence of the RNA product? (All sequences are written according to standard convention 5'-3')

a) GITAGATC
b) GAUCUAC
c) GUAGAUC
d) CTAGAUG

(Depart. Questions book)

28- Which sequence of RNA bases would pair with this partial DNA strand ATG TGA CAG?

a) ATG TGA CAG
b) UAC ACU GUC
c) TUC UCT GTC
d) TAC ACT GTC

(Depart. Questions book)

29- The base sequence of codons number 6 and 7 in the coding strand in a specific gene is CAGCGC. The mRNA produced upon transcription of this gene will contain which sequence?

a) GCGCTC
b) CAGCGC
c) CUGCGC
d) GUCGCG

(Depart. Questions book)

30- Which of the following best describes the key function of helicases during transcription?

a) Relief of tension in the DNA strands
b) Separation of two strands of DNA
c) Synthesis of the RNA primers
d) Re-annealing of two DNA strands after transcription

(Depart. Questions book)

31- What is the role of the promoter region in the regulation of gene expression?

a) Bind negative regulators to inhibit gene expression
b) Recruit DNA polymerase and transcription factors to DNA
c) Allow for alternative splicing and recombination of genetic components
d) Alter gene expression by binding transcription factors

(Depart. Questions book)

32- The region of DNA known as TATA BOX is the site for binding of:

a) DNA polymerase
b) Topoisomerase
c) RNA polymerase
d) DNA ligase

(Depart. Questions book)

33- The large eukaryotic rRNA genes, such as 18S and 28S RNA-encoding genes, are transcribed by which of the following RNA polymerases?

CANCELLED

a) RNA polymerase III
b) RNA-dependent RNA polymerase
c) RNA polymerase I
d) RNA polymerase II

(Depart. Questions book)

34- In eukaryotes, the TATA sequence functions as:

a) An indication of the starting point for replication
b) An indication of the starting point for transcription
c) A termination signal for transcription
d) A termination signal for translation

(Depart. Questions book)

35- Processing of mRNA lacks:

a) Capping
b) Polyadenylation
c) Splicing
d) Trimming

(Depart. Questions book)

36- The bond connecting the initiation nucleotide of the mRNA with the 5′-G Cap structure is a:

CANCELLED

a) 3'-5' phosphodiester bridge
b) 5'-5' triphosphate bridge
c) 3'-3' triphosphate bridge
d) 3'-5' triphosphate bridge

(Depart. Questions book)

37- What sequence feature is thought to protect mRNAs from degradation?

a) Telomeres
b) Poly (A) tail
c) Introns
d) Exons

(Depart. Questions book)

38- mRNA splicing involves:

a) Cutting of exons and joining of introns
b) The action of RNA polymerase
c) The action of DNA ligase
d) Cutting of introns and joining of exons

(Depart. Questions book)

39- A gene encodes a protein with 150 amino acids. There is one intron of 1000bp, a 5'-untranslated region of 100bp, and a 3'-untranslated region of 200bp. In the final processed mRNA, how many bases lie between the start AUG codon and the final termination codon?

a) 450
b) 550
c) 650
d) 1750

(Depart. Questions book)

40- What is the macromolecular complex that associates with introns during mRNA splicing?

a) A splicer
b) Ribosomes
c) Nuclear body
d) Spliceosomes

(Depart. Questions book)

41- A 32-year-old female patient complains of skin rash, and painful swelling of joints. She was diagnosed with systemic lupus erythematosus (SLE) with autoantibodies against small nuclear ribonucleoproteins (snRNPs). This is caused by defective:

a) Capping
b) Polyadenylation
c) Splicing
d) Alternative splicing

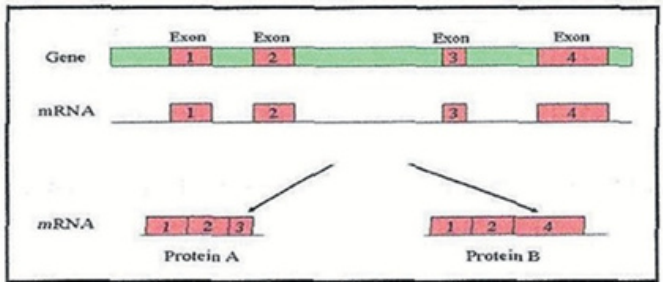
(Depart. Questions book)

42- A two-year-old girl was diagnosed with β-thalassemia. Genetic analysis revealed a nucleotide change at an intron-exon junction. this may result from defective:

a) Capping
b) Polyadenylation
c) Splicing
d) Alternative splicing

(Depart. Questions book)

43- Which best describes the process in the following figure?



a) Capping
b) Polyadenylation
c) Splicing
d) Alternative splicing

(Depart. Questions book)

44- Which of the following is true of RNA synthesis (transcription)?

a) RNA synthesis is always in the 5' – 3' direction.
b) RNA polymerase needs a primer to initiate transcription.
c) The sequence of the mRNA is complementary to the sequence of coding strand
d) The RNA polymerase utilizes ribonucleoside triphosphates (ATP, GTP, CTP and TTP)

(Old Depart. Questions book)

45- In eukaryotic promoters, which of the following describes the 'Pribnow box'?

a) The 5' untranslated region
b) The -10 box
c) The -35 box
d) The termination sequence

(Old Depart. Questions book)

46- Prokaryotic RNA polymerase doesn't synthesize:

a) tRNA.
b) mRNA.
c) rRNA.
d) RNA primer.

CANCELLED

(Old Depart. Questions book)

47- Mature mammalian mRNA would be LEAST likely to contain:

a) a 5′ cap structure.
b) a 3′ poly A tail structure.
c) an exon.
d) an intron.

(Old Depart. Questions book)

48- Which of the following is true about RNA synthesis:

a) Synthesis of RNA is always in the 5' to 3' direction
b) RNA polymerase requires a primer for initiating transcription
c) U is inserted opposite T in transcription
d) New nucleotides are added on the 2'-OH of the ribose sugar.

(Old Depart. Questions book)

49- What sequence feature of mature mRNAs listed below is thought to protect mRNAs from degradation?

a) 3′ Poly (A) tail
b) 5′ G cap
c) Introns
d) Both a and b

(Old Depart. Questions book)

50- Eukaryotic RNA polymerases all have a requirement for a large variety of accessory proteins to enable them to bind promoters and form physiologically relevant transcription complexes; these proteins are termed:

a) Basal or general transcription factors
b) Activators
c) Accessory factors
d) Elongation factors

(Old Depart. Questions book)

51- Which of the following features of eukaryotic mRNA contribute importantly to message half-life?

a) 5' UTR sequences
b) The promoter
c) The operator
d) The poly (A) tail

(Old Depart. Questions book)

52- miRNA is synthesized by:

a. RNAP I
b. RNAP II
c. RNAP III
d. Prokaryotic RNAP

CANCELLED

53- Capping of mRNA occurs by:

a. 7 methyl GTP
b. 7 methyl UTP
c. 7 methyl ATP
d. 7 methyl CTP

54- Methylation of GTP to 7 methyl GTP cap occurs by:

a. SAM
b. SAH
c. Glycine betaine
d. All the above

55- Importance of splicing is:

a. Decrease incidence of mutation
b. Variability of mRNA
c. Protects the 5 end from 5 exonuclease.
d. a and b

56- Changing the coding information of mRNA is called:

a. Capping
b. Polyadenylation
c. Splicing
d. Editing

57- In apo B gene mRNA editing UCA is converted to:

a. UAA
b. UGA
c. UAG
d. All the above

58- mRNA is complementary to the nucleotide sequence of:

a) Coding strand
b) rRNA
c) tRNA
d) Template strand

(Exam 2026)

59- The enzyme responsible for synthesizing mRNA in eukaryotes is:

a) RNA polymerase I
b) RNA polymerase II
c) RNA polymerase III
d) DNA polymerase δ

(Exam 2026)

60- A toxin that blocks the binding of RNA polymerase to the promoter region inhibits:

a) Translation
b) Transcription initiation
c) RNA processing
d) DNA replication

(Exam 2026)

61- The TATA box functions as:

a) Enhancer sequence
b) Termination signal
c) Promoter region
d) Repressor site

(Exam 2026)

62- The type of RNA that carries genetic information from DNA to ribosomes is:

a) tRNA
b) mRNA
c) rRNA
d) snRNA

(Exam 2026)

63- In eukaryotes, the post-transcriptional modification that increases mRNA stability is:

a) Addition of 5' cap
b) Removal of exons
c) Histone methylation
d) DNA replication

(Exam 2026)

64- A 32-year-old female patient was diagnosed with systemic lupus erythematosus (SLE) with autoantibodies against small nuclear ribonucleoproteins (snRNPs). This is caused by defective:

a) Capping
b) Polyadenylation
c) Splicing
d) Alternative splicing

(Exam 2026)

65- Which of the following is synthesized by RNA polymerase II ?

a) tRNA
b) mRNA
c) 5S rRNA
d) 45S rRNA

(Exam 2024)

66- Which of the following general transcription factors has a kinase activity?

a) TFIID
b) TFIIE
c) TFIIF
d) TFIIH

(Exam 2024)

67- The strand of DNA that is transcribed into RNA is known as:

a) Coding strand
b) Template strand
c) Leading strand
d) Lagging strand

(Exam 2024)

68- In transcription, if a nucleotide sequence is present "upstream ", this means:

a) It is before the start site and takes negative numbers.
b) It is after the start site and takes positive numbers.
c) It is before the start site and takes positive numbers.
d) It is after the start site and takes negative numbers.

(Exam 2024)

69- Which of the following post transcriptional modifications occur at the 5'end:

a) Polyadenylation
b) Splicing
c) mRNA editing
d) Capping

(Exam 2024)

70- The number of proteins exceeds the number of genes due to:

a) Capping
b) Polyadenylation
c) Alternative splicing
d) mRNA editing

(Exam 2024)

71- Protection of mRNA from degradation is the function of:

a) Poly (A) tail
b) Introns
c) Telomeres
d) Exons

(Exam 2024)

72- Post-transcription modifications of mRNA include:

a) Trimming
b) Splicing
c) Acetylation
d) Carboxylation

(Exam 2024)

Answers

1-C	2-A	3-C	4-B	5-B	6-B	7-D	8-C	9-C
10-D	11-B	12-C	13-C	14-A	15-C	16-C	17-B	18-D
19-B	20-D	21-B	22-B	23-A	24-D	25-C	26-D	27-B
28-B	29-B	30-B	31-D	32-C	33-C	34-B	35-D	36-B
37-B	38-D	39-A	40-D	41-C	42-C	43-D	44-A	45-B	46-B
47-D	48-A	49-D	50-A	51-D	52-B	53-A	54-A	55-B
56-D	57-A	58-D	59-B	60-B	61-C	62-B	63-A	64-C	65-B
66-D	67-B	68-A	69-D	70-C	71-A	72-B			

Chapter 12

PROTEIN SYNTHESIS (TRANSLATION)

1-Which of the following processes doesn't occur in the eukaryotic cell nucleus?

a) Transcription
b) Protein synthesis
c) Removal of introns
d) DNA replication

(End of module exam 2023)

2-Replacement of Adenine by Guanine results in ..... mutation:

a) Transition
b) Transversion
c) Translocation
d) Frameshift

(End of module exam 2023)

3-Which of the following is not required in protein synthesis?

a) Ribosomes
b) Peptidyl transferase
c) Spliceosome
d) Amino acyl-tRNA synthetase

(End of module exam 2023)

4-The first amino acyl tRNA which initiates translation is .....

a) Methionine-tRNA
b) Tyrosine-tRNA
c) Alanine-tRNA
d) Glycine-tRNA

(End of module exam 2023)

5-Sickle cell anemia is due to:

a) Nonsense mutation
b) Missense mutation
c) Silent mutation
d) Frame shift mutation

(End of module exam 2021 , End of module exam 2023)

6-Together with proteins, rRNA:

a) Provides a site for polypeptide synthesis.
b) Transports amino acids to the ribosome.
c) Transcribes DNA.
d) Translates DNA.

End of module exam 2023

7-Which modification can allow proteins to bind Ca2+ as a step during blood coagulation?

a) Phosphorylation
b) Glycosylation
c) Hydroxylation
d) Carboxylation

(End of module exam 2023)

8-Which enzyme is incorrectly matched to its function?

a) RNA polymerase - transcription.
b) Aminoacyl-tRNA synthetase - attaches an amino acid to a tRNA.
c) Peptidyl transferase - transfers amino group from one amino acid to another.
d) DNA ligase - joins Okazaki fragments.

(End of module exam 2023)

9-The first codon to be translated on mRNA is:

a) AUG
b) GGU
c) GGA
d) AAA

(End of module exam 2021)

10-Which of the following statements is not correct about the genetic code?

a) Each codon consists of three bases
b) A single amino acid can have more than one codon
c) The genetic code is mainly universal
d) A single codon can code for more than one amino acid

(End of module exam 2021)

11-The synthesis of a certain protein does not require one of the following:

a) A ribosome
b) mRNA
c) Amino acids
d) Topoisomerases

(End of module exam 2021)

12-In translation, one of the following statements is incorrect:

a) The termination (stop) codon has no corresponding tRNA
b) ATP is required for amino acid activation
c) There is only one amino-acyl-tRNA synthetase in a cell
d) Peptidyl transferase activity is part of the 60S ribosomal subunit

(End of module exam 2021)

13-One of the following does not cause a mutation:

a) Replication error
b) Chemical mutagens
c) Phosphorylation
d) Irradiation

(End of module exam 2021)

14-Insertion of one or two bases into a gene leads to:

a) Nonsense mutation
b) Frameshift mutation
c) Silent mutation
d) Missense mutation

(End of module exam 2021)

15-In translation, one of the following statements is incorrect:

a) The termination (stop) codon has no corresponding tRNA
b) ATP is required for amino acid activation
c) There is only one amino-acyl-tRNA synthetase in a cell
d) Peptidyl transferase activity is part of the 60S ribosomal subunit

(Final exam 2021)

16-The first amino-acyl -tRNA which initiates translation in eukaryotes is:

a) Formylmethionyl tRNA
b) Methionyl tRNA
c) Tyrosinyl tRNA
d) Alanyl tRNA

(Final exam 2021)

17-The synthesis of a certain protein DOES NOT require one of the following:

a) A ribosome.
b) mRNA.
c) Amino acids.
d) Okazaki fragments

(End of the module 2019, Final exam 2019 - September sitting)

18-Which of the following statements is NOT a characteristic of the genetic code?

a) Each codon consists of three bases.
b) A single amino acid can have more than one codon.
c) The genetic code is mainly universal.
d) A single codon can code for more than one amino acid.

(End of the module 2019)

19-The amino acid sequence of the encoded protein is not changed in:

a) Missense mutation
b) Nonsense mutation
c) Silent mutation
d) Frame shift mutation

(End of the module 2019)

20-Which of the following statements about translation is FALSE?

a) The first codon to be translated on mRNA is AUG.
b) Peptidyl transferase activity is part of the 60S subunit.
c) For protein synthesis, initial activation of amino acid is required.
d) The first ammino acyl transfer RNA which initiates translation in eukaryotes is Formylmethionyl tRNA.

(End of the module 2019)

21-Which of the following statements is NOT a characteristic of the genetic code?

a) Each codon consists of three bases
b) A single amino acid can have more than one codon
c) The genetic code is mainly universal
d) A single codon can code for more than one amino acid

(End of the module 2020)

22-Which of the following is NOT a requirement for protein synthesis?

a) Okazaki fragments
b) Ribosomes
c) Peptidyl transferase
d) Amino acyl-tRNA synthetase

(End of the module 2020)

23-Nonsense codons lead to:

a) Amino acid activation
b) Initiation of protein synthesis
c) Termination of protein synthesis
d) Elongation of polypeptide chains

(End of the module 2020)

24-Codons are present in

a) mRNA
b) Template strand of DNA
c) tRNA
d) rRNA

(Final exam 2019)

25-Translation is the:

a) Synthesis of DNA from mRNA template
b) Synthesis of protein from mRNA template
c) Synthesis of RNA from mRNA template
d) Synthesis of RNA from DNA template

(Depart. book)

26-AUG, the only codon for methionine, is important as:

a) A releasing factor for peptide chains
b) A polypeptide chain terminating codon
c) Recognition site on tRNA
d) A polypeptide chain initiating codon

(Depart. book)

27-Enzymes called amino acyl-tRNA synthetases

a) Synthesize tRNA
b) Attaches amino acids to tRNA
c) Strips tRNA from its amino acid in the process of translation
d) Helps tRNA synthesize amino acids

(Depart. book)

28-Which molecule contains the genetic code?

a) DNA
b) mRNA
c) tRNA
d) rRNA

(Old Depart. book)

29-Which of the following processes is absent in the eukaryotic cell nucleus?

a) Transcription
b) Translation
c) Removal of introns
d) DNA replication

(Depart. Questions book)

30-Which of the following is inapplicable for the genetic code?

a) Specific
b) Degenerate
c) Universal
d) Overlapping

(Depart. Questions book)

31-The genetic code lacks which of the following?

a) Each codon consists of three bases
b) A single amino acid can have more than one codon
c) The genetic code is mainly universal
d) A single codon can code for more than one amino acid

(Depart. Questions book)

32-While the genetic code has 64 codons, there are only 20 naturally occurring amino acids. Consequently, some amino acids are encoded by more than one codon. Which best describes this feature of the genetic code?

a) Degenerate
b) Duplicative
c) Nonoverlapping
d) Overlapping

(Depart. Questions book)

33-The flexibility of the 1st base in the anticodon loop of the tRNA when pairing with the 3rd base of the codon is called:

a) Degeneracy of genetic code
b) Wobble hypothesis
c) The central dogma
d) Alternative splicing

(Depart. Questions book)

34-AUG, the only codon for methionine is important as:

a) releasing factor for peptide chains
b) A polypeptide chain terminating codon
c) Recognition site on tRNA
d) A polypeptide chain initiating codon

(Depart. Questions book)

35-In biosynthesis of proteins the chain terminating codons are:

a) UAA, UAG and UGA
b) UGG, UGU and AGU
c) AAU, AAG and GAU
d) GCG, GCA and GCU

(Depart. Questions book)

36-Transfer RNA transfers:

a) Information from DNA to ribosome
b) Information from mRNA to cytosol
c) Amino acids from cytosol to ribosomes
d) Proteins from ribosomes to cytosol

(Depart. Questions book)

37-Which one of the following is noncompulsory during synthesis of a protein?

a) A ribosome
b) mRNA
c) Amino acids
d) Okazaki fragments

(Depart. Questions book)

38-Which best defines a codon?

a) A protein that begins transcription by breaking apart H- bonds
b) A free-floating base that attaches to an open DNA strand
c) A code word of three bases on mRNA that specifies one amino acid
d) A strong bond between two complementary nitrogen bases

(Depart. Questions book)

39-During cell analysis, their cytoplasm was determined to have high content of aminoacyl tRNA synthetase. This enzyme ensures the following process:

a) Amino acid activation
b) Amino acid synthesis
c) Amino acid inactivation
d) Amino acid degradation

(Depart. Questions book)

40-Which of the following is irrelevant to protein synthesis?

a) Ribosomes
b) Peptidyl transferase
c) Spliceosome
d) Amino acyl-tRNA synthetase

(Depart. Questions book)

41-At the stage of translation in the rough endoplasmic reticulum, Amino acids are joined together by peptide bonds in a specific sequence. The sequence of amino acids in a polypeptide corresponds to the sequence of:

a) mRNA codons
b) tRNA nucleotides
c) rRNA anticodons
d) rRNA nucleotides

(Depart. Questions book)

42-During the process of translation, the formation of the bond between two successive amino acids is catalyzed by:

a) Peptidyl transferase
b) Glucosyl transferase
c) Peptidyl polymerase
d) Aminoacyl tRNA synthetase

(Depart. Questions book)

43-Peptidyl transferase activity is present in:

a) 40 S ribosomal subunit
b) 60 S ribosomal subunit
c) eKF-2
d) Amino acyl tRNA

(Depart. Questions book)

44-Activation of trypsinogen into trypsin in the small intestine is an example of:

a) Trimming
b) Phosphorylation
c) Glycosylation
d) Carboxylation

(Depart. Questions book)

45-Prolyl and lysyl residues should be modified in collagen for its proper activity by which post-translational modification?

a) Phosphorylation
b) Glycosylation
c) Hydroxylation
d) Carboxylation

(Depart. Questions book)

46-What type of mutation is represented in the second strand of DNA?
TACGGCACT
TACGGACT

a) Deletion
b) Insertion
c) Substitution
d) No mutation

(Depart. Questions book)

47-Which of the following is a transition point mutation:

a) Adenine is replaced by cytosine
b) Uracil is replaced by adenine
c) Guanine is replaced by adenine
d) Guanine is replaced by uracil

(Depart. Questions book)

48-Substitution of a thymine base by adenine in DNA is known as:

a) Transposition
b) Transversion
c) Transition
d) Frame shift mutation

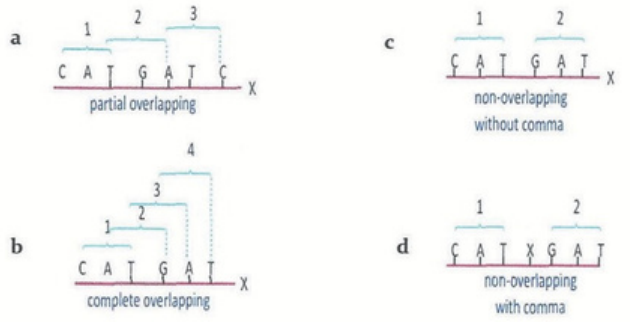
(Depart. Questions book)

49-Insertion or deletion of one or two bases in a gene resulting in:

a) Missense mutation
b) Silent mutation
c) Frame shift mutation
d) Nonsense mutation

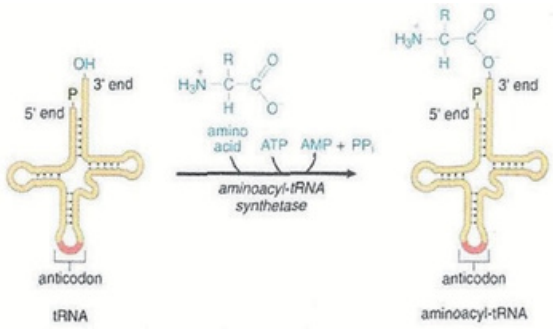
(Depart. Questions book)

50-In the following figure, which best describes the genetic code?



(Depart. Questions book)

51-The following figure describes which step of protein synthesis?



a) Activation of an amino acid
b) Initiation of translation
c) Elongation of a polypeptide chain
d) Termination of translation

(Depart. Questions book)

52-One of the following processes DOES NOT occur in the eukaryotic cell nucleus:

a) transcription
b) translation
c) removal of introns
d) DNA replication

(Old Depart. Questions book)

53-Which of the following statements is NOT a characteristic of the genetic code.

a) Each codon consists of three bases.
b) A single amino acid can have more than one codon.
c) The genetic code is mainly universal.
d) A single codon can code for more than one amino acid.

(Old Depart. Questions book)

54-The first amino acyl transfer RNA which initiates translation in prokaryotes is:

a) MethionyltRNA
b) Formylmethionyl tRNA
c) Tyrosinyl tRNA
d) Alanyl tRNA

(Old Depart. Questions book)

55-Insertion or deletion of one or two bases in a gene results in:

a) Missense mutation.
b) Silent mutation.
c) Frame shift mutation.
d) Nonsense mutation.

(Old Depart. Questions book)

56-The genetic code is NOT characterized by being ONE of the following:

a) Specific
b) Degenerate
c) Universal.
d) Discontinuous

(Old Depart. Questions book)

57-All of the following may lead to mutation except:

a) Replication error.
b) Chemical mutagens.
c) Phosphorylation.
d) Irradiation.

(Old Depart. Questions book)

58-One of the following is a transition point mutation:

a) Adenine is replaced by cytosine.
b) Uracil is replaced by adenine.
c) Guanine is replaced by adenine.
d) Guanine is replaced by uracil.

(Old Depart. Questions book)

59-Phosphorylation of proteins occurs at:

a) Cysteine residues.
b) Serine residues.
c) Histidine residues.
d) Alanine residues.

(Old Depart. Questions book)

60-Which one of the following changes in the coding region of an mRNA (caused by point mutation) would result in translation of a protein identical to the normal protein:

a) UCA: UAA
b) UCA: CCA
c) UCA: UCU
d) UCA: ACA

(Old Depart. Questions book)

61-Which of the following is not a requirement for protein synthesis:

a) Ribosomes
b) Peptidyl transferase
c) Spliceosome
d) Amino acyl-tRNA synthetase

(Old Depart. Questions book)

62-Which of the following statement is correct?

a) Termination codon has no corresponding tRNA.
b) Activated amino acid binds to the 5′ end of the respective tRNA.
c) CTP is required for amino acid activation.
d) There is only one amino acyl-tRNA synthetase in a cell.

(Old Depart. Questions book)

63-Which defines a codon?

a) A protein that begins transcription by breaking apart H bonds.
b) A free-floating base that attaches to an open DNA strand.
c) The genetic code word of three bases on mRNA that specify one amino acid.
d) The strong bond between two complementary nitrogen bases.

(Old Depart. Questions book)

64-Which of the following changes would be expected if a CAUUUG sequence of bases mutated to CACUUG?

a) The amino acid sequence would be shorter than expected.
b) The identity of one amino acid would change.
c) The identity of more than one amino acid would change.
d) The amino acid sequence would remain unchanged.

(Old Depart. Questions book)

65-Degenerated codon differs mostly in:

a) The identities of their second base.
b) The identities of their third base.
c) The wobble position.
d) Both (b) and (c)

(Old Depart. Questions book)

66-The growing polypeptide chain is released from the ribosomes when:

a) A chain terminating codon is reached.
b) A chain terminating tRNA binds to the ribosome.
c) The 7-methyl guanosine cap is reached.
d) The poly A tail is reached.

(Old Depart. Questions book)

67-One of the following is initiation codon

a. UGA
b. UAA
c. UAG
d. AUG

68-Codons that code for the same amino acids are called:

a. Reverse codons
b. Anti-codons
c. Synonym codons
d. All the above

69-In Phosphorylation, Phosphate is added to one of the following amino acids:

a. Phenylalanine
b. Hydroxy proline
c. Tryptophan
d. Tyrosine

70-Carboxylation of protein occurs on the amino acid:

a. Aspartate
b. Glutamate
c. Lysine
d. Alanine

71-Hydroxylation of proteins occurs on the amino acid(s):

a. OH of serine or threonine
b. Amide group of asparagine
c. Glutamate
d. Proline and lysine

72-Acetylation or methylation of protein occur on the amino acid:

a. OH of serine or threonine
b. N-terminal amino acid
c. Glutamate
d. Proline and lysine

73-The amino acid in a protein is replaced by a different amino acid in:

a. Missense mutation
b. None sense mutation
c. Silent mutation
d. None of the above

74-Frame shift mutation occurs with:

a. Insertion of 3 bases
b. Deletion of 3 bases
c. Insertion of 1 or 2 bases
d. All the above

75-Using written convention which one of the following sequences is complimentary to 5TGGCAGCCT3?

a. 5 ACC GTC GGA3
b. 5 ACC GUC GGA3
c. 5AGG CTG CCA3
d. 5 TGG CTC GGA3

76-Amino acids may have more than 1 codon, this is called:

a. Specificity
b. Degeneracy
c. Wobble theory
d. Reading frame

77-All the following can cause serious effects EXCEPT:

a. Mis sense mutation
b. None sense mutation
c. Silent mutation
d. None of the above

78-Which of the following point mutations would NOT produce a change in the protein translated from an mRNA?

a. UCA → UAA.
b. UCA → CCA.
c. UCU → UCA
d. UCA → ACA.

79-Identify the incorrect statement regarding the genetic code:

a) Each codon consists of three bases
b) A single amino acid can have more than one codon
c) The genetic code is mainly universal
d) A single codon can code for more than one amino acid

(Exam 2026)

80-The start codon in mRNA is:

a) UAA
b) UGA
c) AUG
d) UAG

(Exam 2026)

81-A mutation changes the codon UAU (tyrosine) to UAA (stop codon). This is an example of:

a) Missense mutation
b) Nonsense mutation
c) Silent mutation
d) Frameshift mutation

(Exam 2026)

82-Aminoacyl-tRNA synthetase catalyzes the attachment of:

a) mRNA to tRNA
b) Amino acid to tRNA
c) rRNA to tRNA
d) Polypeptide to mRNA

(Exam 2026)

83-Activation of trypsinogen into trypsin in the small intestine is an example of:

a) Trimming
b) Phosphorylation
c) Glycosylation
d) Carboxylation

(Exam 2026)

84-The modification that allows proteins to bind Ca2+ during blood coagulation is:

a) Phosphorylation
b) Glycosylation
c) Hydroxylation
d) Carboxylation

(Exam 2026)

85-During the process of translation, the formation of the bond between two successive amino acids is catalyzed by:

a) Peptidyl transferase
b) Glucosyl transferase
c) Peptidyl polymerase
d) Aminoacyl tRNA synthetase

(Exam 2024)

86-Phenylalanine is coded by two synonym codons, this is called -------

a) Degeneracy ✓ (Marked Answer)
b) Specificity
c) Universality
d) Overlapping

(Exam 2024)

87-Which of the following describes the ability of tRNAs to recognize more than one codon for a specific amino acid:

a) Degeneracy of genetic code
b) Wobble hypothesis
c) The central dogma
d) Alternative splicing

(Exam 2024)

88-The first amino-acyl tRNA which initiates translation is .....

a) Methionine-tRNA
b) Tyrosine-tRNA
c) Alanine-tRNA
d) Glycine-tRNA

(Exam 2024)

89-Peptidyl transferase activity is present in:

a) 40 S ribosomal subunit
b) 60 S ribosomal subunit
c) eEF-2
d) Amino acyl tRNA

(Exam 2024)

90-Which enzyme is incorrectly matched to its function?

a) RNA polymerase - transcription.
b) Aminoacyl-tRNA synthetase - attaches an amino acid to a tRNA.
c) Peptidyl transferase - transfers amino group from one amino acid to another.
d) DNA helicases - unwinding and separation of the two DNA strands

(Exam 2024)

91-Sickle cell anemia is due to:

a) Nonsense mutation
b) Missense mutation
c) Silent mutation
d) Frame shift mutation

(Exam 2024)

92-Presence of synonym codon results from one of the following:

a) Nonsense mutation
b) Sense mutation
c) Silent mutation
d) Missense mutation

(Exam 2024)

93-Replacement of Adenine by cytosine results in ..... mutation:

a) Transition
b) Transversion
c) Translocation
d) Frameshift

(Exam 2024)

Answers

1-B	2-A	3-C	4-A	5-B	6-A	7-D	8-C	9-A
10-D	11-D	12-C	13-C	14-B	15-C	16-B	17-D	18-D
19-C	20-D	21-D	22-A	23-C	24-A	25-B	26-D	27-B
28-B	29-B	30-D	31-D	32-A	33-B	34-D	35-A	36-C
37-D	38-C	39-A	40-C	41-A	42-A	43-B	44-A	45-C
46-A	47-C	48-B	49-C	50-C	51-A	52-B	53-D	54-B
55-C	56-D	57-C	58-C	59-B	60-C	61-C	62-A	63-C
64-D	65-D	66-A	67-D	68-C	69-D	70-B	71-D	72-B
73-A	74-C	75-A	76-B	77- C	78- C	79- D	80- C	81- B
82-B	83-A	84-D	85-A	86-A	87-B	88-A	89-B	90-C
91-B	92-C	93-B						

Chapter 13 REGULATION OF GENE EXPRESSION

1-DNA methylation and histone acetylation are examples for:

a) Gene mutation.
b) Chromosomal rearrangement.
c) Epigenetic regulation of gene expression.
d) Translocation.

(End of module exam 2023)

2-Function of TATA box in eukaryotes is:

a) It defines where transcription is to commence
b) It is a termination signal for translation
c) Responsible for ribosomes recognition
d) It determines how frequently transcription is to occur

(Final exam 2022 - September, Faculty summer revision 2024)

3-Post-translation processing includes:

a) 5' capping
b) 3' polyadenylation
c) Trimming
d) Splicing

(End of the module 2020)

4-One of the following is a post-transcriptional modification of mRNA:

a) Covalent modification
b) Trimming
c) Capping and polyadenylation
d) Glycosylation

(Final exam 2022 - September, Faculty summer revision 2024)

5-Post-transcriptional modifications involves all of the following except:

a) Addition of 7-methylguanosine triphosphate cap
b) Addition of polyadenylate (poly A) tail
c) Insertion of nucleotides
d) Splicing

(End of the module 2020)

6-Post-transcriptional modifications of mRNA do not involve:

a) Capping
b) Polyadenylation
c) Splicing
d) Trimming

(End of module exam 2023, End of the module 2019)

7-Pre-transcriptional regulation includes:

a) Epigenetic mechanisms
b) mRNA stability
c) Trimming
d) Covalent modification

(Depart. Questions book)

8-DNA methylation and histone acetylation are examples of .....

a) Genetic mutation
b) Chromosomal rearrangements
c) Epigenetic phenomena
d) Translocation

(Depart. Questions book)

9-DNA methylation:

a) Is irreversible
b) Occurs on guanine
c) Causes gene silencing
d) Catalyzed by DNA ligase

(Depart. Questions book)

10-The cell responds to methotrexate chemotherapy by:

a) Gene rearrangement
b) DNA methylation
c) Gene amplification
d) Gene diminution

(Depart. Questions book)

11-Diversity of immunoglobulins is achieved by:

a) Gene rearrangement
b) DNA methylation
c) Gene amplification
d) Gene diminution

(Depart. Questions book)

12-The primary difference between enhancers and promoter proximal elements is that enhancers .....

a) Are transcription factors; promoter proximal elements are DNA sequences
b) Enhance transcription; promoter proximal elements inhibit transcription
c) Are at considerable distances from the promoter; promoter proximal elements are close to the promoter
d) Are DNA sequences; promoter proximal elements are proteins

(Depart. Questions book)

13-TATA box:

a) Is transacting element
b) Controls the frequency of transcription
c) Located 40-200 bp upstream the transcription start site
d) Defines where transcription is to start

(Depart. Questions book)

14-Not long ago, it was believed that a count of the number of protein-coding genes would provide a count of the number of proteins produced in any given eukaryotic species. This is incorrect, largely due to the discovery of widespread.....

a) Chromatin condensation control
b) Transcriptional control
c) Alternative splicing
d) Translational control

(Depart. Questions book)

15-Post-transcriptional regulation includes:

a) Epigenetic mechanisms
b) Capping
c) Trimming
d) Basal expression elements

(Depart. Questions book)

16-Posttranscriptional processing of mRNA includes:

a) Replacement of UU with CCA nucleotides at the 3' end
b) Removal of poly A tail
c) Removal of leader sequence
d) Alternative splicing

(Depart. Questions book)

17-Micro RNAs (miRNAs):

a) Initiate translation
b) Are cis-acting elements
c) Reduce expression of the target mRNA
d) Are transacting elements

(Depart. Questions book)

18-IF-4 proteins:

a) Prevent the formation of preinitiation complex
b) Activated under stress
c) Interact with poly A tail
d) Block protein synthesis

CANCELLED

(Depart. Questions book)

19-Trimming is:

a) Transcriptional modification.
b) Pretranscriptional modification
c) Posttranscriptional modification
d) Posttranslational modification

(Depart. Questions book)

20-One of the following is a covalent modification to a peptide chain:

a) Carboxylation
b) Trimming
c) Capping
d) Editing

(Depart. Questions book)

21-One of the following is an example of a posttranslational covalent modification that may regulate protein function:

a) Phosphorylation
b) Trimming
c) Polyadenylation
d) Deamination

(Depart. Questions book)

22-How does nucleosome formation on gemomic DNA affect the initiation and/or elongation phases of transcription?

a) Nucleosomes inhibit access of enzymes involved in all phases of transcription.
b) Nucleosomes recruit histone and DNA modifying enzymes, and the actions of these recruited enzymes affect the access of transcription proteins to DNA.
c) Nucleosomes induce DNA degradation where the DNA contacts the histones.
d) Nucleosomes have no significant effect on transcription.

(Old Depart. Questions book)

23-Which types of molecules interact with eukaryotic mRNA gene core promoter sites to facilitate the association of RNA polymerase II?

a) Termination factors.
b) General, or basal transcription factors (i.e., the GTFs).
c) Elongation factors.
d) GTPases.

(Old Depart. Questions book)

24-Which of the following histone amino acids are typically acetylated?

a) Lysine
b) Arginine
c) Asparagine
d) Leucine

(Old Depart. Questions book)

25-Concerning epigenetic regulation of eukaryotic gene expression via DNA methylation One of the following statements is incorrect:

a) Methylation of cytosine to form 5'methylcytosine by a DNA methyltransferase.
b) The target cytosine is usually located next to a guanine residue, forming a CpG island, which is often near or in the promoter region of a gene.
c) It is associated with inducing the activities of certain genes.
d) It is a mechanism for regulating gene expression during cell differentiation, particularly in fetal development.

(Old Depart. Questions book)

26-Concerning chromatin remodeling One of the following statements is incorrect:

a) Is the changes in the state of the chromatin, to make the desired region accessible to RNAP and other proteins (e.g., transcriptional factors).
b) Histone acetylation is one of the mechanisms of chromatin remodeling.
c) Histone acetylation removes a negative charge from the ϵ-amino group of the lysine, there by reducing the electrostatic interactions between the histones and the negatively charged DNA.
d) The acetyl groups can be removed by histone deacetylases (HDACs) to form inactive chromatin.

(Old Depart. Questions book)

27-Enhancers:

a) Are trans-acting sequences.
b) They act by binding certain proteins that inhibit the binding of transcription factors to the promoter.
c) They can be close to or thousands of base pairs away from the promoter (upstream or downstream) and they can occur on either strand of DNA.
d) Are sites of DNA acetylation

(Old Depart. Questions book)

28-Concerning micro RNAs (miRNAs) One of the following statements is incorrect:

a) They are small non-coding RNA molecules (containing about 22 nucleotides).
b) They function in RNA silencing and post-transcriptional regulation of gene expression.
c) MicroRNAs are produced from either their own genes or from introns.
d) Each miRNA can interact with several hundred mRNAs increasing expression of the target mRNA.

(Old Depart. Questions book)

29-Enhancers in eukaryotic DNA:

a) Must be adjacent to the promoter
b) Can be close to or thousands of base pairs away from the promoter
c) Are DNA elements that permit hormones to regulate replication
d) Repress gene expression

(Exam 2026)

30-In eukaryotic cells, histone acetylation leads to:

a) DNA condensation
b) Gene silencing
c) Gene activation
d) RNA degradation

(Exam 2026)

31-The cell responds to methotrexate chemotherapy by:

a) Gene rearrangement
b) DNA methylation
c) Gene amplification
d) Gene diminution

(Exam 2024)

Answers

1-C	2-A	3-C	4-C	5-C	6-D	7-A	8-C	9-C
10-C	11-A	12-C	13-D	14-C	15-B	16-D	17-C	CANCELLED
19-D	20-A	21-A	22-B	23-B	24-A	25-C	26-C	27-C
28-D	29-B	30-C	31-C					

Chapter 14 CELL CYCLE, APOPTOSIS AND TUMOR NECROSIS GENES

1-DNA replication occurs in the ..... of cell cycle.

a) G1 phase
b) S phase
c) M phase
d) G2 phase

(End of module exam 2023)

2-Release of cytochrome c from the mitochondria during apoptosis occurs in response to:

a) Cellular stress as DNA damage.
b) Caspase 8
c) Caspase 9
d) Caspase activated DNase.

(End of module exam 2023)

3-Which of the following types of cells become permanently in G0, post-mitosis?

a) Embryonic stem cells
b) Hematopoietic cells
c) Hepatocytes
d) Neurons

(Depart. book)

4-Which of the following statements is correct concerning cytochrome C?

a) It is released from the nucleus to the cytosol
b) It activates CADase directly
c) Its release occurs in the intrinsic pathway of apoptosis
d) Its release is promoted by binding of a ligand to TNF or FAS receptors

(Depart. book)

5-Quiescent phase of the cell cycle refers to:

a) S phase
b) M phase
c) G0 phase
d) G1 phase

(Depart. Questions book)

6-Two daughter cells are formed in ..... of the cell cycle.

a) S phase
b) M phase
c) G2 phase
d) G1 phase

(Depart. Questions book)

7-Cell cycle initiation occurs by binding of the cell to:

a) Enzyme
b) Growth factor
c) Hormone
d) Transcription factor

(Depart. Questions book)

8-CDk4 forms complex with:

a) Cyclin A
b) Cyclin B
c) Cyclin D
d) Cyclin E

(Depart. Questions book)

9-Cyclin B forms complex with:

a) CDK1
b) CDK2
c) CDK4
d) CDK6

(Depart. Questions book)

10-Spindle assembly check point of the cell cycle:

a) Check for proper attachment of chromosomes to spindle
b) Check for cell size
c) Check for DNA damage
d) Check for growth factors

(Depart. Questions book)

11-Restriction point of the cell cycle is:

a) To check for DNA Damage
b) To ensure normal cell size
c) Point beyond which cell become committed to enter the S phase
d) For DNA replication

(Depart. Questions book)

12-Genetically programmed cell death is termed:

a) Necrosis
b) Mutation
c) Apoptosis
d) Transversion

(Depart. Questions book)

13-Extrinsic pathway of apoptosis is initiated by:

a) Caspase 9
b) Caspase 8
c) Binding of a ligand to TNF or FAS receptors
d) Cytochrome c

(Depart. Questions book)

14-Caspases belong to the class of:

a) Serine proteases
b) Cystine proteases
c) Aspartate proteases
d) Hydrolases

(Depart. Questions book)

15-Which cellular organelles are involved in the initiation of the intrinsic pathway of apoptosis?

a) Nucleus
b) Mitochondria
c) Endoplasmic reticulum
d) Lysosomes

(Depart. Questions book)

16-One of these is pro-apoptotic gene:

a) Bax
b) BCL-2
c) BCl-x
d) MYC

(Depart. Questions book)

17-One of these is anti-apoptotic gene:

a) Bax
b) Bok
c) Bak
d) Bcl-2

(Depart. Questions book)

18-Extrinsic pathway of apoptosis is initiated by:

a) Cyclins
b) Cyclin-dependent kinases
c) Binding of a ligand to TNF or FAS receptors
d) Cytochrome c

(Exam 2024)

19-Increased expression of Bcl-2 protein results in:

a) Promotion of apoptosis
b) Inhibition of apoptosis
c) DNA repair failure
d) Mitotic arrest

(Exam 2026)

20-One of the following is anti-apoptotic modulator:

a) Bax
b) Bok
c) Bak
d) Bcl-2

(Exam 2026)

Answers

1-B	2-A	3-D	4-C	5-C	6-B	7-B	8-C	9-A
10-A	11-C	12-C	13-C	14-C	15-B	16-A	17-D	18-C
19-B	20-D							

CHAPTER 1 INTRODUCTION & PHYSIO-CHEMISTRY PRINCIPLES

1-Which type of bonding exist between carboxyl and amino group (or present in NaCl)?

a. Polar covalent bond.
b. Nonpolar covalent bond.
c. Ionic bond.
d. Hydrophobic bond.

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2- The interaction between nonpolar compounds as oil in water is called:

a. Covalent bonds.
b. Hydrophobic interactions.
c. Hydrogen bonds.
d. Ionic bonds.

(Depart. book)

3- Attraction between two opposite atom dipoles is called:

a. Ionic bonds.
b. Van der Waal interactions.
c. Polar covalent bonds.
d. Hydrophobic interactions.

(Depart. book)

4- The non-covalent bond include all of the following except:

a. Ionic bond
b. Hydrogen bonds.
c. Peptide bond.
d. Vander waals forces.

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5- The tendency of nonpolar compounds to cluster together in an aqueous medium is known as:

a. Nonpolar covalent bond.
b. Hydrogen bond.
c. Hydrophobic interactions.
d. Van der Waal interactions.

(Depart. book)

6- 1.One of the following bonds occurs when a hydrogen atom comes between 2 electronegative atoms:

a. Hydrogen bond
b. Van der Waals forces
c. Disulfide bond
d. Hydrophobic interaction

7- Occurs when the water pushes 2 non-polar compounds towards each other:

a. Hydrogen bond
b. Van der Waals forces
c. Ionic interaction
d. Hydrophobic interaction

8- Important for the structure and properties of water and DNA:

a. Hydrogen bond
b. Van der Waals forces
c. Ionic interaction
d. Hydrophobic interaction

9- Occurs when 2 atoms or groups are close to each other:

a. Hydrogen bond
b. Van der Waals forces
c. Ionic interaction
d. Hydrophobic interaction

10- Occurs between 2 oppositely charged atoms or groups:

a. Hydrogen bond
b. Van der Waals forces
c. Ionic interaction
d. Hydrophobic interaction

11- When there are 2 methyl groups in water, the possible non-covalent bond between them will be:

a. Hydrogen bond
b. Van der Waals forces
c. Ionic interaction
d. Hydrophobic interaction

12- When there is a carboxyl group (-COO-) close to an amino (-NH3+) in the compound, the possible non-covalent bond between them will be:

a. Hydrogen bond
b. Van der Waals forces
c. Ionic interaction
d. Hydrophobic interaction

13- When a Na atom becomes separated from a K atom by a distance of 0.3 nm in the compound, the possible non-covalent bond between them will be:

a. Hydrogen bond
b. Van der Waals forces
c. Ionic interaction
d. Hydrophobic interaction

14- When the hydrogen in an OH group is attracted to the nitrogen of NH in the compound, the possible non-covalent bond between them will be:

a. Hydrogen bond
b. Van der Waals forces
c. Ionic interaction
d. Hydrophobic interaction

15- One of the following is NOT characteristic for non-covalent bonds:

a. Characterized by no sharing of electrons
b. Weak bonds and easy to be broken down
c. Mainly formed due to electrostatic attraction
d. Characterized by sharing of electrons

16- The weakest non-covalent force is:

a. Hydrogen bond
b. Van der Waals forces
c. Ionic interaction
d. Hydrophobic interaction

17- Formation of transient dipole occurs in:

a. Hydrogen bond
b. Van der Waals forces
c. Ionic interaction
d. Hydrophobic interaction

18- If we put 2 oil droplets in water and after a while they become one larger droplet, the possible force is:

a. Hydrogen bond
b. Van der Waals forces
c. Ionic interaction
d. Hydrophobic interaction

19- Polarity of a covalent bond is due to:

a. Electronegativity
b. Sharing of electrons
c. Electrostatic attractions
d. Formation of transient dipole

Answers

1-C	2-B	3-B	4-C	5-C	6-A	7-D	8-A	9-B	10-C
11-D	12-C	13-C	14-A	15-D	16-B	17-B	18-D	19-A	